Chapter 6
Metabolism: Energy, Heat, Work, and Power of the Body

Abstract The study of energy in the body begins with an overview of heat and heat flow. The analysis of the energy content of food is followed by how energy is stored in the body and used in the form of ATP. Energy usage by the body is described in terms of the metabolic rate under passive and active conditions, and how it is related to body weight. The net energetics of the body includes several modes of passive and active heat loss, and this is related to body temperature.

We cannot function without energy. The processes involved in the energy intake, storage, and use by the body are collectively called the *metabolism*; the discipline describing this area is sometimes called *bioenergetics*. More generally, metabolism is any energy usage by the body, and is the sum of all chemical processes performed by the cells in order to keep the body alive. For a complete picture we need to include input of food and oxygen to the body, energy storage, and loss of energy by the body through the loss of heat and work done by the body, as is shown in Fig. 6.1.

Metabolic processes can be divided into catabolic and anabolic reactions. In *catabolic* reactions complex molecules are broken into simple ones, for purposes such as energy usage. In *anabolic* reactions simple molecules are combined to form complex ones, for purposes such as energy storage.

The body uses food to (1) operate organs, (2) maintain a constant temperature by using some of the heat that is generated by operating the organs (while the rest is rejected), (3) do external work, (4) build a stored energy supply (fat) for later needs, (5) grow to adulthood, and (6) help the fetus grow during pregnancy and then nurse infants. About 5–10% of the food energy intake is excreted in the feces and urine.

We will first consider the basics of the conservation of energy (thermodynamics). Then we will examine the energy content of food and the way it is stored in the body. We will see what the body's metabolic rate needs to be to perform tasks. (This is technically the catabolic rate.) We will then analyze how the body loses energy as heat. These steps are interrelated in a complex feedback and control mode that is discussed in Chap. 13.

For excellent general discussions about metabolism see [10, 11, 19, 20, 35, 57, 89, 106].

© Springer International Publishing Switzerland 2016

I.P. Herman, *Physics of the Human Body*, Biological and Medical Physics, Biomedical Engineering, DOI 10.1007/978-3-319-23932-3_6

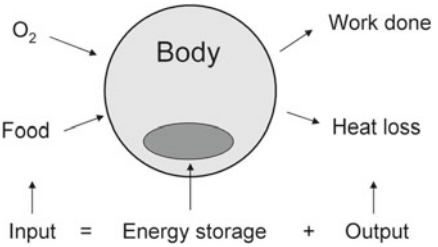


Fig. 6.1 Energy flow into and from the body

6.1 Conservation of Energy and Heat Flow

Let us briefly review some of the basics of the thermodynamics and heat flow physics that we will use in this discussion.

The First Law of Thermodynamics is essentially the conservation of energy in any process. In reference to the body, it can be stated as

$$\Delta U = Q - W, \tag{6.1}$$

where ΔU is the change in stored energy, Q is the heat flow to the body, and W is the mechanical work done by the body. The stored energy decreases, $\Delta U < 0$, when there is heat flow from the body, $Q < 0$, and work done by the body, $W > 0$. This type of work is purely mechanical in nature, such as in moving and lifting items. Heat flow includes heat production from the metabolism (Q_{met}) and heat loss (Q_{loss}) from radiation, convection, conduction, and evaporation. We can express $Q = Q_{\text{met}} + Q_{\text{loss}}$, where metabolic heat production is positive and a negative Q_{loss} indicates heat flow away from the body, so

$$\Delta U = Q_{\text{met}} + Q_{\text{loss}} - W. \tag{6.2}$$

Q_{met} is called the metabolic rate (MR).

Relationships in thermodynamics involve amounts of energies changing in a process at equilibrium and not those changing per unit time, i.e., the kinetics of that process, which involves the rates of energy changes or flows. The study of the metabolism usually involves rates and therefore

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{\mathrm{d}Q_{\text{met}}}{\mathrm{d}t} + \frac{\mathrm{d}Q_{\text{loss}}}{\mathrm{d}t} - \frac{\mathrm{d}W}{\mathrm{d}t} \tag{6.3}$$

is more appropriate. (We need to be careful about signs. The body increases its energy with terms such as $\mathrm{d}Q_{\text{met}}/\mathrm{d}t$ that are positive and loses it with terms such as $\mathrm{d}Q_{\text{loss}}/\mathrm{d}t$ that are negative. The amount of heat flowing from the body is $-\mathrm{d}Q_{\text{loss}}/\mathrm{d}t$, which is a positive quantity.)

Table 6.1 Units of power

1 watt (W) = 1 J/s
100 W = 1.43 kcal/min
1 horsepower (hp) = 746 W = 642 kcal/h
1 kcal/min = 69.7 W = 0.094 hp
1 kcal/h = 1.162 W

All types of energy have the same units, including heat (often expressed in terms of calories) and work (often expressed in terms of joules). One important conversion between units is

$$1 \text{ calorie (cal)} = 4.184 \text{ joule (J)}. \tag{6.4}$$

1 kilocalorie (1 kcal = 1,000 cal) is sometimes called 1 Cal, which is also known as a food calorie. The energy content of food is always expressed in terms of these Cal (kcal) units. The relations between the various units of power (energy/time) are given in Table 6.1.

One important concept in the thermal physics of the body is the *heat capacity* C, which is the energy (or more specifically, the heat) required to raise the temperature T of an object by 1 °C. The heat capacity per unit volume or mass is the *specific heat* c. The heat capacity is an *extensive property* of a given object, while the specific heat is an *intensive property* of a material. (This classification is analogous to that for mechanical properties described in Chap. 4.) If the specific heat is expressed per unit mass, it can be converted to that per unit volume by multiplying it by the mass density ρ. The heat capacity C is the specific heat (expressed per unit mass) $\times$ the total object mass m, so $C = mc$.

The temperature rise ΔT of an object with a heat flow Q to the body is

$$\Delta T = \frac{Q}{mc}. \tag{6.5}$$

For water, c_{water} = 1.0 cal/g-°C = 1.0 kcal/kg-°C. Even though the human body contains much water, the average specific heat of the body is a bit less, c_{b} = 0.83 cal/g-°C = 0.83 kcal/kg-°C. This means that it takes 83 kcal to raise the temperature of a 100 kg person by 1°C. This 83 kcal (83 food calories) is approximately the food energy content of a slice of bread. An obvious question arises: *If most of our metabolized energy becomes heat, why does not our body temperature increase by* 1.0 °C = 1.8 °F *each time we eat and metabolize a slice of bread?* We are very fortunate it does not. (The reason is heat loss by the body.)

The heat capacity of an object describes how its temperature changes with time due to heat flow to and from the object. The thermal conductivity K describes how the temperature varies (ΔT) spatially due to the heat flow between different regions that are separated by a distance Δx. (Conversely, it also describes how much heat flows due to this spatial variation in temperature.) This relation is

$$\frac{1}{A}\frac{\mathrm{d}Q}{\mathrm{d}t} = -K\frac{\mathrm{d}T}{\mathrm{d}x} \sim -K\frac{\Delta T}{\Delta x}. \tag{6.6}$$

The left-hand side is the amount of heat that flows per unit area A per unit time, and is also called the heat flux. The minus sign indicates that heat flows from hotter regions to colder regions. When there is a well-defined distance $d = \Delta x$ between two regions of different but uniform temperature, say due to the thickness of clothing or an air boundary layer, we can define a heat transfer coefficient per unit area $h = K/d$ and then

$$\frac{1}{A}\frac{\mathrm{d}Q}{\mathrm{d}t} = -h\,\Delta T. \tag{6.7}$$

Heat flow due to other mechanisms, such as due to radiation, can often be expressed in terms of (6.6) or (6.7).

One consequence of thermodynamics is that engines that convert chemical energy to heat and use that heat for mechanical work, so-called heat engines, have a limited efficiency to do such useful mechanical work. An ideal heat engine has a maximum efficiency of $\epsilon = 1 - (T_c/T_h)$ when it operates at a temperature T_h and rejects heat to a lower temperature T_c (both expressed in K). Humans operate internally at about T_h= 310 K and reject heat to a $T_c \simeq 293$ K ambient, so ϵ would be 5.5% if we were heat engines. This is much less than the ∼25% efficiency of humans converting chemical energy into mechanical work. This is not a contradiction because we use the chemical energy directly to do mechanical work, as seen in Chap. 5, and do not produce heat in an intermediate step.

6.2 Energy Content of Body Fuel

There is some similarity between metabolic oxidation and combustion, even though the body does not "burn" its fuels in oxygen. It is useful to learn about the combustion of these fuels because combustion tells you the maximum amount of energy that is available from breaking and rearranging bonds. Metabolic oxidation is a bit less efficient, as we will see. The combustion energies (enthalpies) are obtained from "bomb" calorimetry in which the materials are burned in heavy-walled vessels. The resulting energies are useful and approximately correct even though the conditions of this bomb—its operation at constant volume and the actual "bomb" temperature—are different than those in the body. For example, many processes in the body occur at constant pressure and not constant volume.

Glucose. The metabolism of glucose is representative of that of carbohydrates. If we start with 1 mol of glucose (180 g, 6.02×10^{23} molecules) and oxidize it with 6 mol of oxygen molecules (192 g = 134.4 L at standard conditions, given there are 22.4 L/mol at 1 atm. and 0 °C), there are 6 mol of carbon dioxide, 6 mol of water (108 g, 108 mL), and 686 kcal of energy produced. This can be represented by

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + 686\,\text{kcal}. \tag{6.8}$$

The energy produced per mass of fuel is 686 kcal/180 g glucose = 3.80 kcal/g glucose. We also define a *calorific equivalent*, the energy produced per liter of oxygen consumed, which is 686 kcal/134.4 L O_2 = 5.5 kcal/L O_2 here.

We will see that in metabolic oxidation the body combines this 1 mol of glucose with 30–32 mol of ADP and 30–32 mol of the phosphate group P_i to form 30–32 mol of the energy storage molecule ATP.

Palmitic Acid. The oxidation of palmitic acid is representative of that of fatty acids. Oxidation proceeds by

$$CH_3(CH_2)_{14}COOH + 23O_2 \rightarrow 16CO_2 + 16H_2O + 2,397\,\text{kcal}, \tag{6.9}$$

so 1 mol (256.4 g) of palmitic acid is burned by 23 mol (515.2 L) of oxygen to form 16 mol (358.4 L) each of carbon dioxide and water, and 2,397 kcal. The energy released is 2,397 kcal/256.4 g = 9.3 kcal/g and the calorific equivalent is 2,397 kcal/515.2 L O_2 = 4.7 kcal/L O_2. In metabolic oxidation the body uses this 1 mol of palmitic acid to combine 106 mol of both ADP and the phosphate group P_i to form 106 mol of ATP.

In comparing glucose $C_6H_{12}O_6$ with palmitic acid $C_{16}H_{32}O_2$, both have the same 1:2 ratio of C and H, so after oxidation, equal numbers of moles of CO_2 and H_2O are formed in both cases. However, palmitic acid and other fats have much less oxygen per C and H than glucose and other carbohydrates, so a smaller mass of them is consumed per mole of CO_2 and H_2O formed. (The molar mass of palmitic acid is 256 g, and this is much less than the 480 g molar mass of the "equivalent carbohydrate" $C_{16}H_{32}O_{16}$.) Alternatively (and equivalently), we can say that fats are less oxidized, are more reduced, and have greater reducing power than carbohydrates.

The heats of combustion for carbohydrates, fats and protein differ slightly for different types of food. For carbohydrates they range from 3.7 to 4.4 kcal/g and they can be averaged and grouped in different food groups with 3.90, 3.95, 4.00, and 4.20 kcal/g respectively for animal products, meats, dairy products; sugars; fruits; cereals, legumes, starches, and vegetables. The heats of combustion per g for fats and oils varies from ~9.0–9.5 kcal/g, and these group into 9.25, 9.30, and 9.50 kcal/g respectively for dairy products; cereals, vegetables, and fruits; and meat, fish and eggs. The comparison is more complex for nitrogeneous food components, which are generally called proteins in this field. This includes all compounds containing nitrogen and is usually determined from the analyzed mass of nitrogen in the food. For true proteins, sometimes called proteids in this field, there are 6.25 g of proteids per g of N (16% N), which have heats of combustion of ~5.8 kcal/g protein. For other nitrogen-food components, sometimes called non-proteids, the N fraction and heats of combustion are different. For example, for gelatin, creatin (common in animals foods), and asparagin (vegetable foods) the heats combustion are respectively 5.27, 4.27, and 3.45 kcal/g, which is lower than that for proteids. Weighted averaging (using methods addressed in Problem 6.26) leads to average heats of combustion of 5.00, 5.20, 5.65, 5.70, 5.75 and 5.80 kcal/g respectively for the (general) proteins

Table 6.2 Average caloric content of food

Food	Net caloric value (kcal/g)	Bomb calorimetry energy (kcal/g)	Calorific equivalent (kcal/L O_2)	CO_2 production (kcal/L CO_2)	RER (L CO_2/L O_2)
Carbohydrate	4.02	4.10	5.05	5.05	1.0
Protein	4.20	5.65	4.46	5.57	0.80
Ethanol	7.00	7.10	4.86	7.25	0.67
Fat	8.98	9.45	4.74	6.67	0.71

RER is the respiratory exchange ratio.
Using data from [35]

in vegetables (which are 60% proteids); fruits (70% proteids); dairy products and meat; legumes (96% proteids); eggs; and cereals (96% proteids) [68]. The heat of combustion for alcohol is 7.07 kcal/g. Ethanol has a high heat of combustion and, as we will see, provides much caloric input, which explains why it is easy to gain weight by drinking alcohol, such as beer. If your caloric input is mainly from alcohol, you can "starve" to death in the sense that your caloric needs are being met, but not your micronutrient needs, such as vitamins and minerals. (Note that some of the averaged values given here from [68] differ slightly from those listed in Table 6.2 from [35].)

These energies released per unit mass, or heats of combustion, are the maximum energies available, and are really called *bomb calorimetry energies* because they are often measured in laboratory-controlled explosions. The amount available to the body is somewhat less and is defined as the *caloric value*. Table 6.2 shows the average caloric values andcalorific equivalents of the types of basic body fuels.

The caloric values are lower than the bomb calorimetry values because of losses during digestion, which average 2% for carbohydrates, 5% for fats, and 8% for proteins plus an additional loss of 17% of protein energy in the urine [68]. The small losses for ethanol, ∼2%, are in the urine and exhaled air. With these and related corrections, the average caloric value of carbohydrates and proteins is ≃4 kcal/g and that of fats is ≃9 kcal/g, which are known as the Atwater system values [68]. (Each kcal is a food calorie, or Cal, which is called a calorie on food package labels.) Using these values is often a very good approximation, but it is not perfect. Values for the digestible energy from these sources for specific foods are given in [100], along with the food nutrient content. A more general discussion of the energy and nutrient content is in [23], and a more detailed analysis about energy content is in [6, 68]. These caloric values can be compared to common heating and automotive fuels: gasoline, 11.4 kcal/g; coal, 8.0 kcal/g; and pine wood, 18.5 kcal/g. The energy storage densities of different chemical fuels are compared in Table 6.3.

We can put the caloric value in perspective by developing a standard unit: The Donut. We will use 1 Rich Frosted (i.e., chocolate frosted) Entenmann's™ Donut as the standard donut. The package labeling says that a donut contains 18 g of fat, 29 g of carbohydrate, and 2 g of protein (in 2005), so we can estimate the caloric value of each donut, as in Table 6.4. We see that 49 g of the total 57 g mass of the donut has caloric content. Our calculation suggests there are 286 kcal (286 food calories)

Table 6.3 Energy storage density for chemical fuels

Energy storage fuel	Storage density (J/m^3)	Storage density (J/kg)
ATP	1.4×10^8	1.0×10^5
H_2 gas, 10^3 atm.	4.9×10^9	1.2×10^8
Nitroglycerine	1.0×10^{10}	6.3×10^6
Glycine (amino acid)	1.0×10^{10}	6.5×10^6
Wood	1.1×10^{10}	1.9×10^7
Urea	1.4×10^{10}	1.1×10^7
Methanol	1.8×10^{10}	2.2×10^7
Vegetable protein	2.3×10^{10}	1.7×10^7
Acetone	2.4×10^{10}	3.1×10^7
Glucose	2.4×10^{10}	1.6×10^7
Glycogen (starch)	2.5×10^{10}	1.8×10^7
Animal protein	2.5×10^{10}	1.8×10^7
Carbohydrate	2.6×10^{10}	1.7×10^7
Gasoline	2.8×10^{10}	4.4×10^7
Butane	3.0×10^{10}	4.9×10^7
Fat	3.3×10^{10}	3.9×10^7
Cholesterol (lipid)	4.2×10^{10}	3.9×10^7
H_2 solid (10^5 atm.)	7.2×10^{10}	1.2×10^8
Diamond	1.2×10^{11}	3.3×10^7

Note that 10^{10} J/m^3 = 2.39 kcal/cm^3 and 10^7 J/kg = 2.39 kcal/g.
Using data from [38]

Table 6.4 Caloric value of 1 rich frosted Entenmann's™ donut (in 2005)

18 g fat	×9 kcal/g	= 162 kcal
29 g carbohydrate	×4 kcal/g	= 116 kcal
2 g protein	×4 kcal/g	= 8 kcal
49 g total		= 286 kcal

per donut. The package says there are 280 kcal per donut, which we will take as the caloric content of the standard donut. These values are very consistent with each other because we have used average caloric values for the fat, carbohydrate, and protein. Also, we have rounded off the caloric values and they have provided values of the mass of each body fuel in grams rounded off to the nearest integer (as required by the US FDA (Food and Drug Administration)). (For more on rounding off see Problems 6.11 and 6.12.) Approximately 57% of the calories (162 kcal/286 kcal) come from fat. We will soon determine how much physical exertion is needed to remove the "fattening" consequences of eating a standard donut. Also, we note that during the writing of this second edition (2015), the Rich Frosted Entenmann's™ Donut had 300 kcal. We will explore this aspect of our constantly changing world

in Problem 6.15. In any case, we will still assume the standard donut has 280 kcal. (Problems 6.13 and 6.14 address other potential standard donuts.)

The energy released per L O_2 (calorific equivalent) is fairly constant for all body fuels, ranging from ~4.5 to 5.5 kcal/L O_2. For a "mixed diet" the calorific equivalent can be estimated to be 4.83 kcal/L O_2 and the CO_2 production to be 5.89 kcal/L CO_2.

The *respiratory exchange ratio* (RER) (or *respiratory quotient* (RQ)) is another way to characterize metabolic processes. It is the number of moles of CO_2 produced/number of moles O_2 used, and is a measure of how much carbon dioxide needs to be released in respiration relative to how much oxygen the body needs to bring in by respiration. We see that RER = 6 L CO_2/6 L O_2 = 1.0 for glucose oxidation and 16 L/23 L = 0.7 for palmitic acid oxidation. It is typically 0.8 for protein oxidation.

6.2.1 Metabolizable Energy

We should be a bit more careful about defining how much of the energy from food is being metabolized and actually used, even though these relatively fine distinctions in body energetics may not be overly significant to us here. The rate of *apparently digested energy* is the difference between the rate of intake of dietary energy and the rate of loss of energy in the feces. The rate of obtaining *metabolizable energy* (ME), $d(ME)/dt$, is the difference between the rate of intake of dietary energy and the sum of the rates of loss of energy in the feces, urine, and combustible gas. This is the actual rate at which energy is being made available to the body [11].

Table 6.5 shows the caloric content and components of several types of food. The caloric content per unit mass depends on the relative amounts of carbohydrates, proteins, and fats, and ethanol for alcoholic drinks, and the amounts of water and "ash." For comparison, our standard donut has 280 kcal, or 280 kcal/57 g = 4.91 kcal/g. Table 6.6 shows the enthalpy of combustion for different types of milk; this constitutes consumed and not necessarily metabolizable energy. The enthalpy available from fat, protein, and carbohydrates is quite different for each source. With a 42% fat/18% protein diet, about 89.3% of the consumed food is typically metabolizable—meaning that this is the fraction of the energy content of food that is metabolizable energy. About 5.8% of the consumed energy is lost in the feces, 4.5% in the urine, and 0.4% lost as methane. The difference between the energy content using the Atwater factors and the consumable energy can be much higher than this in some foods. For example, it is ~30% for almonds [76].

We spend relatively little time and energy in metabolizing food because of its high caloric value and digestibility (and so we have much time to engage in activities other than eating and digesting food). We often affect both by food processing, so the true caloric value or ME of a food depends not only on its heat of combustion and the metabolic losses described earlier, but on how we process it before consuming it [100]. Processing of foods by cooking them increases our caloric and nutrient intake after we eat them compared to foods that are eaten raw [14, 21, 56]. In some cases, caloric intake can be increased compared to raw foods by mechanical processing,

Table 6.5 Components and energy (kcal) of edible parts of common foods

Food, serving size	Mass (g)	Energy (kcal)	Carb.[a] (g)	Protein (g)	Fat (g)	Water (%)
Grains and cakes						
Bread, white, 1 slice	25	67	12	2	1	37
Oatmeal, regular, 1 cup prepared	234	145	25	6	2	85
Yellow cake, chocolate frosting, 1 piece	64	243	35	2	11	22
Cheesecake, 1/6 of 17 oz cake	80	257	20	4	18	46
Dairy						
Milk, whole, 1 cup	244	150	11	8	8	88
Butter, salted, 1/4 lb stick	113	813	Tr[b]	1	32	16
Cheddar cheese, 1 oz	28	114	Tr	7	9	37
Cottage cheese, 4%, 1 cup	225	233	6	28	10	79
Ice cream, chocolate, 1/2 cup	66	143	19	3	7	56
Eggs, raw, 1 large	50	75	1	6	5	75
Meat and fish						
Chicken, meat only, roasted, 1/2 breast	86	142	0	27	3	65
Beef, ground, 79% lean, broiled, 3 oz	85	231	0	21	16	56
Salmon, broiled, 3 oz	85	184	0	23	9	62
Fruits, vegetables, nuts, and oils						
Apple, raw, unpealed, 1 whole	138	81	21	Tr	Tr	84
Apricots, raw, without pits, 1 whole	35	17	4	Tr	Tr	86
Apricots, dried, sulfured, 10 halves	35	83	22	1	Tr	31
Orange, peeled, 1 whole	131	62	15	1	Tr	87
Carrots, raw, $7\frac{1}{2}$ in long	72	31	7	1	Tr	88
Potato, baked, with skin	202	220	51	5	Tr	71
French fries, medium portion	134	458	53	6	25	35
Peanuts, dry roasted, 1 cup	146	854	31	35	73	2
Walnuts, 1 cup chopped	120	785	16	18	78	4
Canola oil, 1 cup	218	1,927	0	0	218	0
Beverages						
Cola, 12 fl oz	370	152	38	0	0	89
Beer (regular), 12 fl oz	355	146	13	1	0	92
Gin, vodka, whiskey, 86 proof, 1.5 fl oz	42	105	Tr	0	0	64
Wine, red, 3.5 fl oz	103	74	2	Tr	0	89

[a]Carbohydrate.
[b]Trace.
Using data from [41]

Table 6.6 Food content of milk

Species	Enthalpy of combustion (kcal/g)	Distribution of enthalpy (%)			Dry matter (%)
		Fat	Protein	Carbohydrate	
Human	0.69	54	7	39	12.4
Cow	0.71	48	26	26	12.4
Goat	0.69	50	22	27	12.0
Horse	0.51	23	22	54	10.5
Seal (northern fur)	5.09	88	11	0.1	61.0

Using data from [11]

such as by pounding, grinding, cutting the food into small pieces. The energy cost of metabolizing food decreases by ~10% when it is processed into smaller components pieces (such as processing peanuts into peanut butter) [14, 56]. Cooking tubers (plant structures that expand to store energy, such as potatoes (stem tubers) and sweet potatoes and yams (root tubers)) converts starches from semicrystalline structures to amorphous materials that are easily hydrolyzed to sugars. Cooking meat increases caloric intake because it denatures proteins, making them more digestible, and eases the separation of muscle fibers, increasing the area exposed for digestive activities. Similarly, the digestibility of cooked eggs (~93%) exceeds that of raw eggs (~58%) [31, 32]. Cooking also reduces the energy needed to fight diseases that could be caused by eating some raw foods. The way food processing affects its mechanical properties is described in Chap. 4.

The average retention time of food in a human's digestive track is 46 h; the first appearance of the residues of food appears in about one third this mean time and the last in about four times this time.

Very roughly 10 kcal of energy are needed to produce, ship and process each 1 kcal of food energy (and this amount is very different for different types of food) [56, 80].

6.2.2 *Energy Storage*

Energy can be retained or secreted (as milk) by the body as the enthalpy of tissues R (which includes fat, protein, and CHO). (The enthalpy (or heat content) is the maximum thermal energy that is obtainable at constant pressure.) The rate at which enthalpy is retained $\mathrm{d}R/\mathrm{d}t$ is the difference between the rate of metabolizable energy input and heat production through the metabolism

$$\frac{\mathrm{d}R}{\mathrm{d}t} = \frac{\mathrm{d}(\mathrm{ME})}{\mathrm{d}t} - \frac{\mathrm{d}Q_{\mathrm{met}}}{\mathrm{d}t}. \tag{6.10}$$

Table 6.7 Calorimetric experiments each averaged over several runs (in kcal/day)

Energy	Eating, without exercise	Eating, without exercise (bicycle)	Fasting
Intake energy	2,659	4,340	0
Feces energy	107	176	0
Urine energy	134	138	105
Change in body protein	−16	−57	−463
Change in body fat	176	−484	−1,892
Heat produced	2,270	4,554	2,187
Discrepancy	12	−13	−63

Using data from [11], from [6]

(This ignores lactation.) We combine this with (6.3) to give

$$\frac{\mathrm{d}R}{\mathrm{d}t} = \frac{\mathrm{d(ME)}}{\mathrm{d}t} + \frac{\mathrm{d}Q_{\mathrm{loss}}}{\mathrm{d}t} - \frac{\mathrm{d}W}{\mathrm{d}t} - \frac{\mathrm{d}U}{\mathrm{d}t}. \tag{6.11}$$

When food is not eaten ($\mathrm{d(ME)}/\mathrm{d}t = 0$), $\mathrm{d}R/\mathrm{d}t$ is negative and its magnitude is the heat of catabolism of body tissues. As more food is eaten, $\mathrm{d}R/\mathrm{d}t$ increases, and less stored (or retained) energy is used and there is increased heat production; this increased heat production is the *specific dynamic or thermogenic effect of food*. When energy retention is zero over several days (so no net stored energy is used or formed and $\mathrm{d}R/\mathrm{d}t$ averages to zero), the input dietary energy ME is that required for *maintenance*. For greater values of ME, energy is stored in the body.

Table 6.7 shows the results of calorimetric experiments for people who are fasting (for whom stored energy in proteins and fats is turned into heat), eating who do no exercise (who have net storage of body fat), and eating who do exercise (for whom a small amount of stored energy is turned into heat, even though there is more food intake).

Figure 6.2 shows the retention of energy and the heat production (thermogenic effect) versus food intake. A fraction of the metabolizable energy from food intake

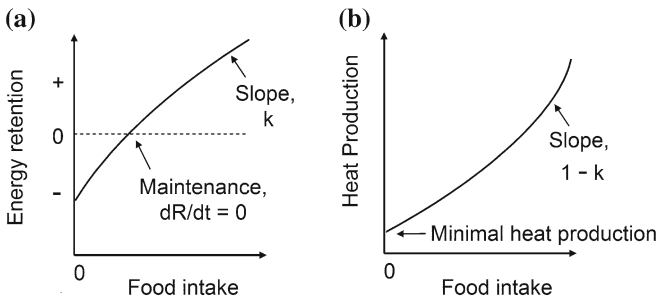


Fig. 6.2 Energy retention and heat production in the body versus food intake (based on [11])

is retained in the body, K, and a fraction is lost as heat, $1 - K$. The slope of the retention curve k is

$$k = \frac{\mathrm{d}R/\mathrm{d}t}{\mathrm{d}(\mathrm{ME})/\mathrm{d}t} \tag{6.12}$$

and the slope of the heat production curve is $1 - k$. K is an averaged value of k. Because the curves in Fig. 6.2 are not linear, k and K are usually not equal. k, also called the *efficiency of the utilization of metabolizable energy*, varies with the amount of food consumed. Below maintenance, k is $\simeq$0.90—where it represents the efficiency of using stored energy—and above it is $\simeq$0.75—where it really represents the efficiency of forming stored energy. This means that below maintenance only about 10% of food goes to heat and above it about $\simeq$25% goes into heat production—and so $\simeq$75% goes into making us fatter (unless we are building up our muscles). These numbers are for average diets. For carbohydrates, $k = 0.94$ below maintenance and 0.78 above it. For fats, $k = 0.98$ below maintenance and 0.85 above it. For proteins, $k = 0.77$ below maintenance and 0.64 above it. The body is more efficient in using stored energy than in depositing fat and proteins. Proteins are metabolized less efficiently than are carbohydrates and fat [11].

We have seen how metabolized carbohydrates, fats, and proteins contribute to our energy balance. However, the details of how each is used by the body is different [56], and so, aside from providing the energy you need, what you eat makes a difference. Proteins are used building and maintaining tissues but usually not directly as a fuel. Some fat is stored in the liver, some is burned, some is stored in muscles as glycogen, as we will soon see, and the rest is stored in fat cells. These cells are throughout the body under your skin (subcutaneous fat), in muscles and other organs, such as belly fat (visceral fat), which is thought to be particularly bad for one's health. Glucose and fructose are important basic products in the metabolism of carbohydrates. Glucose can come from the decomposition of starches, and sucrose (table sugar) and lactose (milk sugar), which are both one half glucose. It is used directly for fuel or for glycogen, and insulin converts excess levels of blood glucose to fat (see below). Another common basic sugar is fructose, which is paired with glucose in sucrose (table sugar). If there is too much fructose to be metabolized by the liver, the liver converts the rest into fat, much into triglycerides (see Chap. 8) that are stored in the liver or released into the blood.

The glucose levels in your blood can become too high when you eat large quantities of foods containing much glucose or components that easily decompose into glucose (such as starches); these foods often called *high glycemic index foods*. Moreover, carbohydrates are metabolized even faster and glucose is released faster into the blood stream more after you eat many processed foods (that have been divided into smaller pieces, have had fiber removed from them, and so on). The pancreas responds to lower glucose levels that are too high by producing insulin, which converts excess glucose into fat for fat cells, much of it forming as visceral fat. Sometimes when glucose levels are very high the insulin level overshoots, and too much glucose is removed and you feel hungry. With long-term high levels of insulin production, the various stages of type 2 diabetes can begin and progress, first with the body

becoming less sensitive to insulin and then with insulin production decreasing [56]. When blood glucose and glycogen levels are too low, fat is released for energy use. This is controlled by a range of hormones that are controlled by a variety of factors, including stress and exercise. The existence and subsequent release of visceral fat is thought to increase triglyceride levels. (See Chap. 8 for more on this and on foods that may help contribute to the potential for heart disease (such as red meat) or lessen it (fish, nuts).)

This leads us to a discussion about how energy is locally stored and used.

6.3 Energy Storage Molecules

6.3.1 How ATP Is Produced and Used as an Energy Source

Catabolism. ATP, adenosine triphosphate (a-duh'-nuh-seen), is the basic unit of energy storage in the body and it enables the rapid release of energy. *Why does the body convert food fuel to ATP and not directly oxidize carbohydrates, fatty acids, and proteins?* The use of ATP is more controllable. Also, the unit of energy provided by ATP is small enough to be useful.

The structure of ATP is shown in Fig. 6.3. It consists of a five-carbon sugar, ribose, that is linked to the aromatic base, adenine—forming adenosine, and three phosphate groups. The two bonds linking the phosphate groups are unstable, high-energy bonds; the leftmost interphosphate bond in Fig. 6.3 splits in the hydrolysis of ATP

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{inorganic phosphate} + \text{energy}, \tag{6.13}$$

where ADP is adenosine diphosphate. This is the basic catabolic process for energy release, and was discussed as the driving force in the myosin power stroke in generating force in muscles in Chap. 5. The energy released, or more precisely the free

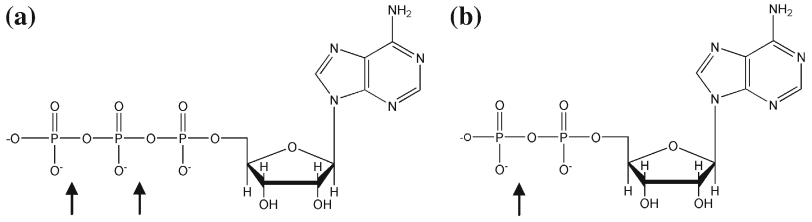


Fig. 6.3 The structures of (**a**) ATP and (**b**) ADP, showing that ATP has one more inorganic phosphate group P_i than does ADP. From *left* to *right* are the (**a**) three or (**b**) two phosphate groups, the five-carbon sugar, ribose, and adenine, which is the double-ringed structure. The two unstable, high energy bonds linking the phosphate groups in ATP and the one similar unstable bond in ADP are denoted by *arrows*

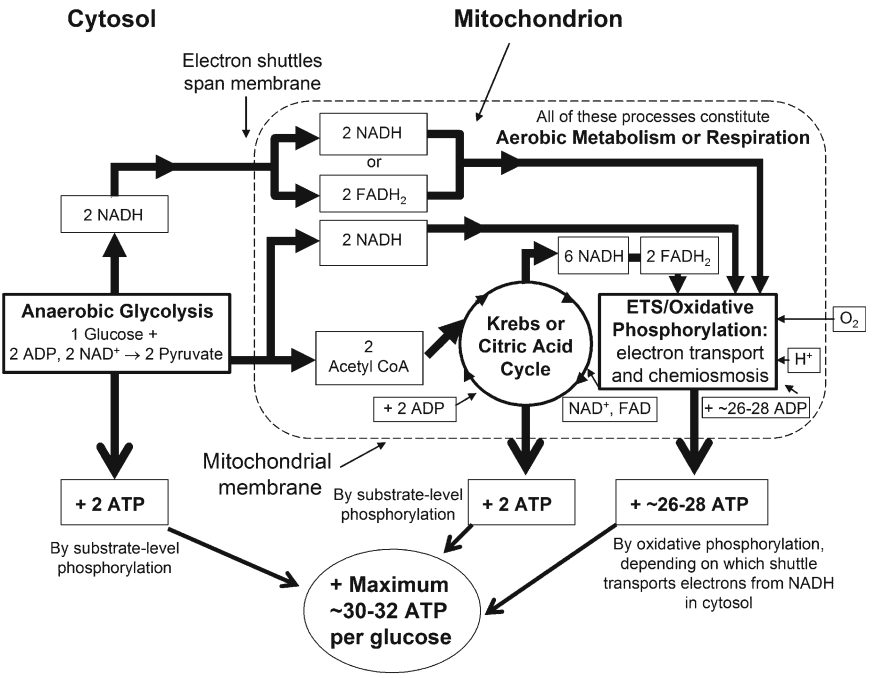


Fig. 6.4 Schematic of the processes involved in the resynthesis of ATP from ADP and inorganic phosphate by anaerobic glycolysis in the cytosol (the intracellular fluid) and aerobic metabolism (or respiration) in the mitochondrion (based on [20, 66, 75])

energy, ranges from 7 to 14 kcal/mol of ATP, depending on conditions (see Problem 6.32). Under typical cellular conditions it can be $\simeq$12–14 kcal/mol [55, 63, 75].

Anabolism. After hydrolysis, ADP needs to be combined with a phosphate group to reform ATP for later use. On the average, each ATP molecule is recycled this way roughly every two minutes (see Problem 6.38). *How does the body use food sources like glucose to do this?* This occurs by a series of chemical steps that can proceed to a limited extent without oxygen (*anaerobic glycolysis*) and to a greater extent with oxygen (*aerobic metabolism or respiration*) [20, 35, 57, 75, 106]. This utilizes glucose in the blood stream and glycogen, $(C_6H_{12}O_6)_n$—a branched-chain polymerized sugar consisting of glucose molecules as monomers linked together with glycosidic (oxygen) bonds. Glycogen is stored in muscle cells, where it is used directly, and in the liver, where is it broken down into glucose by glycogenolysis, which is then delivered by the blood to the other cells. In aerobic metabolism, the products of the anaerobic steps are further metabolized (in the presence of oxygen) to complete the metabolism of stored glucose by a complex series of steps collectively called the *Krebs cycle* and the *electron transfer system (ETS)*; they are shown in Fig. 6.4 and described in a bit more detail later. The overall result is that in aerobic metabolism 1 mol of glucose can produce approximately 30–32 mol of ATP (depending on the details of the membrane shuttle mechanism). (In fact, 1 mol of carbohydrates from

muscle glycogen can produce approximately 31–33 mol of ATP, but energy is used in forming glycogen from glucose; we will consider only the metabolism of glucose.) Of these, 2 mol are produced by anaerobic processes and 28–30 additional moles are produced when there is sufficient oxygen. The bottom line is that the hydrolysis of these, say for now, 30 mol of ATP provided from glucose

$$30\,\text{mol ATP/mol glucose} \times 14.0\,\text{kcal/mol ATP} = 420\,\text{kcal/mol glucose}. \tag{6.14}$$

This should be compared to the energy from the combustion of glucose

$$180\,\text{g/mol glucose} \times 3.8\,\text{kcal/g glucose} = 686\,\text{kcal/mol glucose}. \tag{6.15}$$

This means that the efficiency of the body's usage of glucose to form available energy in the form of ATP is 420 kcal/686 kcal = 61%; for 32 ATP formed by glucose, this gives 448 kcal/686 kcal = 65%. This efficiency range of ≃61–65% is for conditions in muscle cells. For other specific conditions the free energy of ATP can be a bit lower. (This is explored in Problems 6.32 and 6.33.) In any case, this efficiency is pretty good, although it is not perfect. (It is much better than the efficiency of typical heat engines, ∼10–20%.) The overall efficiency of using glucose to do mechanical work is much lower, because of other efficiency factors that we will discuss later.

6.3.2 How ATP Is Actually Used by the Body

ATP is the ultimate source of energy for muscular motion, but it is not always the locally stored source. The body has a mechanism for using it directly, and then several levels of mechanisms for transferring energy from other molecules to the formation of ATP (from ADP), which is then used directly. There are four levels of steps, each of which can be used for successively longer times, although at successively lower levels of activity (Table 6.8) [35, 66, 75].

Table 6.8 Estimated power and energy available from the body, for a 70 kg man with 30 kg of muscle, assuming 10 kcal/mol of ATP

System	Maximum power (moles of ATP/min)	Maximum capacity (total moles of ATP)
Phosphagen (ATP-PC) system	3.6	0.7
Anaerobic glycolysis	1.6–2.5	1.2
Aerobic metabolism from glycogen	1.0	90.0

Using data from [35]

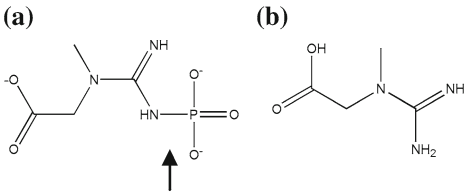


Fig. 6.5 The structures of (**a**) phosphocreatine (PCr) and (**b**) creatine (Cr), showing that PCr has an inorganic phosphate group P_i. The unstable, high-energy bond linking the phosphate group in PCr is denoted by an *arrow*. The enzyme that catalyzes the breakdown of PCr to form ATP, which is the transfer of the phosphate group from PCr to ADP to form ATP (6.17), is called creatine kinase

Step 1. Normally there is enough ATP in living skeletal muscles to supply energy for about 8 twitches. This may be enough for about 3 s or for about half of a 50 m dash. The energy comes from (Fig. 6.3)

$$\text{ATP} + \text{H}_2\text{O} \rightarrow \text{ADP} + \text{P}_\text{i} + \text{H} + 14\,\text{kcal/mol}. \tag{6.16}$$

Step 2. Now more ATP is needed. It is resynthesized from the local phosphocreatine (or creatine phosphate) (PCr) (Fig. 6.5) reservoir by

$$\text{ADP} + \text{PCr} \rightarrow \text{ATP} + \text{Cr}, \tag{6.17}$$

where Cr is creatine. This reaction is driven strongly to the right because it has an equilibrium constant >20. Muscle has enough PCr to provide ATP for about ~100 twitches, or for the first 50 m (~8–10 s) of a sprint.

Together, steps 1 and 2 (reactions (6.16) and (6.17)) constitute the *phosphagen system* or the ATP-PC system. It is the source of the highest peak power, with about 4 mol ATP used by a person every minute. It can be used for short times, 8–10 s, and at most for 100 m dashes or for limited jumping. PCr can be reformed from Cr and P_i by using ATP itself. For intense levels of exercise, this can occur only after the activity, and usually occurs by aerobic metabolism.

When more energy is needed than can be supplied by the phosphagen system alone, it is produced from glycogen that is stored in the muscle and from the metabolism of glucose and fatty acids in the blood stream. Two high-energy molecules are produced as intermediates during this formation of ATP (1) NADH (from NAD^+, nicotinamide adenine dinucleotide, a derivative of the vitamin niacin) and (2) $FADH_2$ (from FAD, flavin adenine dinucleotide, derived from riboflavin, a B vitamin). The modes of using these energy sources to form ATP are different for light and heavy activity.

Step 3. During "heavy" activity people need ATP fast and there is not enough oxygen for aerobic metabolism, so only anaerobic glycolysis (the Embden–Meyerhof pathway) occurs. One mole of glucose-1-phosphate (obtained from 1 mol of carbohydrates from glycogen) can convert only 3 mol of ADP to ATP. In this process 1 mol of ATP is used to make fructose-1, 6-diphosphate from fructose-6-phosphate (which is

produced from glucose-1-phosphate), and 2 mol of ATP are formed in each of the two subsequent steps of glycolysis, leading to a net of 3 mol of ATP. Metabolism of blood glucose yields a net of only 2 mol of ATP because it takes 1 mol of ATP to convert it into glucose-6-phosphate (see Problem 6.35). This all occurs in the intracellular fluid in the muscle cell (the *cytosol*).

This degradation of the glycogen or glucose is only partial, and without sufficient oxygen the potential of producing the other 28–30 mol of ATP by the other steps in aerobic metabolism is totally lost. At the end of anaerobic glycolysis, energy is still stored in the 2 pyruvic acid molecules ($C_3H_4O_3$) (pyruvate ions in solution) formed from each glucose molecule, which are converted to 2 lactic acid molecules ($C_3H_6O_3$) (lactate ions in solution) if they cannot be used in the Krebs cycle in aerobic metabolism. In addition, 2 NADH molecules are formed per glucose molecule in anaerobic glycolysis, and the energy from these high-energy molecules is also wasted because they are not metabolized by the ETS in aerobic metabolism (see Fig. 6.4). (However, the NADH can remain until the oxygen debt is recovered.)

The advantage of this *anaerobic glycolysis or glycogen-lactic acid system* is that it provides a medium level of power, ~2.5 mol ATP/min, which is ~60% of that of the phosphagen system. It provides power for about 600 twitches, for an intermediate time ~1.3–1.6 min—enough for a 400 m run. (The maximum anaerobic peak power is 2.1 hp for men and 1.7 hp for women. These values decrease dramatically after 25 years of age.) Anaerobic glycolysis has disadvantages. In addition to being inefficient, it produces lactic acid, which causes discomfort from acidosis, and causes fatigue. Because the pyruvate-lactate reaction is reversible, the lactic acid from anaerobic glycolysis can be converted to form pyruvate when oxygen is available during the recovery from heavy exercise (but primarily in the liver and not the muscle).

Step 4. During "light" exercise there is enough oxygen for aerobic metabolism to complete the oxidation of the carbohydrates. Aerobic metabolism consists of two complex processes in addition to anaerobic glycolysis: the Krebs cycle (which is also known as the tricarboxylic acid (TCA) cycle or the citric acid cycle) and the electron transfer system (ETS) (which is also known as the cytochrome pathway or oxidative phosphorylation) (Fig. 6.4). They both occur in the cell mitochondrion. Only the ETS directly requires oxygen, but without oxygen even the benefits of the Krebs cycle are lost due to back reactions. (By the way, Hans Adolf Krebs shared the Nobel Prize in Physiology or Medicine in 1953 for his discovery of the citric acid cycle.)

The 2 mol of NADH from anaerobic glycolysis are shuttled in through the mitochondrion membrane. Two moles of pyruvate from anaerobic glycolysis serve as the starting material for the Krebs cycle. Before entering this cycle, each pyruvate produces an acetyl CoA molecule, which enters the Krebs cycle, and 1 NADH, so 2 acetyl CoA and 2 NADH are produced per glucose molecule. For each pyruvate, the Krebs cycle produces 1 ATP, 3 NADH, and 1 $FADH_2$, along with 2 CO_2 molecules, which are waste products, and so for each glucose molecule the Krebs cycle produces 2 ATP, along with 6 NADH and 2 $FADH_2$ molecules.

This means that for each mole of glucose, a total of 10 mol of NADH and 2 mol of $FADH_2$ enter the ETS, where they are converted to ATP and H_2O. (Alternatively, if the 2 mol of NADH from anaerobic glycolysis shuttled through the mitochondrion

membrane are cycled through the membrane as 2 mol of $FADH_2$, a total of 8 mol of NADH and 4 mol of $FADH_2$ enter the ETS.) There are several series of alternative pathways in which these high-energy molecules are converted to ATP, which lead to effectively 2.5 ATP per NADH and effectively 1.5 ATP per $FADH_2$, or $2.5 \times 10 + 1.5 \times 2 = 28$ mol of ATP. There are also the 2 mol of ATP directly from anaerobic glycolysis and the 2 mol of ATP directly form the Krebs cycle, for a total of 32 mol of ATP. Overall, a total of 30–32 mol of ATP are produced from 1 mol of glucose (depending on the membrane shuttle mechanism). (This estimate of 30–32 mol of ATP per mole of glucose is cited in more current work [57, 75]; older estimates concluded that of 36–38 mol ATP are produced per mole of glucose [55].)

ATP is also formed by the oxidation of fatty acids and some amino acids and proteins. (Fats and protein can be metabolized only in the presence of oxygen.) The role of protein metabolism is very minor during rest and does not contribute more than 5–10% of the total energy supply during normal exercise. Each fatty acid molecule is activated using 2 ATP molecules, and is then metabolized in successive passes through a metabolic cycle within the mitochondrion in which two carbon atoms are lost in each pass to produce a molecule of acetyl-CoA, NADH, and $FADH_2$ by β oxidation, until acetyl-CoA is left. For the 16-carbon chain palmitic acid, there are seven complete passes and so one mole of palmitic acid produces 8 mol of acetyl-CoA (including the one remaining after β oxidation), 7 mol of NADH, and 7 mol of $FADH_2$. Each mole of acetyl-CoA is then oxidized in the Krebs cycle to give 3 mol of NADH, 1 mol of $FADH_2$, and 1 mol of ATP. The NADH and $FADH_2$ enter the ETS system. For palmitic acid, this means that 31 mol of NADH and 15 mol of $FADH_2$ enter the ETS system. Overall, $2.5 \times 31 + 1.5 \times 15 = 100$ mol of ATP are formed in the ETS. Including the other 8 mol of ATP, 108 moles of ATP are formed per mole of palmitic acid. Including the 2 ATP moles used for activation, a net of 106 moles of ATP are formed [75].

A healthy man with proper training can provide ~50 mL oxygen/kg-min. This aerobic system delivers the lowest peak power, 1 mol ATP/min ~25% of the phosphagen system, but for a very long time. There is enough glycogen for ~10,000 twitches. It is useful for long-distance, endurance activities, including marathon running. The glycogen in the muscle is depleted after several hours (~1.5–4.0 h) of this level of activity. (Glucose in the blood can also be converted into pyruvic acid and then be used in the cell mitochondria in the presence of oxygen to form more ATP. This is an efficient, long-term process.)

Negligible oxygen is stored in the body. It must be brought in by the lungs and transferred to the blood in the arteries continuously. The oxygen is transferred to the cells, leaving oxygen-depleted blood in the veins. We can track the conservation of oxygen molecules in the body. The rate of body consumption of O_2, $\mathrm{d}V_{O_2}/\mathrm{d}t$ equals the product of the cardiac output (blood flow rate, Q_t, see Chap. 8) and the difference in the oxygen content (oxygen partial pressure) in the arteries and veins, $p_a - p_v$:

$$\frac{\mathrm{d}V_{O_2}}{\mathrm{d}t} = Q_t(p_a - p_v). \tag{6.18}$$

Table 6.9 Estimated energy available from the body (per kg and also, in parentheses, total), for a 70 kg man with 30 kg of muscle, assuming 10 kcal/mol ATP

System	Muscular amounts (in mmol/kg muscle[a]) (total)	Useful energy (in kcal/kg muscle) (total)
Phosphagen (ATP-PC) system		
ATP	4–6 (120–180)	0.04–0.06 (1.2–1.8)
PC	15–17 (450–510)	0.15–0.17 (4.5–5.1)
Total: ATP + PC	19–23 (570–690)	0.19–0.23 (5.7–6.9)
Anaerobic glycolysis		
ATP Formation	33–38 (1,000–1,200)	0.33–0.38 (10.0–12.0)
Aerobic metabolism		
From stored glycogen	13–15 g (400–450)	
ATP formation	2,800–3,200 (87,000–98,000)	28–32 (870–980)

[a]Unless otherwise noted.
Using data from [35]

During aerobic exercise, $\mathrm{d}V_{O_2}/\mathrm{d}t$ increases linearly with Q_t. For a person with average fitness, the maximum oxygen use and blood flow rates are $(\mathrm{d}V_{O_2}/\mathrm{d}t)_{max} \approx 2.8$ L/min (L of oxygen gas at an atmosphere) and $(Q_t)_{max} \approx 19$ L/min, and for a highly fit person they are ≈4 L/min and 25 L/min, respectively.

Table 6.9 gives the concentrations and total amounts of energy storage molecules and the energy available from them for a 70 kg man with 30 kg of muscle. Women have about the same concentrations of ATP and PC per kg muscle as do men, but have less overall muscle. The typical specific chemical energy resources stored in the body are listed in Table 6.10.

Table 6.11 lists the energy system that is primarily used in various sports activities. This is explored in more detail in Table 6.12, which gives the fraction of each system used in different sport activities. These fractions are given for shorter to longer distances in ice speed skating, swimming, and running in Tables 6.13, 6.14 and 6.15.

Table 6.10 Body stores of fuel and energy, for a 65 kg (143 lb) person with 12% body fat

	Amount (g)	Energy (kcal)
Carbohydrates		
Liver glycogen	110	451
Muscle glycogen	500	2,050
Glucose in body fluids	15	62
Carbohydrates total	625	2,563
Fat		
Subcutaneous and visceral	7,800	73,320
Intramuscular	161	1,513
Fat total	7,961	74,833

Using data from [106]

Table 6.11 Energy systems used in sports

Mostly phosphagen system
100-m dash
Jumping
Weightlifting
Diving
Football dashes
Phosphagen and glycogen–lactic acid systems
200-m dash
Basketball
Baseball home run (running around the bases)
Ice hockey dashes
Mainly glycogen–lactic acid system
400-m dash
100-m swim
Tennis
Soccer
Glycogen–lactic acid and aerobic systems
800-m dash
200-m and 400-m swim
1,500-m skating
Boxing
2,000-m rowing
1,500-m/1-mile run
Aerobic systems
10,000-m skating
Cross-country skiing
Marathon run (26.2 miles, 42.2 km)
Jogging

Using data from [45]

In each case, short distance sprints use the ATP-PC and anaerobic systems, while the longer distance events primarily use the aerobic system.

The metabolic steps involved in these systems are all regulated by a feedback and control mechanism (Chap. 13), and are followed by a recovery step. Table 6.16 gives the recovery time for returning resting energy reserves and O_2 and for reducing lactic acid after exercise. Anaerobic exercise is limited by the maximal lactic acid tolerance, which is about 2.0–2.3 g/kg muscle, and so this is 60–70 g for 30 kg of muscle. You can learn a bit about the recovery step by running as fast as you can for as long as you can. You will then be huffing and puffing, breathing in air as fast as possible. This is part of recovery. (Consult with your physician before attempting this demonstration.)

Table 6.12 Percent emphasis of energy systems

Sport or activity	ATP-PC and anaerobic glycolysis	Anaerobic glycolysis and aerobic	Aerobic
Aerobic dance	5	15–20	75–80
Baseball	80	15	5
Basketball	60	20	20
Diving	98	2	Negligible
Fencing	90	10	Negligible
Field hockey	50	20	30
Football	90	10	Negligible
Golf	95	5	Negligible
Gymnastics	80	15	5
Ice hockey			
Forward, defense	60	20	20
Goalie	90	5	5
Lacrosse			
Goalie, defense, attacker	50	20	30
Midfielders, man-down	60	20	20
Rowing	20	30	50
Soccer			
Goalie, wings, strikers	60	30	10
Halfbacks or sweeper	60	20	20
Stepping machine	5	25	70
Tennis	70	20	10
Field events, in track and field	95–98	2–5	Negligible
Volleyball	80	5	15
Walking	Negligible	5	95
Wrestling	90	5	5

Using data from [35, 36]

Table 6.13 Percent emphasis of energy systems for a range of distances in ice speed skating

Activity	ATP-PC and anaerobic glycolysis	Anaerobic glycolysis and aerobic	Aerobic
Ice speed skating			
500 m	80	10	10
1,000 m	35	55	10
1,500 m	20–30	30	40–50
5,000 m	10	25	65
10,000 m	5	15	80
In-line skating			
>10 km	5	25	70

Using data from [35, 36]

Table 6.14 Percent emphasis of energy systems for a range of distances in swimming

Swimming	ATP-PC and anaerobic glycolysis	Anaerobic glycolysis and aerobic	Aerobic
50 m	90	5	5
100 m	80	15	5
200 m	30	65	5
400 m	20	40	40
1,500 m	10	20	70

Using data from [35, 36]

Table 6.15 Percent emphasis of energy systems for a range of distances in running

Running	ATP-PC and anaerobic glycolysis	Anaerobic glycolysis and aerobic	Aerobic
100, 200 m	95–98	2–5	Negligible
400 m	80	15	5
800 m	30	65	5
1,500 m (or mile)	20–30	20–30	40–60
3,000 m (or 2 miles)	10	20	70
5,000 m (or 3 miles)	10	20	70
10,000 m (or 6 miles)	5	15	80
Marathon	Negligible	5	95

Using data from [35, 36]

Table 6.16 Recovery times after exhaustive exercise

	Minimum	Maximum
Restoration of phosphagen (ATP + PC)	2 min	5 min
Muscle glycogen replenishment	5–10 h	24–46 h
Liver glycogen replenishment	Unknown	12–24 h
Restoration of O_2 in plasma and myoglobin	10–15 s	1 min
Duration of fast component of O_2 recovery	3 min	6 min
Duration of slow component of O_2 recovery	30 min	1 h
Reduction of lactic acid in blood and muscle	30–60 min[a]	1–2 h[a]

[a]Faster recovery with exercise and slower recovery with rest.
Using data from [35]

6.4 Metabolic Rates

There are variations among people's metabolic rates (MR) due to their different weights, genes, etc. Metabolic needs over the years have changed, in part due to changing body heights and weights, and levels of activity [34]. First, we will discuss the minimum metabolic rates for people and then address how activity increases the metabolic rate. The metabolic rate can also depend on food intake. For example, the MR is known to decrease under fasting conditions and this change counters some of the expected effects of dieting. Although widely called metabolic rates, these are more precisely catabolic rates. Sometimes the MR is called the total energy expenditure, TEE, which might make it clearer that caloric intake that is not expended as energy is stored by the body.

6.4.1 Basal Metabolic Rate

The basal metabolic rate (BMR) is that of an inactive, awake body. The BMR for a 70 kg person is about 1,680 kcal/day ≃70 kcal/h ≃81 W. This means that even at rest each of us gives off almost the same amount of heat as a 100 W incandescent light bulb. As we will see later, the heavier the person the higher the BMR. Approximately 85% of people of the same gender and weight have a BMR within 6–10% of the mean for their classification (Table 6.14).

The exact numbers one finds for the BMR sometimes vary with the source. This may be due in part to the exacting conditions used to define the BMR and the lack of specification of the person's gender, age, weight, etc. To have your BMR measured you must:

(1) Have eaten no food for at least 12 h
(2) Have had a night of restful sleep and no strenuous activity thereafter
(3) Be resting completely in a reclining position for at least 30 min
(4) Be experiencing no excitement from psychic or physical factors
(5) Be in a room with a temperature from 20 to 27 °C (68 to 80 °F).

The BMR is greater than the metabolic rate during sleeping (when there is minimal digestion of food). Because of the stringent nature of BMR testing, "background" or "minimal" metabolic rates are sometimes tested under somewhat less exacting conditions and these measured rates are not cited as BMRs, but are termed differently although not always in a uniform way. The basal energy expenditure (BEE) is often defined to be equivalent to the BMR. The resting metabolic rate (RMR) is similar to the BMR, but is measured under somewhat less restrictive conditions—early in the morning after an overnight fast and 8 h of sleep—so it is easier to achieve and thus it is often used. The resting energy expenditure (REE) is usually measured after 4 h of sitting and is a bit larger than the BMR. The BMR and metabolic rates

Table 6.17 Metabolism of a resting person

System	Percentage of BMR	Met. rate (kcal/min)	Organ mass (kg) for a 65 kg man
Liver and spleen	27	0.33	–
Brain	19	0.23	1.40
Skeletal muscle	18	0.22	28.0
Kidney	10	0.13	0.30
Heart	7	0.08	0.32
Remainder	19	0.23	–
		Sum = 1.22	

Using data from [19]

during activities are often determined by measuring oxygen intake, because they are proportional to each other for aerobic activities.

The functioning of several organs contributes to this BMR in a resting person, as is seen from Table 6.17. Three-fourths of the metabolic activity takes place in organs with a total mass of 5 kg, which is ∼8% of the total body mass. We will estimate these metabolic rates of the heart and lungs in Chaps. 8 and 9.

Human brains and gastro-intestinal systems (which are not listed in Table 6.17) have about same mass and metabolic rates, and so they require about the same energy per unit mass, ∼11–12 W/kg. Our brains are much larger and our gastro-intestinal systems are much smaller than those in primates and other mammals as expected from allometric mass scaling (though our livers and kidneys are of the expected sizes). Our smaller than expected, and consequently less energy demanding, gastro-intestinal tracts are sufficient because humans can easily acquire and eat high caloric-value foods that we can digest rapidly, in part because they do not have extremely high fiber content and can be processed (i.e., cooked). Our brains are larger than expected, and consequently more energy demanding. Perhaps, because of the smaller gastro-intestinal tracts, humans can energetically "afford"to have very large brains [3, 56] (and, in balance, still have overall basal metabolic needs consistent with those expected from the allometric scaling of the BMR with body mass, which we will soon discuss).

The BMR is a function of mass, height, and gender, as we will see later in this chapter. It is also a function of body temperature. It changes by about 10% (some say 13%) per 1 °C in body temperature. Consequently, the BMR increases by 30% (or 39%) when the body temperature increases from the normal 37 °C (98.6 °F) to 40 °C (104.0 °F) and decreases by 30% (or 39%) when the body temperature decreases from normal to 34 °C (93.2 °F).

Table 6.18 BMR determined for several mammals

Species	Mass	BMR (kcal/day)
Mouse	20 g	3
Reference woman (25 years)	55 kg	1,260
Reference man (25 years)	65 kg	1,500
Elephant	5,000 kg	70,000

See, for example [66]

Scaling of the BMR

Before addressing how the BMR and MR depend on gender, weight, age, etc. we will temporarily expand our discussion of the BMR from that for humans to that for all mammals. The larger the animal the higher the metabolic (catabolic) rate, which is as expected. Moreover, there is a way that the BMR scales with a physical attribute of size, specifically the body mass m_b.

Table 6.18 shows that the BMR of mammals increases sublinearly with m_b. Many think the variation for mammals is as the 3/4 power:

$$\text{BMR} = cm_b^{3/4}, \tag{6.19}$$

where $c \approx 90\,\text{kcal/kg}^{3/4}\text{-day}$. This is known as *Kleiber's Law* [51–53], and is an example of an allometric relationship, as described in Table 1.13. It is valid from mice to elephants and some say to whales. (Steinbeck's novel was valid only from mice to men. Regarding whales: there is great uncertainty about the BMR of whales. How do you measure it?) While this scaling relation was actually developed for comparing different species, it works fairly well intraspecies for humans, from children to adults. Others claim that the BMRs of mammals increase as mass to the 2/3 power. Those believing in Kleiber's Law point out that this other scaling rule is off by a factor of two from mice to cows.

Both dependences give essentially the same prediction over the mass range of humans, so it matters little which we use for analysis. Adherents of both rules claim that the available BMR data support them. In any case, it is interesting to examine the physical reasons put forth to support the $m_b^{2/3}$ and $m_b^{3/4}$ scaling rules [27, 42]. There is both some sound and some fairly questionable physical reasoning in both sets of arguments.

Supporting Kleiber's $m^{3/4}$ Scaling Rule

The metabolic rate, and the BMR in particular, are supposed to scale as the maximum power output of muscles [65, 66]. Power usage is proportional to Fv, where F is the muscle force and v is the speed of muscle contraction. The muscle force $F = \sigma(\text{PCA})$, where σ is the muscle force per unit area and PCA is the muscle cross-sectional area. Therefore,

$$\text{BMR} \propto Fv = \sigma(\text{PCA})v. \tag{6.20}$$

Studies show that for muscles of all species and size, v and σ do not vary substantially, so

$$\text{BMR} \propto \text{PCA}. \tag{6.21}$$

If the limbs of mammals, which of course contain many of these muscles, have a characteristic width d and length L, this area and the mass of the mammal scale m_b as

$$\text{PCA} \propto d^2 \tag{6.22}$$

$$m_\text{b} \propto d^2 L. \tag{6.23}$$

To relate the BMR to the mass, we need to relate L to d. Four arguments have been put forth suggesting that L scales with $d^{2/3}$. First, experimental data on primates show that L varies as d this way. Second, this scaling law also relates the height of a tree L with the width of the tree trunk d, and so this scaling law seems to be universal. Third, $L \propto d^{2/3}$ is also the scaling law for the condition for which a long vertical column of height L and width d buckles under its own weight when displaced from the vertical orientation.

A fourth argument is based on self-similarity, using the expression we derived earlier, (4.46). The end of the beam of length L and lateral dimension d in Fig. 4.40 deflects down by a distance $y(L) = -FL^3/3YI_\text{A}$ when a force F is applied at its end. Even though the area moment of inertia is different for different cross-sections—$I_\text{A} = \pi d^4/64$ for cylinders (from $I_\text{A} = \pi a^4/4$ for a circular cross-section of radius $a = d/2$ using (4.42)) and $d^4/12$ for rectangular solids with a square cross-section (from $I_\text{A} = wh^3/12$ for a rectangular cross-section of width w and height h—here with $d = w = h$, using (4.41))—it makes no difference which we use because both have the same d^4 dependence. (We will formally use the circular cross-section.)

This downward angle of deflection θ is $\sim|y|/L$, giving

$$\theta = \frac{|y|}{L} = \frac{FL^2}{3YI_\text{A}}. \tag{6.24}$$

Self-similarity suggests that θ would be same for all species, and consequently equal to a constant. If F is due to gravity, then $F = m_\text{b}g = \rho V g$, where ρ is the beam mass density and V is its volume $= (\pi d^2/4)L$. If this force acted entirely at the end, then

$$\theta = \frac{(\rho((\pi d^2/4)L)g)L^2}{3Y(\pi d^4/64)} = \frac{16}{3}\frac{\rho g}{Y}\frac{L^3}{d^2}. \tag{6.25}$$

If θ were independent of L and d, we again get $L \propto d^{2/3}$.

In any case, given that $m_\text{b} \propto d^2 L$ and $L \propto d^{2/3}$, we see that $m_\text{b} \propto d^2(d^{2/3}) = d^{8/3}$. Therefore $d \propto m_\text{b}^{3/8}$. With BMR $\propto$ PCA and PCA $\propto d^2$, we find that BMR $\propto d^2$. Using $d \propto m_\text{b}^{3/8}$, we see that

$$\text{BMR} \propto (m_\text{b}^{3/8})^2 = m_\text{b}^{3/4}, \tag{6.26}$$

which is Kleiber's Law.

There are many gaps in this reasoning. For example, the power needed to operate skeletal muscles is actually a small fraction of the BMR (Table 6.17), so the scaling laws based on this power may not represent those of the whole BMR.

Supporting the $m^{2/3}$ Scaling Rule

The previous argument assumed that the BMR scaled as the metabolic power needs. This line of reasoning says that the BMR is limited by the rate of heat loss from the mammal. A mammal has a cross-sectional area that scales as L^2, where L is its dimension, while its volume and therefore mass m_b scale as L^3. Heat loss scales as the available surface area A, which clearly varies as $m_\text{b}^{2/3}$, and so

$$\text{BMR} \propto m_\text{b}^{2/3}. \tag{6.27}$$

BMR Scaling in Humans

The BMR in humans depends on mass as well as other factors, such as gender, height, and age. The BMRs for 20–39 year old men in moderate temperature climates ranges between 1,350 and 2,000 kcal/day for common ranges of heights and weights [34]. The Harris–Benedict equations are commonly used to characterize measured human BMRs:

$$\text{For men:} \quad \text{BMR} = 66.4730 + 13.7516 m_\text{b} + 5.0033 H - 6.75505 Y, \tag{6.28}$$

$$\text{For women:} \quad \text{BMR} = 655.0955 + 9.5634 m_\text{b} + 1.8496 H - 4.6756 Y, \tag{6.29}$$

where the BMR is in kcal/day, m_b is the body mass in kg, H is the height in cm, and Y is the age in yr. This can also be expressed as

$$\text{For men: BMR} = 71.2 m_\text{b}^{3/4}[1 + 0.004(30 - Y) + 0.010(S - 43.4)], \tag{6.30}$$

$$\text{For women: BMR} = 65.8 m_\text{b}^{3/4}[1 + 0.004(30 - Y) + 0.010(S - 43.4)], \tag{6.31}$$

where S is the specific stature $= H(\text{in cm})/m_\text{b}^{1/3}(m_\text{b}$ in kg). These relations track averages, and not necessarily how the BMR changes when a given individual becomes heavier or older.

Another study re-examined the BMR in term of age groups and gender, as presented in Table 6.19. Table 6.20 gives the BMRs for adults of given heights as predicted by Table 6.19. The person's mass is chosen to make the BMI (or the Quételet's index Q) equal to m_b/H^2 (m_b is the body mass (in kg) and H is the height (in m)) equal to 22 for men and 21 for women.

Table 6.19 BMR (kcal/day) for different age groups

Age group (years)	BMR (males)	BMR (females)
Under 3	$59.5m_b - 30$	$58.3m_b - 31$
3–10	$22.7m_b + 504$	$20.3m_b + 486$
10–18	$17.7m_b + 658$	$13.4m_b + 693$
18–30	$15.1m_b + 692$	$14.8m_b + 487$
30–60	$11.5m_b + 873$	$8.1m_b + 846$
Over 60	$11.7m_b + 588$	$9.1m_b + 658$

m_b is the body mass in kg.
Using data from [11], using [90]

Table 6.20 BMR (kcal/day) for adult men and women of different ages, assuming the BMI (or Quételet's index Q) is 22 for men and 21 for women and the relations in Table 6.19

Height (m) (also ft, in)	Mass (kg) (also lb)	Age (yr)		
		18–30	30–60	Over 60
Men				
1.5 (4′11″)	49.5 (109)	1,440	1,450	1,150
1.6 (5′3″)	56.5 (124)	1,540	1,530	1,250
1.7 (5′7″)	63.5 (140)	1,650	1,620	1,350
1.8 (5′11″)	71.5 (157)	1,770	1,710	1,450
1.9 (6′3″)	79.5 (175)	1,900	1,800	1,560
2.0 (6′7″)	88.0 (194)	2,030	1,900	1,670
Women				
1.4 (4′7″)	41.0 (90)	1,100	1,190	1,030
1.5 (4′11″)	47.0 (104)	1,190	1,240	1,090
1.6 (5′3″)	54.0 (119)	1,290	1,300	1,160
1.7 (5′7″)	61.0 (134)	1,390	1,360	1,230
1.8 (5′11″)	68.0 (150)	1,500	1,420	1,310

Using data from [11], from [33]

These relations show that younger people have higher BMRs than older people, which is true because they have more lean body mass. Similarly, tall thin people have higher BMRs and overweight people have lower BMRs. Pregnant women have higher BMRs than predicted above, as do people with fevers, those under stress, and those in hot and cold conditions. The BMR is lower than is predicted here for people who are fasting, starving, and malnourished.

During pregnancy, basic dietary energy needs increase by an average of roughly 200–250 kcal/day, with most of the increase occurring in the second half of gestation [18, 58, 73, 85]. Of the total extra estimated caloric needs a woman has during pregnancy, about half goes to increased metabolic needs, as expected from a larger body mass, and about half to building protein and fat in added tissue (for the fetus, uterus, blood, placenta and breasts) [18]. Consequently, a very large fraction of this

increase can be considered an increase in BMR and a smaller fraction can be attributed to body activity, which causes the MR to exceed the BMR. In developed countries the extra caloric needs are ~90,000 kcal for a total weight gain of ~13.8 kg (30.4 lb). (An earlier estimate of the average extra energy cost was ~80,000 kcal, with a weight gain during pregnancy of ~12.5 kg and median infant birth weight of ~3.3 kg [33].) In developed countries, the BMR increases by ~5, 11, and 24% during the first, second and third trimesters of pregnancy [18]. The USDA recommends no increase in the caloric intake rate in the first trimester of pregnancy and 340 and 450 kcal/day increase in the second and third trimester. In postpartum, lactation increases energy needs by ~500 kcal/day [73], and again this increase can be considered an increase in overall BMR. This increase in caloric intake can be smaller in those with more stored fat.

Tissue growth is important during pregnancy, normal growth from infancy through adolescence, and "catch-up" growth in malnourished children. During pregnancy the added tissue is estimated to be ~5–8% protein and ~25–31% fat, with the balance being water [18, 85]. The energy cost to produce new tissue can be determined from the heats of combustion for protein, fat, and carbohydrate are 5.7, 9.3, and 4.3 kcal/g, which are the Atwater factors excluding losses as to urine and feces. For tissue growth in children it is estimated to be roughly 5 kcal per g of (dry) tissue growth; in malnourished children it is 4.4 kcal/g [94]. This extra energy cost is only <5% of the total metabolic needs from age 1 through adolescence including the growth spurt [17], but can be very significant during fetus growth (as noted above) and from infancy until 12 months.

See the problems at the end of the chapter that address the BMR and tissue growth for pregnant women.

The variations in the BMR and MR among humans are discussed below.

6.4.2 Metabolic Rates During Common Activities

Fortunately, we spend most of our time under conditions less restrictive than those needed to measure the BMR. Metabolic rates during activity are commonly expressed in several ways. (1) The total metabolic rate (MR) (including that due to the BMR) is expressed either as (a) a cumulative value (BMR + the additional MR due to activities), (b) a factor times the BMR—the factor being called the activity factor f or the physical activity ratio (PAR) (which is commonly used in the health community), or (c) a factor times the RMR, with that factor called the MET, the metabolic equivalent (which is commonly used in the exercise community). The RMR is more frequently used as the reference because it is much easier to measure than the BMR. (For our purposes we can assume that the MET and the slightly larger f are equal.) (2) Only the net increase in metabolic rate over the BMR (or RMR) is given, such as MR−BMR.

Of course, each method should provide the same total MR. It is not surprising that the published values of MR expressed in these different ways are often somewhat inconsistent; this is true for several reasons. The exact metabolic rates may not be known very well, the details of the specific activity may not be well defined, the differences in rates among people of different genders, ages, and weights may not have been accounted for properly, and the exact differences between the reference metabolic rates may not be clear. For example, the differences in the BMR and RMR may not have been considered well. Also, sometimes the MET is defined relative to the RMR as defined earlier, right after sleep, and sometimes it is referenced to the MR for sitting. One more subtle reason is that the BMR varies as $m_b^{3/4}$, while the increase in metabolic rate due to activity is thought to depend linearly on m_b (and this may not always be exactly true). Consequently, scaling factors, such as the activity factor, cannot be rigorously correct. Nevertheless, examining the total MR is intriguing and we can learn much from it, even amidst these potential inconsistencies.

We see from Table 6.21 that the heavier the activity, the higher the metabolic rate and the greater the need for oxygen, and of course this is why we then need to breathe faster and harder. In fact, the air/oxygen consumption is proportional to the power exerted and the rate of heat being produced, if the power is low enough for only aerobic processes to be important (Tables 6.11, 6.12, 6.13, 6.14 and 6.15). We can compare the MRs in this table with our BMRs. These values are only approximate

Table 6.21 Approximate total metabolic rates (MR) and oxygen consumption for different levels of activity for an average 70 kg person

Activity	Equivalent heat production (kcal/h)	(W)	O_2 consumption (L/min)
Very low level activity			
Sleeping	71	83	0.24
Sitting at rest	103	120	0.34
Standing relaxed	108	125	0.36
Light activity			
Walking slowly, 5 km/h	228	265	0.76
Moderate activity			
Cycling, 15 km/h	344	400	1.13
Moderate swim	400	465	1.32
Heavy activity			
Soccer	500	580	1.65
Quite heavy activity			
Climbing stairs, 116 steps/min	589	685	1.96
Cycling, 21 km/h	602	700	2.00
Basketball	688	800	2.28
Extreme activity			
Racing cyclist	1,400	1,600	4.62

From [9] and [19]

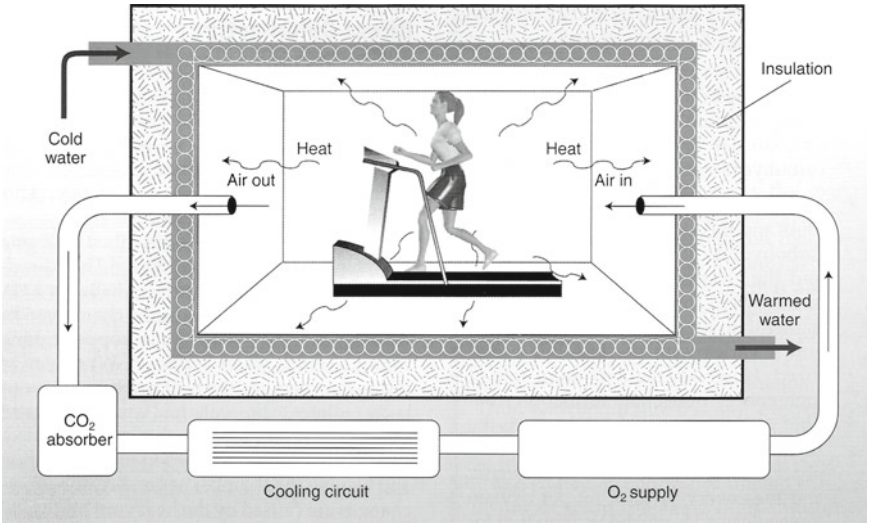


Fig. 6.6 In a calorimetric chamber for direct calorimetry, heat generated by the person is transferred to the air and chamber walls, and the heat produced—and therefore the metabolic rate—are determined by the measured temperature change in the air and the water flowing through the chamber (From [106]. Used with permission)

and depend on the specific conditions. For example, young adults expend ~20 kcal/h more when using nontraditional standing desks than when sitting at traditional desks [87], which is a bit larger than the difference between the MRs for standing and sitting listed in this table.

The metabolic rate during exercise can be measured by using *direct calorimetry* (Fig. 6.6) or *indirect calorimetry* (Fig. 6.7). In direct (or "bomb") calorimetry the heat produced by a person exercising in a thermally insulated room is measured. The metabolic rate is determined assuming that ~40% of the energy liberated during metabolism is used to produce ATP, while the remaining 60% is converted to heat—which is being measured. The subject is in an enclosed, thermally insulated chamber. Heat produced by the subject's body eventually heats water flowing in tubing within the chamber walls and this water temperature is measured. Constructing such a chamber is relatively expensive and the time response of this method is slow, so metabolic rates during intense activity cannot be measured well.

In indirect calorimetry the rates of O_2 consumption, dV_{O_2}/dt, and CO_2 production, dV_{CO_2}/dt, are determined, so it is simpler to implement, as depicted in Fig. 6.7. They are determined from the rates of inspiring and expiring volumes of air and from the fractions of O_2 and CO_2 in both the inspired and expired air (Problem 6.10). If the calorific equivalents of carbohydrates and fats were the same, the rate of heat energy produced would be $dV_{O_2}/dt \times$ the calorific equivalent. Because they are not the same, the respiratory exchange ratio RER is measured (Table 6.2) to determine the relative amounts of carbohydrates and fats being metabolized. The RER can be expressed as



Fig. 6.7 Gas collection for indirectly measuring metabolic rates (From [106]. Used with permission)

$(\mathrm{d}V_{CO_2}/\mathrm{d}t)/(\mathrm{d}V_{O_2}/\mathrm{d}t)$. An RER = 1.0 indicates that muscles are using only glucose or glycogen (carbohydrates). An RER = 0.71 indicates that only fats are being used. Values in between these limits indicate that both fats and carbohydrates are being used (Problem 6.10). This approach assumes that no proteins are being metabolized, which is not true for exercise lasting several hours, during which up to 10% of the metabolic energy can come from protein metabolism.

The aerobic capacities of normally active 18–22 year old college students are 44–50 mL/kg-min (of oxygen) for men and 38–42 mL/kg-min for women. For trained athletes this capacity increases to about 60 mL/kg-min and to 70–80 mL/kg-min and higher for elite male long distance runners. Poorly conditioned adults can have aerobic capacities below 20 mL/kg-min. The aerobic capacity decreases with age, by about 10% per decade, starting in the mid-teens for women and mid-20s for men. Among normally active men it decreases from about ∼47.7 mL/kg-min at 25 years of age, to ∼43.1 mL/kg-min at 35, ∼39.5 mL/kg-min at 45, ∼38.4 mL/kg-min at 52, ∼34.5 mL/kg-min at 63, and ∼25.5 mL/kg-min at 75 years of age.

All of these changes in body metabolic rates are linked to corresponding changes within cells, organs, and systems in the body. Table 6.22 lists the power output of various cells and tissues at various levels of activity.

As we discuss metabolic rates, we need to keep in mind that there are significant variations in the metabolism between people with the same body characteristics and activity levels. The usual variation in BMR is ±6–10%; this is presumably also true for the MR (for the same average activity levels). Table 6.23 shows a striking example in which the metabolic rates of two young women differ by 50%. Because most of this difference is due to metabolism during sleep at night, potential differences in activity levels do not seem very important.

Table 6.22 Estimated power output (metabolic rates) and power densities of biological cells and tissues (human, unless stated otherwise)

Organelle, cell, or object	Power output (picowatts)	Volume (μm^3)	Power density (W/m^3)
Myosin muscle motor crossbridge	0.000001	5×10^{-7}	2×10^6
Platelet (resting)	0.003–0.09	3	$0.1–3.0 \times 10^4$
Red blood cell	0.008	94	8.5×10^1
E. coli bacterium (basal)	0.05	2	2.5×10^2
Mitochondrian organelle	0.1–1.1	1	$0.1–1.1 \times 10^6$
Platelet (activated)	0.7–7.0	3	$0.2–2.3 \times 10^6$
Skin cell	1–3	1,000	$1.0–3.1 \times 10^3$
Skeletal muscle (resting)	1–10	2,000	$0.5–4.9 \times 10^3$
Typical tissue cell (basal)	30	8,000	3.8×10^3
Intestine/stomach cell	46–52	8,000	$5.6–6.5 \times 10^3$
Neuron cell (basal)	70–110	14,000	$5.0–7.9 \times 10^3$
Heart muscle cell (typical)	87–290	8,000	$1.1–3.6 \times 10^4$
Skeletal muscle cell (max., voluntary)	113	2,000	5.7×10^4
Kidney cell	155–346	8,000	$1.9–4.3 \times 10^4$
Neuron cell (maximum)	255–330	14,000	$1.8–2.4 \times 10^4$
Typical tissue cell (maximum)	480	8,000	6.0×10^4
Skeletal muscle cell (max., tetanic)	2,300	2,000	1.2×10^6
Honeybee flight muscle cell	3,400	1,000	3.4×10^6
Heart muscle cell (maximum)	3,500–5,000	8,000	$4.4–6.3 \times 10^5$
Human brain	15–25 W	1.4×10^{-3} m^3	$1.1–1.8 \times 10^4$
Human body (basal)	100 W	0.1 m^3	1.0×10^3
Human body (maximum)	1,600 W	0.1 m^3	1.6×10^4
Gasoline-powered automobile	200,000 W	10 m^3	2.0×10^4
The Sun	3.92×10^{26} W	1.41×10^{27} m^3	0.28

Note that 1 W = 0.86 kcal/h.
Using data presented in [38]

Table 6.23 Food intakes and heat production for two 23-year-old women of comparable physical size

	"Large eater"	"Small eater"	Ratio
Body mass (kg)	54.2	52.7	1.03
Lean body mass (kg)	45.1	43.6	1.03
Average food intake (kcal/d)	2,370	1,550	1.54
Heat production (kcal/d)	2,170	1,390	1.55
Heat production at night (kcal/d)	1,720	1,100	1.56

Using data from [11], from the result of Warwick as quoted by [39]

Metabolic Rates During Locomotion

Figure 6.8 shows that the rate of heat production versus walking speed is roughly linear, meaning that the energy cost per distance traveled is independent of speed. Above 3 m/s (7 mph) this curve becomes very superlinear. The energy cost of walking above that of standing is ~2.13 J/kg-m at 1.2 m/s (2.6 mph), ~2.49 J/kg-m at 1.8 m/s (4.0 mph), and ~3.93 J/kg-m at 2.4 m/s (5.4 mph).

At high speeds, why do we prefer to run than walk? It is very hard to keep walking at these faster speeds because of the increased power needed relative to jogging and running (Fig. 3.26). In fact at 2.4 m/s (5.4 mph) the metabolic power needed to run is 14% less than that needed to walk. Phrased in terms of the energy needed per unit mass

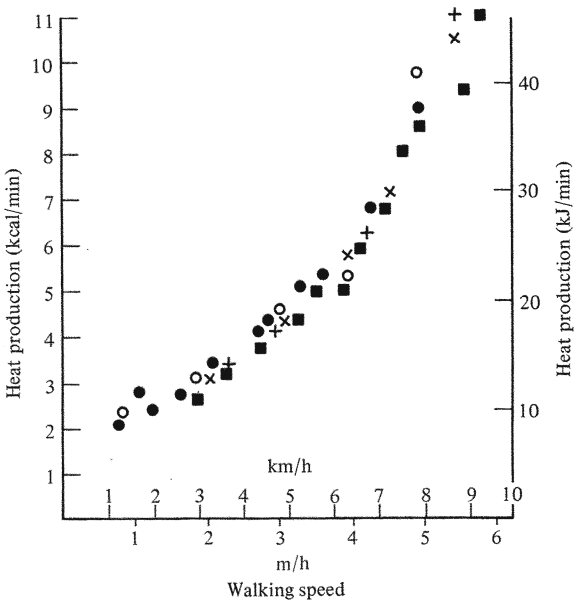


Fig. 6.8 Heat production (metabolic needs) of walking (From [11], as summarized by [77])

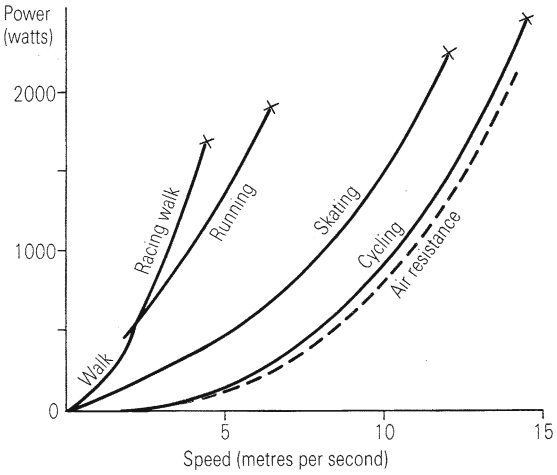


Fig. 6.9 Metabolic needs versus speed for several types of locomotion, extrapolated to world record speeds for 5 km races (for running) or 10 km races (for the others). The *dashed line* shows the power needed to overcome air resistance only (From [4], based on data from [83]. Used with permission)

to move horizontally per meter, this is a decrease from ∼3.93 J/kg-m (0.94 cal/kg-m) to ∼3.40 J/kg-m (0.81 cal/kg-m). Figure 3.26 shows that the metabolic cost of locomotion per distance traveled is relatively independent of speed for a wide range of speeds when using the optimal gait, both for humans and horses. The energy needed to run on level ground is ∼4.2 J/kg-m for untrained runners, independent of speed, which is approximately double that for walking at the most energy economical walking speed of 4 km/hr. This energy is ∼6% lower than this value for trained runners [60]. (Although it costs the same energy to walk a mile quickly and to run a mile, running provides better cardiovascular training.) The oxygen intake is about 5.8 mL/kg-min per mph running speed. Figure 6.9 shows metabolic needs versus speed for several types of locomotion.

Concentric muscle contractions are said to do positive work, while eccentric contractions do negative work (see Fig. 3.56). The metabolic cost of positive work is greater than an equal amount of negative work. The average extra energy needed to move horizontally ∼2.8 J/kg-m (relative to standing) is much less than that extra needed to move vertically ∼30 J/kg-m. Obviously, walking uphill requires more power than on level terrain; walking downhill requires less power than on level terrain. The work and power required going uphill is about 2.7× the increase in potential energy and the concomitant rate of change of this energy; this means the efficiency of doing this type of work is about $1/2.7 = 36\%$. (As we will soon see, this says the average extra energy needed to move vertically is 26 J/kg-m, which is consistent with the value of ∼30 J/kg-m that was just given.) When moving downhill, less power is needed than that needed on level terrain by ∼50% of the rate of decrease of potential energy. All of the gained potential energy is not recovered because muscular work is needed to prevent us from falling downhill. Problem 6.65 explores the power needed in walking (or running) and gradients.

Table 6.24 The Bruce protocol for an exercise test on a treadmill

Stage	Speed (mph)	Grade (%)	Cumulative time (min)
I	1.7	10	1–3
II	2.5	12	4–6
III	3.4	14	7–9
IV	4.2	16	10–12
V	5.0	18	13–15

Using data from [35]

Let us address the use of treadmills to assess the cardiovascular and cardiorespiratory capacity of healthy people and those with heart disease. The patient is wired with EKG (electrocardiogram) leads and EKG scans are continually taken as the patient walks/runs on a treadmill (see Chaps. 8 and 12 for more about EKGs). The treadmill speed and/or the uphill grade is increased after set periods of time (called stages), until either the complete cycle ends, the patient can no longer endure the workout, or an irregularity appears. In the Bruce Protocol, Table 6.24, the speed and grade increase with each stage. In another common protocol, the Naughton Protocol, the speed is fixed at 3.0 mph, and the grade increases from 2.5% in Stage I to 20.0% in Stage VIII, in steps of 2.5%, with each stage lasting 2 min. Such tiltable treadmills are also used to determine the efficiency of skeletal muscles in human locomotion. Oxygen consumption rates—and therefore metabolic rates—are compared for an ensemble of people moving at a given speed first on a level treadmill and then on a tilted treadmill. The efficiency is about 30%.

As an example, we will determine the metabolic power and rate of oxygen consumption needed during Stage V of the Bruce protocol for a 75 kg male. The total rate is the sum of the BMR (or more precisely the MR for standing), the power needed to move horizontally, and the additional power needed to move uphill. The second term is ~2.8 J/kg-m × (horizontal distance moved per unit time). This is 2.8 W $m_b v$, where m_b is the body mass in kg and v is the speed of walking in m/s (or really jogging for this faster speed). The third term comes from how fast the potential energy $m_b gh$ changes with time, which is $m_b g(\mathrm{d}h/\mathrm{d}t)$, and the "inefficiency" factor of 2.7, and so it is 2.7 $m_b g(\mathrm{d}h/\mathrm{d}t)$. A person moving a distance x along a ramp at angle θ elevates a distance $x \sin\theta$ (draw a diagram to prove this), so this third term becomes 2.7 $m_b g \sin\theta(\mathrm{d}x/\mathrm{d}t)$. With $v = \mathrm{d}x/\mathrm{d}t$, this becomes 2.7 $m_b g v \sin\theta = 26$ W $m_b v \sin\theta$. Combining the second and third terms gives $(2.8 + 26 \sin\theta) m_b v$ in W or $(2.4 + 22.8 \sin\theta) m_b v$ in kcal/h (using 1 W = 0.86 kcal/h). The speed is 5.0 mph = 2.23 m/s (using 1 mph = 0.447 m/s). A grade of 18% corresponds to an angle of $(0.18)90^\circ = 16.2^\circ$, and so $\sin\theta = 0.28$. For this 75 kg person, the second term is (2.4)(75)(2.23) = 401 kcal/h, and the third term is (22.8)(0.28)(75)(2.23) = 1,068 kcal/h. The burden of moving uphill rapidly is by far the largest contribution to the metabolic rate. The sum of these two terms is 1,469 kcal/h. Adding to this the MR for walking of about 100 kcal/min, the MR in Stage V is about 1,570 kcal/h, which is pretty high. (Of course, the patient is at this stage for no longer than 3 min.) Assuming

Table 6.25 The effect of wind on energy needs of walking and running

Activity, speed (m/s) (or mph)	Wind speed w (m/s) (or mph)	Oxygen consumption (L/min)	Heat production (W)	Increase in energy/w^2 (W-s^2/m^2)
Walking, 1.25 (2.80)	0.0 (0.0)	0.878	294	
	10.0 (22.4)	1.192	399	≃1.05
	14.1 (31.5)	1.505	504	≃1.05
Walking, 2.08 (4.65)	0.0 (0.0)	1.649	552	
	10.0 (22.4)	2.221	744	≃1.91
	14.1 (31.5)	2.792	935	≃1.91
Running, 3.75 (8.39)	0.0 (0.0)	2.836	950	
	10.0 (22.4)	3.243	1,086	≃1.36
	14.1 (31.5)	3.710	1,243	≃1.36
Running, 4.47 (10.00)	0.0 (0.0)	3.010	1,008	
	10.0 (22.4)	3.710	1,243	≃2.52
	14.1 (31.5)	4.525	1,516	≃2.52

Note that the last column is the increase in heat production divided by the square of the wind speed for the particular walking or running condition.
Using data from [11], calculated using [86]

only aerobic metabolism, the rate of oxygen usage is this metabolic rate divided by the calorific equivalent. If only carbohydrates are being burned, the calorific equivalent is 5.05 kcal/L O_2 and we see that in Stage V dV_{O_2}/dt = (1,570 kcal/h)/(5.05 kcal/L O_2) = 311 L O_2/h =5.2 L O_2/min is needed.

Why is it hard to run into the wind? The answer is that it requires much more metabolic power (heat production, or equivalent oxygen consumption). Table 6.25 shows that for walking and running this power increases as the square of the wind speed. A fast walk into the wind can require as much power as a relatively slow run with no wind. Even still air affects metabolic needs; air resistance accounts for about 8% of the metabolic needs for middle distance runners and up to 15% for sprinters. (We examine this effect of drag in Chap. 7.)

MR Scaling with Metabolic Equivalents and Activity Factors

The actual heat production, or metabolic rate (MR), is higher than the BMR because of activity (Table 6.21). We can relate the MR and BMR by

$$\text{MR} = f(\text{BMR}), \tag{6.32}$$

where f is the activity factor.

The metabolic equivalent (MET) is another very closely related measure of activity. An average body consumes (3.5 mL/kg-min) × MET. Typical values of MET are given in Tables 6.26, 6.27, 6.28, 6.29 and 6.30. It is often said that MET $\sim f/1.5$, because the activity factor f is referenced to the smaller BMR, but it is not likely that this ratio is as large as 1.5, especially for light activities. (In comparing values for walking with those in [47], this ratio seems to be closer to 1.2 for moderate activities.) Examples of f [47], include 1.05 for lying at rest, 1.20 for knitting, 1.48 for playing cards, 2.36 for working on a car assembly line, 3.02 for playing pool, 3.66 for dusting, 4.22 for dancing the waltz, 5.03 for weeding, and 5.08 for drill competition.

As noted earlier, there is a seeming inconsistency in our discussion of the BMR, MR, and activity factors. Among humans, the BMR varies slower than linearly with m_b in a way that is roughly but not exactly the same as the Kleiber's Law $m_b^{3/4}$ dependence. Adopting the activity factor (or PAR) as the ratio of MR and BMR for a given activity presumes that the extra metabolic rate due to that activity also varies roughly as $m_b^{3/4}$; however, the extra metabolic rate due to locomotion is proportional to m_b. How the energy we expend on different activities depends on body mass is not well understood yet.

Table 6.26 Typical MET values for self care

Self-care	MET
Rest, sitting, standing, eating, conversation	1.0
Dressing, undressing, washing hands and face, propelling wheelchair	2.0
Walking 2.5 mph	3.0
Showering	3.5
Walking downstairs	4.5
Walking 3.5 mph	5.5

Using data from [106]

Table 6.27 Typical MET values for housework

Housework	MET
Handsewing	1.0
Machine sewing, sweeping floor	1.5
Polishing furniture	2.0
Peeling potatoes, scrubbing, hand washing clothes, kneading dough	2.5
Scrubbing floors, cleaning windows, making beds	3.0
Ironing, mopping, wringing wash by hand, hanging wash by hand	3.5
Beating carpets	4.0

Using data from [106]

Table 6.28 Typical MET values for occupations

Occupational	MET
Sitting at desk, writing, riding in car, watch repair	1.5
Typing	2.0
Welding, radio assembly, playing musical instrument	2.5
Parts assembly	3.0
Bricklaying, plastering	3.5
Heavy assembly work, wheeling wheelbarrow (115 lb, 2.5 mph)	4.0
Carpentry	5.5
Mowing lawn with hand mower, chopping wood	6.5
Shoveling	7.0
Digging	7.5

Using data from [106]

Table 6.29 Typical MET values for exercise

Exercise (level conditions)	MET
Walking 2 mph	2.5
Cycling 5.5 mph	3.0
Walking 2.5 mph, cycling 6 mph	3.5
Walking 3 mph; calisthenics	4.5
Cycling 9.7 mph; swimming, crawl, 1.0 ft/s	5.0
Walking 3.5 mph	5.5
Walking 4 mph	6.5
Jogging 5 mph	7.5
Running 7.5 mph; cycling 13 mph	9.0
Swimming, crawl, 2.0 ft/s	10.0
Running 8.5 mph	12.0
Running 10 mph; swimming, crawl, 2.5 ft/s	15.0
Running 12 mph; swimming, crawl, 3.0 ft/s	20.0
Running 15 mph; swimming, crawl, 3.5 ft/s	30.0

Using data from [106]

Daily-Averaged Activity Factor

The activity factor could refer to a given activity, as above, or to the daily average; we will call the daily average f_{av}. (The term physical activity level (PAL) is commonly used in the health community.) There is no universal agreement on the actual values of f_{av}. They depend on the type of work, amount of exercise, and access to automated conveniences. United Nations studies suggest for adult men $f_{\text{av}} \sim 1.55$ for light occupational work, 1.78 for moderate occupation work, and 2.10 for heavy occupational work. This is consistent with another evaluation, presented differently, with it being about 1.2 for those during best rest, ~1.6 for sedentary office workers, ~1.8 for those engaged in manufacturing or farming, ~2 for substinence farmers

Table 6.30 Typical MET values for recreation

Recreational	MET
Painting, sitting	1.5
Playing piano, driving car	2.0
Canoeing 2.5 mph, horseback riding (walk)	2.5
Volleyball (recreational), pool (billiards)	3.0
Bowling, horseshoes	3.5
Golf, cricket	4.0
Archery, ballroom dancing, ping pong (table tennis), baseball	4.5
Tennis	6.0
Horseback riding (trot), folk dancing	6.5
Skiing, horseback riding (gallop)	8.0
Squash	8.5
Fencing, basketball, football	9.0
Gymnastics, handball, paddleball	10.0

Using data from [106]

and modern-day "hunter-gatherers" (who hunt and gather food, as in the Hanza tribe in Tanzania) [7], and ~2.5 during training for endurance events, like the marathon [56]. "Desirable" values of f_{av} of ~1.65 for adolescents and adults to 59 years old in developed countries (and ~1.8 in developing countries), and ~1.5 for those 60 and older have also been suggested [33, 47]. Other studies suggest that the MR is much lower, with $f_{av} \sim 1.3$ for low activity (sedentary), ~1.5 for intermediate activity (some regular exercise), and ~1.7 for a high level activity (regular activity or physically demanding job). For a survival diet that includes the energy needed to eat and digest food and essential hygiene, but not that needed to earn a living or prepare food, f_{av} has been estimated to be 1.27 [34].

The highest of these values for heavy occupational work, $f_{av} \sim 2.10$, is still lower than that estimated for a Scottish coal miner in the 1950 s before the occupation became more automated. The peak metabolic rate of the coal miner was 4.3 × BMR during working hours and averaged to a still very high 2.5 × BMR during the entire day, as is calculated in Table 6.31. Most occupations are physically less demanding now and in general modern day lifestyles require much less physical activity. Household work is much less physically demanding with the widespread use of clothes washing machines, vacuum cleaners, dishwashers, and food processors. The time spent on household chores each week has decreased by about half in the last 50 years [5].

Typical Americans walk ~1/4 mile/day and Western Europeans ~2/3 mile/day, much less than the ~5–9 miles/day for modern day hunter-gatherers [7]. Male hunter-gatherers have a body mass of ~51 kg, walk about 9 miles each day, and have an f_{av}~2,650 kcal/day (MR)/1,170 kcal/day (BMR) = 2.26, with exceeds that of ~81 kg Western males, 3,050 kcal/day/1,685 kcal/day = 1.81. These hunter-gatherers use about twice as many kcal per kg body mass on activities,

Table 6.31 Analysis of the metabolic power needs of a Scottish coal miner in the early 1950s before mechanization

Activity	Time		MR/BMR = f	Product $F \times f$
	(min)	(Fraction of day, F)		
In bed	501	0.348	1.00	0.348
Nonoccupational				
Sitting	331	0.230	1.51	0.347
Standing	19	0.013	1.71	0.022
Walking	129	0.089	4.67	0.416
Washing and dressing	43	0.030	3.14	0.094
Gardening	17	0.012	4.76	0.057
Cycling	21	0.015	6.28	0.094
Occupational				
Sitting	130	0.090	1.60	0.144
Standing	18	0.012	1.71	0.021
Walking	58	0.040	6.38	0.255
Hewing	11	0.007	6.38	0.045
Timbering	59	0.041	5.43	0.223
Loading	104	0.072	6.00	0.432
Totals	1,440	1.000	–	2.497

MR is the metabolic rate.
Using data from [11], recalculated using the data of [40]

$(2,650-1,170)/51 = 29.0$ kcal/kg, than typical Westerners, $(3,050-1,685)/81 = 16.9$ kcal/kg. Not surprisingly, they also have a lower BMI, ~20.3 versus ~25.6 kg/m^2, and less body fat, ~13.5 versus ~22.5% [56, 62, 82].

Still, sustained human MRs usually do not exceed 2.0–2.5 × BMR, except for those in high-performance careers, such as endurance athletes. This may constitute a metabolic ceiling for humans [29, 46, 78]. This limit is exceeded by Tour de France bicyclists who have average activity factor of ~4.3, but not by hunter-gatherers.

Why do women normally give birth 38–40 weeks after conception? The reason seems to relate to fetal size at that time. Some think that it is limited by the hip and birth canal sizes relative to the size of the fetal head (which is large even though the brain size at birth is only 30% of final size). If hip sizes were larger, perhaps locomotion in women would require more energy (and this would not be desirable), but there is evidence that this is not the case [29]. The end of pregnancy might be signaled by a crossover in mother metabolic rates and fetal metabolic needs [29, 30]. There is some indication that a pregnant woman's total MR stabilizes at about 2.1× her pre-pregnancy BMR sometime during the second half of gestation, which would be consistent with an MR ceiling of 2.0–2.5 × BMR [29, 46, 78]. If this were really a limit, the mother may not be able to sustain enough caloric intake to account for the rapidly growing caloric needs late in pregnancy when the fetus starts growing

very fast. If this crossover occurs ≃40 weeks into gestation, this might be why the hormonal signals leading to birth occur then and not later. (The growing metabolic needs of the fetus can be appreciated by noting that the metabolic requirements for infants increase from 450 to 1000 kcal/day from birth to 12 months [33]. This includes caloric needs to support the growing tissue in the baby, but it also includes metabolic needs supporting baby motion and organ function that would not occur for a fetus and excludes other metabolic needs of the mother during pregnancy.)

Mechanical Work and Power

The first law of thermodynamics (6.1) shows that stored energy can be used to supply heat or work. For people, this is mechanical work and not any other type (such as thinking deeply about your homework). Mechanical work is the force you apply to an object × the distance you push or pull it.

In discussing metabolic rates we need to be careful about defining types of work and energy [107]. Figure 6.10 shows that in lifting an object, which we can call *external work*, work is also done about the joints of the body to lift the center of mass of the body itself, which is *internal work*. Also, concentric muscle contractions are said to do positive (mechanical) work, while eccentric contractions do negative work. Walking on level ground requires equal amounts of positive and negative work. Walking uphill requires relatively more positive work, while walking downhill requires relatively more negative work. The work efficiencies of positive and negative work are different.

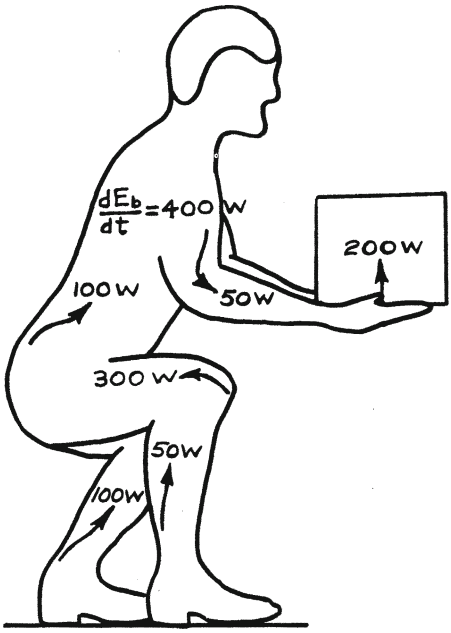


Fig. 6.10 "Internal" versus "external" work in lifting an object (From [107]. Reprinted with permission of Wiley)

Table 6.32 Mechanical power and efficiency of mechanical work

Task	Mechanical power output		Metabolic rate (W)	Efficiency (%)
	(hp)	(W)		
Cycling (typical)	0.15	112	505	19
Tramming	0.12	90	525	17
Shoveling sand	0.024	17.5	570	3

Using data from [9]

Table 6.33 Efficiencies of mechanical work

Exercise activity or task	Efficiency (%)
Cycling, level	24–34
Cycling, uphill	19
Ice skating	11
Rowing	10–20
Stepping machine	23
Swimming, freestyle	2.9–7.4
Walking, horizontal	20–35
Walking, incline	21–43

Using data from [35]

The term *efficiency* can mean different things, so in general we need to be careful and see what is being defined [105, 107]. *Metabolic or muscle efficiency* is the total mechanical work done by all muscles divided by the metabolic work done by the muscles. Another measure of efficiency is the *mechanical efficiency*, which is the total mechanical work done (internal + external) divided by the metabolic energy cost in excess of the resting metabolic cost. A more restrictive parameter is the *work efficiency*, which is the external mechanical work divided by the metabolic cost in excess of the zero-work metabolic cost. Inefficiency results from incomplete conversion of metabolic energy to mechanical energy and from neurological inefficiency in controlling this mechanical energy.

For our purposes we will simply define the efficiency, ϵ, as the rate of doing mechanical work divided by the metabolic rate. Tables 6.32 and 6.33 show that it rarely exceeds 20%, and it is usually much less than this. Cycling is relatively efficient, with ϵ approaching 22% for racing cyclists. Tramming is also efficient. (This involves pushing railroad cars loaded with coal down a railroad track.) Shoveling sand, a more common activity, is only 3% efficient. For comparison, steam engines have an efficiency of 17% and gasoline engines have $\epsilon = 38\%$.

Where does the balance of the energy go? The balance goes into the production of heat. This means that 80–100% of the final product of your metabolism is heat; this number is usually closer to 100%. In any case, the ~500 kcal mechanical work done by a physical laborer during a day translates into dietary needs of at least 2,500 kcal over the BMR, because the mechanical efficiency is usually $<20\%$.

The rate of oxygen intake is limited by the rate the lungs can take up oxygen and the rate the heart can distribute this oxygen around the body for aerobic metabolism.

Table 6.34 World record running speeds

Distance (m)	Average running speed (m/s)
100	10.44
200	10.42
400	9.26
1,500	7.28
10,000	6.34
42,195[a]	5.72
100,000	4.46

[a]Marathon, 26 miles, 385 yd.
As of 2014

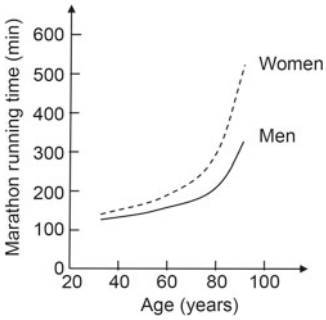


Fig. 6.11 Variation of best marathon times with age (based on [99])

One example of this is the decrease in human running speed with increasing distance (beyond ~200 m), which is due to the limitations in oxygen intake. This is clear from the world-record average running speeds at various distances shown in Table 6.34. World-class times in running each of these events increases with advancing age, with peak performance maintained until ~35 years of age and then it decreases, first slowly until ~50–60 years of age and then faster, with faster decreases in endurance events than in sprints. This is seen in Fig. 6.11 for those running marathons [99]. Still, the winning times in the 1896 Olympics (the first modern Olympic games, in Athens) are currently achieved by world-class runners who are approximately 60 years old [99]. Peak performance in swimming also decreases with advancing age, but at comparable rates for sprint and endurance events (Fig. 6.12) [28, 99].

Table 6.35 shows that the maximum power of mechanical work a well-conditioned person can do decreases with the duration of the task. For short times we can do mechanical work at a higher rate because of anaerobic metabolic processes; longer-term work requires aerobic processes.

How to "Burn" Off Food

Let us say you have just eaten a "standard" donut. You feel guilty and you want to burn off those extra calories. What can you do?

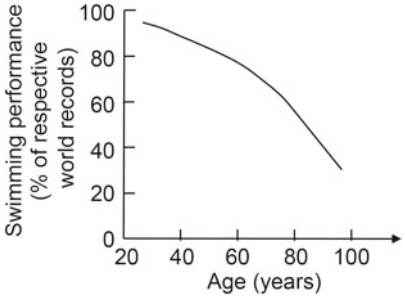


Fig. 6.12 Variation of best swimming times with age, for 50 and 1500 m races, for men and women (based on [99])

Table 6.35 Approximate maximum mechanical power from well-conditioned humans

Power		Duration	Total energy	
(hp)	(W)		(kJ)	(kcal)
2	1,500	6 s	9.0	2.2
1	750	1 min	45	10.8
0.35	260	35 min	546	130
0.2	150	5 h	2,700	645
0.1	75	8 h day of a laborer	2,160	516

Using data from [9]. Also see [105]

If you are sitting at rest you are naturally burning off ∼103 kcal/h (70 kg man). If you decide to play basketball your metabolic rate increases to ∼688 kcal/h, so you will be increasing your metabolic rate by ∼585 kcal/h. You will burn off that standard 280 kcal donut in

$$\frac{280\,\text{kcal}}{585\,\text{kcal/h}} = 0.48\,\text{h} = 29\,\text{min}. \tag{6.33}$$

That donut will cost you a half an hour of real up-tempo basketball.

Let us say you want to "walk off" that donut. The metabolic rate during slow walking is ∼228 kcal/h, which exceeds that of sitting at rest by ∼125 kcal/h. To walk off that donut you would have to walk for

$$\frac{280\,\text{kcal}}{125\,\text{kcal/h}} = 2.24\,\text{h} = 2\,\text{h}\ 14\,\text{min}. \tag{6.34}$$

Which is a little longer than most after-dinner strolls.

Why do people put on weight when they get older? One reason is the decrease in BMR with age (Tables 6.19 and 6.20). The activity level f_{av} often decreases with age. Also, sometimes people eat more (snacking) (see Problem 6.88). We now examine how caloric intake and activity combine to determine body weight.

6.4.3 Weight Gain and Loss

If your caloric intake rate exceeds your metabolic rate, you gain weight. If it is less, you lose weight. You can control your caloric intake by controlling what you put in your mouth. You have some control of your metabolic rate, by being more or less active in general and by exercising more or less. (In a daring example of his devotion to science and his readers, while writing the first edition of this text the author showed how easy it is to gain 10 lb in 6 months merely by exercising less and by snacking more at night. He needed no dietary supplements to accomplish this feat! The author considered the reverse experiment of weight loss through more exercise and less snacking. While writing this second edition, he was still considering this experiment.) As we will see later in this chapter, your metabolic rate is related to your body temperature and loss of body heat. For example, if you are very cold, you shiver and this increases your MR.

We will examine some very simple models of weight gain and loss here and in the problems. *How much weight in the form of body fat can you actually lose in a week? How is body weight controlled by salt intake? How does your body weight vary due to changes in caloric intake, age, and specific stature?* Our models are not only simple, they are overly simplistic—and yet it is a bit difficult to make them more realistic. For example, increases in body weight due to added muscle increase the BMR much more than those due to added fat. How can this be included in the model?

Is it reasonable to apply the relations characterizing how the average metabolic rate depends on body weight, age, and stature by using Kleiber's Law, Harris–Benedict equations, etc. to determine the steady-state weight and weight changes for a given individual? It is not clear that it is reasonable. As noted earlier, these are averages over sets of people and not trends for a given person. The body controls metabolic rates differently for different individuals in response to stimuli such as dieting—making it easier for some, while harder for others, to lose weight. Still, for now, let us assume that these relations also apply for a given individual and they can then be used to determine how the steady-state weight of a person changes with food intake, activity level, and age. *For a sudden change in metabolic intake or activity to a new level, how fast does the body reach its new steady state?* General changes are depicted in Fig. 6.13.

If your weight gain or loss involved only fat, the caloric value of fat of 9 kcal/g means that an increase or decrease in metabolic input by 4,090 kcal would result in a gain or loss of a pound (of fat). If your weight change involved only protein (4 kcal/g), a change in 1,820 kcal would change body mass by a pound. According to Wishnofsky's rule, when you increase or decrease your metabolic input by 3,500 kcal you gain or lose about a pound [108]. Consequently, Wishnofsky's rule suggests the gain or loss typically averages to about 3/4 fat and 1/4 muscle.

Because of the different muscle, fat, and fluid masses in men and women and in younger and older people, this long-used rule is now not thought to be really that universal. This average energy content of weight change is really significantly

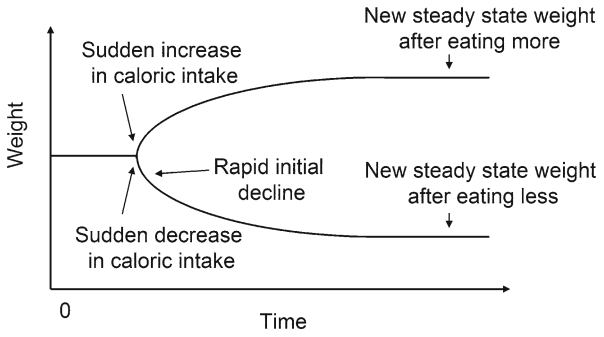


Fig. 6.13 Schematic of the time evolution of weight gain or loss, for a small sudden increase or decrease in metabolic intake (which is then maintained at this new level). Similar changes are expected for a small sudden decrease or increase in activity level, respectively

smaller for men (2,480 kcal/lb) than in women (3,380 kcal/lb). Also, it is about 10% larger for older people (2,610 kcal/lb, 70-year-old men; 3,510 kcal/lb, 70-yr-women) than younger people (2,360 kcal/lb, 35-yr-men; 3,260 kcal/lb, 35-yr-women) [79]. (All for these have been scaled to the same 90 kg, 170-cm tall person.)

Let us say you increase or decrease your food intake by 1,000 kcal/day. If it all went into increasing or decreasing your fat reserves, this would lead to a change of (1,000 kcal/day)/(9 kcal/g fat) or about 110 g of fat a day, which corresponds to 1/4 lb per day or roughly 2 lb per week. If you are on a diet, a decrease in caloric intake of 1,000 kcal/day is very large, so you can never expect to lose more than 2 lb a week. (Of course, regular exercise would also accelerate weight loss.)

Why do some diets claim faster weight loss, especially early on? Some diets may really involve the loss of body water, with very little loss of fat, carbohydrate, or protein. (Boxers who need to lose weight quickly to make a weight division often follow such diets.) Also, weight loss in a legitimate diet can be very fast early on. In the first days of dieting, burning 1,000 kcal involves burning glycogen in the small glycogen pool. Burning glycogen consumes 4.1 kcal/g glycogen, but because of the concomitant loss of water associated with the glycogen, only about 1 kcal is consumed per total g lost. Therefore, 1,000 kcal/(1 kcal/g) = 1,000 g is lost, which is about 2 lb. Later when fat is oxidized instead, this same deficit of 1,000 kcal results in a mass loss of only 110 g (1/4 lb). This suddenly slower rate of weight loss (in the approach to the new steady-state weight) can easily discourage dieters.

To paraphrase the advice given by Mr. Micawber to Charles Dickens' David Copperfield, "Daily caloric intake 2,000 kcal, daily energy expenditure 2,100 kcal, result happiness. Daily caloric intake 2,000 kcal, daily energy expenditure 1,900 kcal, result misery." (This is true at least for those way beyond their college years.)

Energy intake and expenditure clearly affect body weight in this direct way. However several factors that are not very well understood seem to affect energy intake and expenditure and consequently body weight [56]. Genes are one of several factors in causing obesity [101]. Some genes regulate appetite, and with the ready availability

of digestible high-caloric food those with genes that increase appetite can gain weight more easily. Chronic stress increases the steady-state level of the hormone cortisol, which increases the appetite for highly palatable inexpensive calorie-dense food, such as "junk food" [1]. Increased exercise improves cardiovascular fitness (and your mortality rate from many diseases [12]), but it does not always greatly decrease body weight [13, 25, 81]. The net increase in average metabolic rate due to exercise lowers your steady-state weight, but not necessarily by enough to sustain a large steady-state weight loss. Increasing muscle mass by exercising, at the expense of fat, will increase your BMR and lead to some weight loss. However, increased exercise is sometimes accompanied with an increased desire to eat. Older obese people seem to lose weight when they have a controlled diet, whether or not they exercise, and do not lose weight with increased exercise alone or with no diet or exercise [102]. Broader studies of obese people suggest that combined diet and exercise leads to more weight loss than either one alone [48]. If you get very little sleep your daily metabolic rate increases, which you would expect to decrease your weight. However, it appears that your body convinces you to eat more food, so much so that it overcompensates for your increased metabolic rate and you actually gain weight [8, 61]. Surprisingly the MR (or TEE) of westerners and hunter-gatherers are roughly the same for the same body mass (or fat-free body mass), so the increased caloric intake by westerners is more likely important in causing obesity in some than is their suspected, but not proved, decreased energy expenditure [82].

Other aspects of metabolism, diet, and weight control are also not well understood. While exercise appears to help with weight control directly by increasing the metabolic rate, it may indirectly affect it by how the body controls the routes and efficiency of food metabolism. In any case, exercise is usually not a substitute for controlling the amount and type of food eaten [67]. There is evidence that a high-protein–low-glycemic-index diet helps maintain weight loss, while a low-protein–high glycemic-index diet does not [54]. Moreover, the body converts brown fat (not the usual, more abundant white fat) to heat apparently without fueling metabolic activities; it appears to be a source of heat for babies when they are cold, before their shivering mechanism is in place. Adults appear to have some brown fat also. It is possible that the creation and metabolism of brown fat is important in controlling the body energy balance in some adults, even under conditions when metabolic rates are usually slower, such as when people are inactive and when it is cool (lower temperatures, but not low enough to cause shivering) [74].

There is some evidence that eating specific foods can affect body weight, though the reasons for this are not always clear. Deciding to include peanuts and tree nuts, such as almonds, as part of your diet may help control or even lower body weight [64]. Nuts are filling due to their high protein and fiber, and perhaps their high fat content, and this satiating effect lowers later caloric intake by ∼55–75% of the energy reportedly provided by nuts. A smaller fraction of caloric content in whole nuts is digested compared to that in nut butter and oil, with possibly ∼10–15% or more of caloric content lost in the feces. The usable energy content of almonds may be ∼4.8 kcal/g, compared to the 6.0-6.1 kcal/g value obtained using the Atwater factors

[76]. Also, there is some evidence that chronic consumption of nuts may increase the BMR, perhaps by roughly 10%. This is still not certain.

Inexpensive calories are available to many in developed countries, and this availability has increased in recent centuries. For example, the numbers of calories available per capita in England and Wales increased from and estimated ∼2,230 kcal in 1700 to ∼2,980 kcal in 1910 [34]. A major problem in maintaining desirable body weights is that cheap calories (in tasty junk food) are too available.

For most people in developed countries getting enough daily calories from calorie-dense foods is not an issue, but this is not true worldwide and during wars, when transporting enough calories to soldiers and supplying calories to civilians can be very important. Seasonal changes in body weight occur in people who do seasonal physical labor and who do not have access to a steady source of high-calorie foods. For example, farmers in Gambia lose weight (∼10 lb or ∼5 kg) during the wet, planting season and, if the crop is successful, regain it during the dry, harvesting and resting season [56, 84, 93]. Those with careers involving much physical labor, such as farmers, carpenters, and professional athletes, must decrease their daily caloric intake after they retire or they will gain much weight, unless they exercise much more [56].

Why have people become taller with each generation in many developed countries? When the caloric intake of children is limited, much of it is used for basic metabolic activities and extra activities (which might involve much physical labor) and to fight off disease, and less than needed is used for growth. With more caloric intake (and sometimes less physical labor), more energy can be used for growth, and people can grow to the height determined by their genes [56]. (With too much caloric intake, they also become fatter.)

From a health viewpoint, the ideal BMI is sometimes cited as being 25 kg/m^2, but this seems to be true only for men with heights of ∼1.75 m (5′9″) and it is larger, ∼26 kg/m^2, for shorter men (<1.68 m, 5′6″) and smaller, from 22 to 24 kg/m^2, for taller men (>1.83 m, 6′) [34]. The BMIs of American men in their forties and fifties increased from ∼22 to over 27 kg/m^2 from 1870 to 2000, so now many men have a BMI over the current 25 kg/m^2 threshold for being overweight [34, 56]. (Only part of this increase can be attributed to observed increases in height and how the BMI presumes that body mass ideally scales quadratically with height (see Problem 1.42).) The BMI seems to affect lifespan. Mortality for Norwegian men and women aged 50-64 is pretty constant for those with BMIs between 21 and 29, but it increases rapidly for those with BMIs that are progressively smaller than 21 (too underweight) or progressively larger than 29 (too overweight) [34, 103].

During starvation, the body first utilizes all available carbohydrates—mostly glycogen in the liver and muscles—and this satisfies the body's energy needs for about half a day. Then fat and protein are depleted by the body (see Fig. 6.14); fat reserves are about 100× the initial carbohydrates reserves. Then with no fat remaining, protein is further depleted until death. Death occurs when about half of the proteins have been depleted because proteins are needed for cellular function.

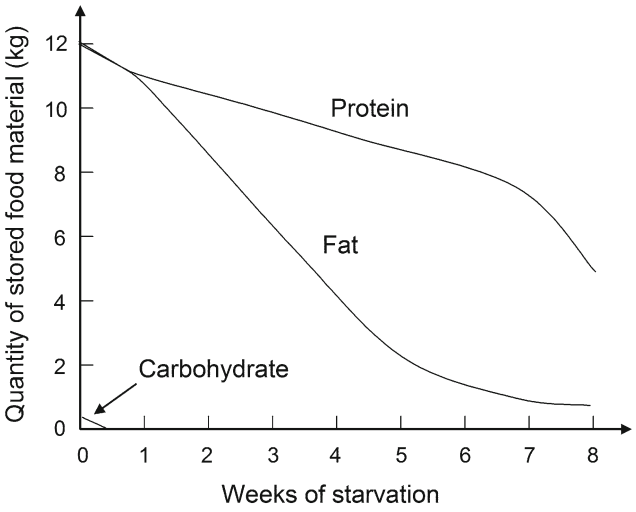


Fig. 6.14 Effect of starvation on energy supplies in the body (based on [45])

6.5 Loss of Body Heat

Why is the loss of body heat important? Virtually all of our metabolism goes into making heat, because very little energy usually goes into useful mechanical work. Consider the BMR of an $m_b = 70\,\text{kg}$ person, which is ~1,680 kcal/day. This translates to 70 kcal/h. Using (6.5), the rate that body temperature would increase with time if all the heat stayed within the body is

$$\frac{dT}{dt} = \frac{1}{m_b c}\left(\frac{dQ}{dt}\right) = \frac{\text{BMR}}{m_b c}, \tag{6.35}$$

where the rate of heat production $dQ/dt \approx$ BMR and the average human specific heat is $c_b = 0.83\,\text{kcal/kg-°C}$. (To be consistent with the literature, we will usually express this in terms of degrees Kelvin in this section, which is 0.83 kcal/kg-K.) This heat capacity is $C = 0.83\,\text{kcal/kg-°C} \times 70\,\text{kg} = 58\,\text{kcal/°C}$. The rate of temperature increase equals

$$\frac{dT}{dt} = \frac{70\,\text{kcal/h}}{58\,\text{kcal/°C}} = 1.2\,\text{°C/h} = 2.1\,\text{°F/h}. \tag{6.36}$$

Our body temperature would go up very fast without effective modes of heat loss, even with some production of mechanical work ($0.8\,\text{MR} < dQ/dt < \text{MR}$).

The specific heat of the body is often cited as 0.829 kcal/kg-K. This is a bit higher than that calculated using the specific heat for each body component, which are (in

Table 6.36 Thermophysical characteristics of body tissues and organs and other materials

Organ or tissue	Thermal conductivity K (W/m-K)	Specific heat c_v (MJ/m^3-K)	Density (approximate) ρ (kg/m^3)
Skin—very warm	2.80	3.77	1,000
Skin—normal hand	0.960	3.77	1,000
Skin—cold	0.335	3.77	1,000
Subcutaneous pure fat	0.190	1.96	850
Muscle—living	0.642	3.94	1,050
Muscle—excised, fresh	0.545	3.64	1,050
Bone—average	1.16	2.39	1,500
Bone—compact	2.28	2.70	1,790
Bone—trabecular	0.582	2.07	1,250
Blood—water at 310 K	0.623	4.19	993
Blood—plasma (Hct = 0%) at 310 K	0.599	4.05	1,025
Blood—whole (Hct = 40%)	0.549	3.82	1,050
Heart—excised, near fresh	0.586	3.94	1,060
Liver—excised, near fresh	0.565	3.78	1,050
Kidney—excised, near fresh	0.544	4.08	1,050
Abdomen core	0.544	3.89	1,050
Brain—excised, near fresh	0.528	3.86	1,050
Brain—living	0.805	–	–
Lung—excised, bovine	0.282	2.24	603
Whole body (average)	–	4.12	1,156

Note that 1 MJ/m^3-K = 239 kcal/m^3-K and 1,000 kg/m^3 = 1 g/cm^3.
Using data from [38]

kcal/kg-K) 0.507 for lipid, 0.299 for protein, 0.201 for mineral matter, 0.272 for carbohydrate, and 1.00 for water. These calculated values are 0.798 kcal/kg-K for a fat-free man (containing 72% water, 22% protein, and 6% minerals by weight), 0.765 kcal/kg-K for a thin man (containing 12% body fat), and 0.652 kcal/kg-K for an obese man (containing 50% body fat). The heat capacities of different body tissues are shown in Table 6.36 and those of other common materials in Table 6.37.

Table 6.37 Thermophysical characteristics of materials

Material	Thermal conductivity K (W/m-K)	Specific heat c_v (MJ/m^3-K)	Density ρ (kg/m^3)
Air	0.009246	0.00119	1.18
Cotton fabric at 310 K	0.0796	0.0267	160
Rubber	0.156	2.41	1,200
Ethanol at 310 K	0.163	1.96	789
Teflon	0.399	2.20	2,180
Concrete	0.934	1.93	2,310
Glass, plate	1.09	1.94	2,520
Ice at 249 K (−42 °C)	2.21	1.76	913
Sapphire (normal to c-axis) at 310 K	2–20	2.89	3,970
Stainless steel	13.8	3.68	7,910
Aluminum	204	2.45	2,710
Silver	405	2.59	10,500
Diamond, natural	2,000	1.82	3,510

Note that 1 MJ/m^3-K = 239 kcal/m^3-K and 1,000 kg/m^3 = 1 g/cm^3.
Using data from [38]

6.5.1 Modes of Heat Loss

There are four modes of heat loss. Their absolute magnitude and relative importance depend on clothing, environment, surroundings, etc. (Figs. 6.15 and 6.16). For general discussions of heat loss in people see [10, 11, 19, 35, 106].

1. Radiation loss, also known as black body radiation, is the thermal radiation emitted by an object in thermal equilibrium. At rest, about 54–60% of energy loss is typically through thermal radiation.
2. Convection and conduction of air from the body account for ∼25% of heat loss.
3. The evaporation of sweat accounts for ∼7% of heat loss.
4. The evaporation of water through breathing accounts for ∼14% of this loss.

During very heavy activity, heat loss increases from the resting value of ∼1.5 to ∼15 kcal/min. The amount due to radiation remains about the same, 0.8 kcal/min, but it decreases to about 5% of the total. The loss to conduction and convection increases from ∼0.3 to 2.2 kcal/min and this represents about 15% of the total loss. Losses due to the evaporation of sweat increase from ∼0.3 to 12.0 mL/min and they constitute about 80% of the heat loss.

The principles of these modes of heat loss are totally determined by physics; the absolute magnitudes of the actual loss can be modified by biological processes.

In simple models of heat loss, people are often treated as upright cylinders, while four-legged animals are modeled as horizontal cylinders. In particular, a typical man is modeled as a cylinder that is 1.65 m high with a 0.234 m diameter. (Recall the

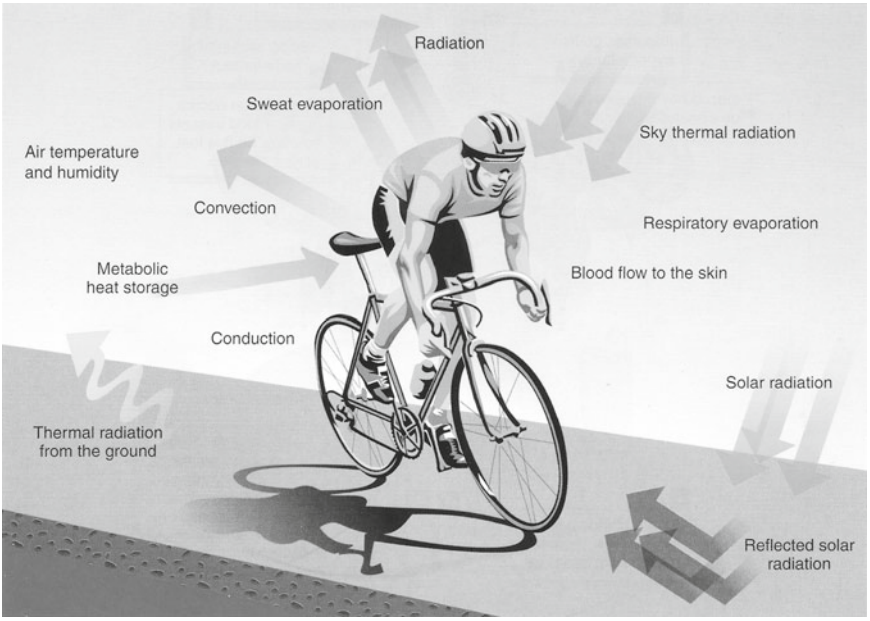


Fig. 6.15 Overall body modes of loss of heat and nonmetabolic gain of energy (From [106], adapted from [43]. Used with permission)

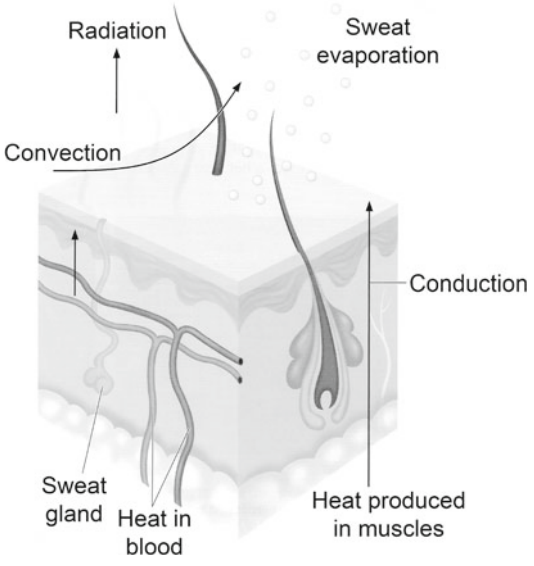


Fig. 6.16 More microscopic view of modes of heat loss by removal from the skin. When the skin temperature exceeds the environmental temperature, heat is removed by sweat evaporation, while when it is less than the environmental temperature there is net heat loss only by sweat evaporation. The skin is heated by arterial blood and by thermal conduction through subcutaneous (*below* the skin) tissue (From [106]. Used with permission)

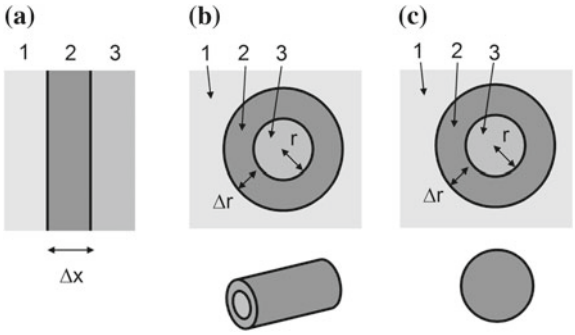


Fig. 6.17 Thermal flow between media 1 and 3, across medium 2, in different geometries: (**a**) *rectangular* slab, (**b**) *cylindrical* shell, (**c**) *spherical* shell. In common cases, Δr is much smaller than r

comment in the introduction that to zero order a physicist might model a cow as a sphere and that our models here may get a bit more complex here; a cylinder is indeed more complex than a sphere. See Fig. 1.33 for a model of a man composed of many cylindrical components.)

We will focus mostly on heat loss in one dimension. For example, the heat conduction equation (6.6) is one dimensional, and describes heat flow between two flat interfaces separated by a thickness Δx (Fig. 6.17a). Mathematically, heat flow radially through a thin cylindrical shell of thickness Δr on a cylinder of radius r (Fig. 6.17b) is identical to that across flat interfaces if the Δx in (6.6) is replaced by $r \ln(1 + \Delta r/r)$. For conduction through a thin spherical shell of thickness Δr on a sphere of radius r (Fig. 6.17c), it needs to be replaced by $r\,\Delta r/(r + \Delta r)$. In both cases the replaced parameter reduces to Δr when $r \gg \Delta r$, i.e., when the new interface approximates a flat plane.

Heat Loss by Radiation

Bodies in thermal equilibrium emit a specific flow of energy per unit surface area and time depending on their temperature. They also receive a flow of thermal radiation from the outside world over their surface area that depends on the temperature of the surroundings.

All objects at a temperature T emit such thermal radiation spread over a range of wavelengths (or equivalently frequencies) according to the Planck black body distribution for intensity, which is

$$I(\lambda, T)\,\mathrm{d}\lambda = \frac{2hc^2\epsilon/\lambda^5}{\exp(hc/\lambda k_\mathrm{B} T) - 1}\,\mathrm{d}\lambda, \tag{6.37}$$

where ϵ is a parameter known as the emissivity, h is Planck's constant (6.626×10^{-34} J-s), k_B is Boltzmann's constant (1.381×10^{-23} J/K), and T is the temperature in degrees Kelvin (K) ($T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273^\circ$). ($I(\lambda, T)$ is actually the intensity per unit wavelength range. $I(\lambda, T)\mathrm{d}\lambda$ is the intensity covering the wavelength range from λ to $\lambda + \mathrm{d}\lambda$, where $\mathrm{d}\lambda \ll \lambda$. Intensity has units of energy/area-time.)

Two features of this distribution are important to us. First, it peaks at a wavelength λ given by Wein's Law

$$\lambda_{\text{peak}}(\text{in } \mu\text{m}) = \frac{2{,}898}{T \text{ (in K)}}. \tag{6.38}$$

This means that thermal radiation from a person or the inside wall of a house peaks at 10 μm (mid-infrared light) because $T \approx 300$ K, while that from the sun peaks near 0.6 μm (600 nm, visible light) because the temperature of the sun is ~5,000 K. (It should not be very surprising that the wavelength of maximum eye detection is near this peak.) Second, the total energy flow per unit area and time (the energy flux) is determined by integrating (6.37) over all wavelengths to obtain the radiation flux, R, (which has units of energy/area-time)

$$R = \epsilon\sigma T^4, \tag{6.39}$$

where σ is the Stefan–Boltzmann constant $= 5.67 \times 10^{-8}$ W/m^2-K^4. This is known as Stefan's Law. At skin temperature (34 °C), the thermal emission is 505 W/m^2 for $\epsilon = 1$.

Radiation incident on an object can be reflected by it, absorbed by it, or transmitted through it. The emissivity is the fraction of energy incident on the object that is absorbed. For a perfect black body it is 1. Most objects have a somewhat lower emissivity because of reflection and transmission, and an emissivity that actually varies with wavelength (and sometimes temperature). We are interested in the emissivity in the infrared (Table 6.38), because black body radiation from people and walls peaks near 10 μm, and that in the visible, because it is needed to understand how thermal radiation from the sun affects people and objects. In calculations the infrared

Table 6.38 Infrared emissivities

Fresh snow	0.89
Ice	0.96
Dry or wet sand	0.89
Concrete	0.95
Moist soil	0.97
Grass surface	0.96
Red brick	0.92
Wood	0.90
White paint	0.93
Aluminized paint	0.55
Galvanized iron	0.28
Aluminum foil	0.08
Human skin	0.95–0.99

Using data from [11, 70, 71], and other sources

emissivity of human clothing and skin is generally taken to be 0.95. (The emissivity at 10 μm is the same for all colors of skin, though in the visible it is different.)

The energy loss per unit time due to thermal radiation from the body is

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{loss}} = RA = \epsilon_{\text{skin}}\sigma T_{\text{skin}}^4 A_{\text{skin}}, \tag{6.40}$$

where ϵ_{skin} is the skin (or clothing) emissivity, T_{skin} is the body skin temperature, and A_{skin} is the body surface area. (Remember that $\mathrm{d}Q/\mathrm{d}t < 0$ means that the body loses energy and becomes colder. The left-hand side of equations such as this are positive.) The correct temperature to use is that of the body skin (34 °C = 307 K), which is a bit lower than that of the core of the body (37 °C = 310 K). The average body surface area is $A_{\text{skin}} \approx 1.85\,\text{m}^2$, so the rate of body heat loss is

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{loss}} = (1)(5.67 \times 10^{-8}\,\text{W/m}^2\text{-K}^4)(307\,\text{K})^4(1.85\,\text{m}^2) = 932\,\text{W}, \tag{6.41}$$

assuming the emissivity ϵ_{skin} is 1. Ignoring all sources of heat gain and other sources of heat loss, this radiative cooling alone would cool the body at a rate of 14°C/h, as is seen by using (6.35) and (6.36). Fortunately, this loss of heat is balanced by gains!

The body also receives thermal radiation from its surroundings. The analysis is somewhat different for a person who is indoors, i.e., in an enclosed space, and one who is outdoors, where there are no enclosures and there can be direct heating by the sun during the day.

Receiving Thermal Radiation Indoors

Let us first assume that the body receives radiation from a source at room temperature of the same area $A_{\text{room}} = A_{\text{skin}}$. Then the rate of heat gain in a 24 °C = 297 K environment (T_{room}) is

$$\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{gain}} = \epsilon_{\text{room}}\sigma T_{\text{room}}^4 A_{\text{room}} \tag{6.42}$$

$$= (1)(5.67 \times 10^{-8}\,\text{W/m}^2\text{-K}^4)(297\,\text{K})^4(1.85\,\text{m}^2) = 816\,\text{W}, \tag{6.43}$$

assuming these temperatures and an emissivity of 1. (We will call the temperature in the environment T_{room}, whether or not the person is indoors or outdoors.) Then the net radiative heat loss by the body is given by

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{net}} = -\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{loss}} - \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{gain}} = 932\,\text{W} - 816\,\text{W} = 116\,\text{W}, \tag{6.44}$$

which we will now call $-(\mathrm{d}Q/\mathrm{d}t)_\mathrm{r}$. This depends on the temperature of the environment. Because the body emissivity is a bit smaller than 1, this is a slight overestimate, but it is still consistent with the body's usual metabolic rate.

This commonly used analysis is actually flawed. The walls of the room have a much larger area, A_room, than that of the person, which would seem to increase the thermal radiation incident on the person relative to that emitted by the person. However, only part of the radiation from this room enclosure impinges on the person—most of it hits the other parts of the enclosure, and this decreases the thermal gain by the body. We have also ignored the likelihood that the emissivities of the walls of the room (or environment) and the person are likely different. These effects are included by replacing $\epsilon_\mathrm{room} A_\mathrm{room}$ in (6.42) by $(\epsilon_\mathrm{room} A_\mathrm{room})_\mathrm{effective}$, by using Christiansen's equation

$$(\epsilon_\mathrm{room} A_\mathrm{room})_\mathrm{effective} = \frac{\epsilon_\mathrm{skin} A_\mathrm{skin}}{1 + (A_\mathrm{skin}/A_\mathrm{room})(\epsilon_\mathrm{skin}/\epsilon_\mathrm{room} - 1)} \tag{6.45}$$

(which we will not derive here). This expression approaches $\epsilon_\mathrm{skin} A_\mathrm{skin}$ as the room area becomes very large relative to the body, because only a small amount of radiation from the room walls hits the body. The prediction of (6.44) is therefore reasonable. (Of course, this analysis needs to be modified when all the "walls" are not at the same temperature, such as when there are cold windows.) A poorly insulated house feels cold, even when there are no cold drafts leaking inside. With cold walls and windows, the air near our bodies may be warm, but the walls are cold and so thermal radiation from them is relatively low (Problem 6.96).

In general, the net radiative heat loss is

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_\mathrm{r} = \epsilon_\mathrm{skin}\sigma T_\mathrm{skin}^4 A_\mathrm{skin} - \epsilon_\mathrm{room}\sigma T_\mathrm{room}^4 A_\mathrm{room} \tag{6.46}$$

and in the limit of a large room we use $(\epsilon_\mathrm{room} A_\mathrm{room})_\mathrm{effective}$ and this becomes

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_\mathrm{r} = \epsilon_\mathrm{skin}\sigma \left(T_\mathrm{skin}^4 - T_\mathrm{room}^4\right) A_\mathrm{skin}. \tag{6.47}$$

This can be used directly (and exactly) or it can be approximated by "linearizing" it. This will also allow us to compare it with other modes of heat loss. With $\delta = T_\mathrm{skin} - T_\mathrm{room}$, then $T_\mathrm{room} = T_\mathrm{skin} - \delta$ and

$$\begin{aligned}
-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_\mathrm{r} &= \epsilon_\mathrm{skin}\sigma \left(T_\mathrm{skin}^4 - T_\mathrm{room}^4\right) A_\mathrm{skin} \\
&= \epsilon_\mathrm{skin}\sigma (T_\mathrm{skin}^4 - (T_\mathrm{skin} - \delta)^4) A_\mathrm{skin} && (6.48)\\
&= \epsilon_\mathrm{skin}\sigma (4\delta T_\mathrm{skin}^3 - 6\delta^2 T_\mathrm{skin}^2 + 4\delta^3 T_\mathrm{skin} - \delta^4) A_\mathrm{skin} && (6.49)\\
&\simeq \epsilon_\mathrm{skin}\sigma (4\delta T_\mathrm{skin}^3) A_\mathrm{skin} = \epsilon_\mathrm{skin}\sigma (T_\mathrm{skin} - T_\mathrm{room})(4T_\mathrm{skin}^3) A_\mathrm{skin}, && (6.50)
\end{aligned}$$

where the last approximation follows because $|\delta| \ll T_{\text{skin}}$. So we see that

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{r}} \simeq (4\sigma T_{\text{skin}}^3)\epsilon_{\text{skin}} A_{\text{skin}}(T_{\text{skin}} - T_{\text{room}}) \tag{6.51}$$

$$= (4 \times 5.67 \times 10^{-8}\,\text{W/m}^2\text{–K}^4 \times (307\,\text{K})^3)\epsilon_{\text{skin}} A_{\text{skin}}\,(T_{\text{skin}} - T_{\text{room}})\,. \tag{6.52}$$

Also, we find

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{r}} = h'_{\text{r}}\epsilon_{\text{skin}} A_{\text{skin}}(T_{\text{skin}} - T_{\text{room}}) = h_{\text{r}} A_{\text{skin}}(T_{\text{skin}} - T_{\text{room}}), \tag{6.53}$$

where h'_{r} is the *first power radiant heat transfer coefficient*

$$h'_{\text{r}} = 4\sigma T_{\text{skin}}^3 = 5.6\,\text{kcal/m}^2\text{–h–}\,^\circ\text{C} \tag{6.54}$$

for $T_{\text{skin}} = 34\,^\circ\text{C}$. In (6.53) we have also combined $h'_{\text{r}}\epsilon_{\text{skin}}$ into one factor $h_{\text{r}} = 4\sigma\epsilon_{\text{skin}} T_{\text{skin}}^3$ to cast it into a form similar to those we will use for heat transfer by convection, conductivity, and evaporation.

This linearization method is quite general and is applicable here because $|T_{\text{skin}} - T_{\text{room}}| \ll T_{\text{skin}}$. However, because of the fortuitous factorization

$$T_{\text{skin}}^4 - T_{\text{room}}^4 = (T_{\text{skin}} - T_{\text{room}})(T_{\text{skin}}^3 + T_{\text{skin}}^2 T_{\text{room}} + T_{\text{skin}} T_{\text{room}}^2 + T_{\text{room}}^3), \tag{6.55}$$

(6.53) becomes exact with $4T_{\text{skin}}^3$ replaced by $(T_{\text{skin}}^3 + T_{\text{skin}}^2 T_{\text{room}} + T_{\text{skin}} T_{\text{room}}^2 + T_{\text{room}}^3)$, and so

$$h_{\text{r}} = \sigma\epsilon_{\text{skin}}(T_{\text{skin}}^3 + T_{\text{skin}}^2 T_{\text{room}} + T_{\text{skin}} T_{\text{room}}^2 + T_{\text{room}}^3) \simeq 4\sigma\epsilon_{\text{skin}} T_{\text{skin}}^3. \tag{6.56}$$

Receiving Thermal Radiation Outdoors

There are generally no wall-type enclosures outdoors that emit significant thermal radiation, but there is thermal (infrared) radiation from the ground (asphalt, grass, snow cover, etc.), buildings, trees, etc. There is also heating by rays directly incident from the sun and scattered sun light; this depends on the latitude, time of year, cloud cover, etc. The heat gained by the body per unit area and time is

$$\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{gain}} = (1-\rho)(R_{\text{S,D}} + R_{\text{S,I}} + R_{\text{S,R}}) + \alpha(R_{\text{IR,U}} + R_{\text{IR,D}}), \tag{6.57}$$

where the first term describes heating by the sun (mostly visible light) and the second term longer wavelength, infrared light. The solar light can be that directly from the

Table 6.39 Reflectance in the visible

From	Reflectance
Clipped grass	0.24
Dry sand	0.40
Dry tar macdam	0.12
Dry concrete	0.24
Fresh snow	0.85
Black human skin	0.18 (0.10[a])
White human skin	0.35 (0.42[a])

[a]From [10] for very black and very white human skin.
Using data from [11]

Table 6.40 Solar radiation from clear skies on a fleeced sheep with its sides at right angles to the solar beam, in W/m^2

Solar elevation (angle, °)	$R_{S,D}$	$R_{S,I}$	$R_{S,R}$	R_S	$(1-\rho)R_S$
5	14	29	5	48	36
10	79	34	12	125	94
15	150	43	19	212	159
20	208	48	27	283	212
25	252	51	35	338	254
30	285	52	44	381	286
45	331	55	67	453	340
60	344	53	86	483	362
75	348	52	97	497	373
90	352	52	101	505	379

The direct, indirect reflected components and the total fluxes are, respectively, $R_{S,D}$, $R_{S,I}$, $R_{S,R}$, R_S, and $(1-\rho)R_S$. The atmospheric transmission coefficient was taken to be 0.7.
Using data from [11], from [24]

sun ($R_{S,D}$), indirect scattered or diffused light from the sky ($R_{S,I}$), or that reflected from the terrain ($R_{S,R}$). A fraction ρ of that light is reflected from the body (see Table 6.39). The contribution of each solar term is seen in Table 6.40. The infrared light can be coming "up" from the terrain ($R_{IR,U}$) or "down" from the sky ($R_{IR,D}$); this contribution is analogous to the thermal emission from the room walls in the previous section. The component of long wavelength, infrared radiation from the sky is seen in Table 6.41. A fraction α of this infrared light is absorbed by the body (and this includes reflection and angle averaging effects); this is typically $\sim 1/2$. Typically more of this radiant heat (solar plus infrared) is absorbed by the body than is emitted by the body during the day; the reverse is typically true at night.

Heat Loss by Convection and Conduction

Convection is the transport of heat through mass flow of the medium. Conduction is the transport of heat through a medium without mass flow (involving electrons,

Table 6.41 Incoming long wavelength radiation from clear skies and the equivalent radiation temperature

Air T (°C)	Incoming radiation (W/m^2)	Radiant T (°C)
−30	67	−88
−20	108	−65
−10	155	−45
0	208	−28
10	266	−12
20	331	3
30	404	17

Using data from [11], calculated using [98]

vibrations, local molecular motion, etc.). In gases and liquids both mechanisms can contribute. Convection is the larger of the two and it is not always simple to differentiate between the two. In solids only conduction contributes.

Convection

The convection heat flux consists of that from forced convection (air streams) and natural or free convection (which is due to movement near the body induced by thermal expansion caused by temperature gradients). In most general form, the convective heat transfer coefficient per unit area (in W/m^2-°C) is

$$h_{\text{c-forced convection}} = \frac{akw^n}{\upsilon^n L^{1-n}}, \tag{6.58}$$

where w is the wind speed in m/s, L is the characteristic dimension of the object in m—which is the diameter of the cylinder in some models, υ is the kinematic viscosity of air in m^2/s, and a, k, and n are parameters. This is used in

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{c}} = h_{\text{c-forced convection}}(T_{\text{skin}} - T_{\text{air}}), \tag{6.59}$$

which is derived [11] using scaling arguments involving the Reynolds number, a parameter that is discussed in Chap. 7. The natural convective heat loss per unit area A (in W/m^2) from a man of height L (in m) has been predicted to be

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{c-natural convection}} = \frac{1.69}{L^{0.25}}(T_{\text{skin}} - T_{\text{air}})^{1.25}, \tag{6.60}$$

which scales with temperature differently than (6.58) and (6.59) for forced convection. Usually, only the larger of forced or natural convection is considered in the analysis, and it is modeled in the form

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\mathrm{c}} = h_{\mathrm{c}}(T_{\mathrm{skin}} - T_{\mathrm{air}}), \tag{6.61}$$

where h_{c} is the convective heat transfer coefficient per unit area and T_{air} is the air temperature very near the person, which is usually the same as T_{room}. With still air and normal clothing, $h_{\mathrm{c}} \sim 2.3\,\mathrm{kcal/m^2\text{-}h\text{-}^\circ C}$, which is $< h_{\mathrm{r}}$ (the radiant heat transfer coefficient). The coefficient h_{c} decreases with heavier clothing and increases with less and lighter clothing; when you are nude you get cold faster because of increases in h_{c}. It also increases with air speed, as is seen in Table 6.42. This leads to the *wind chill factor*. For a naked person

$$h_{\mathrm{c}} = 8.3w^{0.5} \tag{6.62}$$

in $\mathrm{W/m^2\text{-}^\circ C}$, where w is the air speed in m/s. In another formulation [19],

$$h_{\mathrm{c}} = 10.45 - w + 10w^{0.5} \tag{6.63}$$

in $\mathrm{kcal/h\text{-}m^2\text{-}^\circ C}$, for speeds between 2 m/s ($\simeq$4.5 mph) and 20 m/s ($\simeq$45 mph), where again w is in m/s. Both clothed and bare regions on a body need to be considered. The body "feels" the wind chill temperature, which is colder and is

$$T_{\mathrm{wc}} = 13.12 + 0.6215T - 11.37w^{0.16} + 0.3965Tw^{0.16}, \tag{6.64}$$

where w is the wind speed in km/h and T the temperature in °C. (This is the relation as revised in 2001.) Also see Table 6.43. (In contrast, humidity makes you feel even

Table 6.42 The convective heat transfer coefficient (h_{c}) and air insulation [$1/(h_{\mathrm{c}} + h_{\mathrm{r}})$] of the skin–air interface for a naked man

Wind speed (m/s)	Convective heat transfer coefficient ($\mathrm{W/m^2-^\circ C}$)	Air insulation ($\mathrm{m^2-^\circ C/W}$)
0.1	2.6	0.123
0.2	3.7	0.109
0.4	5.2	0.093
0.6	6.4	0.084
0.8	7.4	0.077
1.0	8.3	0.072
2.0	11.7	0.058
3.0	14.4	0.50
4.0	16.6	0.045
5.0	18.6	0.041

The radiant heat transfer coefficient h_{r} is $5.5\,\mathrm{W/m^2\text{-}^\circ C}$.
Using data from [11]

Table 6.43 Wind chill factor temperature, in °C, as a function of wind speed and temperature; the temperature is given in the first row for the 0 km/hr wind speed

Wind speed (km/hr)	Wind chill factor temperature (°C)					
0	10	0	−10	−20	−30	−40
20	7.4	−5.2	−17.9	−30.5	−43.1	−55.7
40	6.0	−7.4	−20.8	−34.1	−47.5	−60.9
60	5.1	−8.8	−22.6	−36.5	−50.3	−64.2
80	4.4	−9.8	−24.0	−38.2	−52.4	−66.6
100	3.9	−10.6	−25.1	−39.6	−54.1	−68.6

Using the scale revised in 2001 by the US National Weather Service

warmer than expected from the ambient temperature, as is described by the heat index discussed later in this chapter.)

With $T_{\text{room}} = T_{\text{air}}$, the total heat flux due to (net) radiation and convection comes from summing (6.53) and (6.61):

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{total}} = -\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{r}} - \frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{c}} = (h_{\text{r}} + h_{\text{c}})(T_{\text{skin}} - T_{\text{room}}), \quad (6.65)$$

which gives

$$T_{\text{skin}} - T_{\text{room}} = -\frac{(1/A)(\mathrm{d}Q/\mathrm{d}t)_{\text{total}}}{h_{\text{r}} + h_{\text{c}}} \quad (6.66)$$

Convection Within the Body We have been discussing the convection of heat from the outside of the body to air by the convection of air. There is also of convection *within* the body by the flow of blood. This does not change the average temperature of the body, but the distribution of temperature within the body. Equation (6.5) says that heat flow $Q = mc(\Delta T)$ is needed into a body of mass m and specific heat c to heat the mass by ΔT and heat flow of this magnitude away from the body is needed to cool it. This means that if blood flows into a body part, such as the foot in Problem 6.107, at a rate of F_m (mass per unit time) with a temperature T_{blood} and leaves it (at the same rate, of course) at $T_{\text{blood}} + \Delta T_{\text{blood}}$, it will draw away heat from the part at a rate

$$-\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{blood flow}} = F_m c(\Delta T_{\text{blood}}). \quad (6.67)$$

Therefore heat is left in the body part by the blood when $\mathrm{d}Q/\mathrm{d}t > 0$ and the blood gets colder, $\Delta T_{\text{blood}} < 0$. Of course, this transfer of heat between the blood and local tissues occurs by thermal conduction.

This analysis has not addressed what $(dQ/dt)_{blood}$ and ΔT_{blood} actually are, but only how they are related. To determine them, we would have to consider the rates of thermal conduction between the blood in the arteries, capillaries, and veins and the surrounding tissue and the actual flow rates of blood in each of these vessels. If the flow is very fast, there may not be enough time for much heat flow to occur.

If the arteries flowing into a body part, such as a hand or foot, are physically displaced from the veins that return the blood to the heart, there is minimal transfer of heat from arteries to veins. If they are instead very near to each other, there can be substantial heat transfer from the warmer blood flowing in the arteries to the cooler blood flowing in the veins. (The venous blood is cooler because of heat loss to the local tissues which can be much colder than the core.) This is called *countercurrent heat exchange*, and is shown in Fig. 6.18 [69]. It lowers overall heat loss at the body part—particularly for extremities and surface regions—and therefore lowers overall heat loss in the blood and the body, because it decreases the difference in temperature between the blood in the body part and the tissue (and locally, outside the body). As part of its temperature control system (Chap. 13), the body can decide whether to minimize heat loss, by using countercurrent cooling, or to maximize it, by diverting more blood flow nearer to the skin. Countercurrent heat exchange is modeled in Problem 6.114.

Figure 6.19 shows how this occurs in the forearm through two different sets of veins. If it is cold at the surface of the body, blood returns to the heart through veins that are very close to the arteries, so heat loss is minimized through countercurrent heat exchange. Because this minimizes the warming of the hands, this is one way

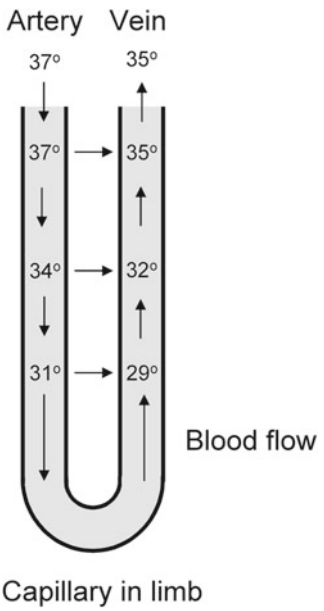


Fig. 6.18 Example of countercurrent heat flow from an artery to a nearby vein. Without this flow the temperature drop along the artery would be even greater under cold conditions because there would be more heat flow to the environment. All temperatures are in °C. Also see Fig. 6.19a (based on [69])

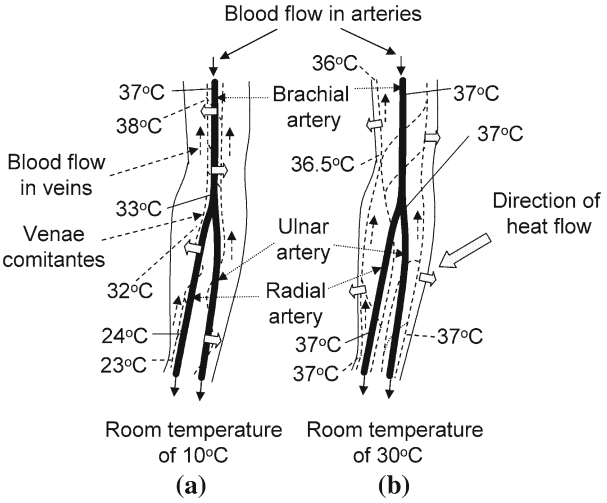


Fig. 6.19 Countercurrent heat exchange in the (*right*) arm for blood flowing through the brachial artery in (**a**). At the higher room temperatures (30 °C) in (**b**) and during exercise the venous return flow occurs through superficial vessels to enable heat loss to the room. At the lower temperature (10 °C) in (**a**) this return path is shifted to deep veins, which helps retain heat in two ways: Heat from the arteries is lost to the venous blood before it is lost to the room, which is countercurrent heat exchange (Fig. 6.18), and little heat is lost to the room from these veins because they are deep. Note that in the lower arm the temperature in the venae comitantes is 1 °C lower than that in the artery. The venae comitantes are the veins accompanying the arteries in the limbs. Different types of *arrows* are used to denote blood flow in arteries (*solid*, *narrow*), blood flow in veins, (*dashed*, *narrow*), and heat flow (*wide*, *gray*) (based on [16, 49])

you can get excessively cold hands. If it is warm, blood flow near the surface and heat transfer and cooling are maximized. Blood is directed to these "inner" and "outer" veins by vasodilating and vasoconstricting muscles surrounding the appropriate veins (to dilate and constrict them, respectively), as controlled by the temperature sensor and the temperature regulatory system (Chap. 12).

Counterflow heat exchange is also used to regulate the temperature of the testes. The testes are kept outside the core of the body because fertility is improved at a few degrees below core temperature. Before blood enters the testes region it often undergoes counterflow heat exchange so that cooler blood enters this region. This prevents excessive heating of the testes by warm blood, as well as the excessive loss of body heat from this "thermally" exposed region.

A counterflow multiplier mechanism is also used by the body in diffusion in the kidney to overcome large gradients in concentrations [69, 95].

Conduction

As shown in (6.6), the heat flow during thermal conduction is the product of the thermal conductivity K and the temperature gradient

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right) = K\frac{\mathrm{d}T}{\mathrm{d}x} \sim K\frac{\Delta T}{\Delta x}. \tag{6.68}$$

For a well-defined distance $d = \Delta x$ between two regions of different but uniform temperature, say due to the thickness of clothing or an air boundary layer, we can define a heat transfer coefficient per unit area $h_\mathrm{t} = K/d$ and

$$-\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right) = h_\mathrm{t}\Delta T, \tag{6.69}$$

as in (6.7). This looks similar to the form for heat loss due to (net) radiation (6.53) and forced convection (6.59).

Let us determine how much heat a person loses when he puts his hand and forearm in very cold water (4°C) for 2 min [2]. His hand and forearm can be modeled as a cylinder with a length of $L = 30$ cm and an average radius of 3 cm. Let us assume the body has a $\Delta r = 3$ mm thick layer of fat with thermal conductivity $K = 0.2$ W/m-°C that separates the cylindrical core (of radius $r = 2.7$ cm, at 37°C) from the cold water. The contact area between the arm and water is $A = 2\pi rL$. Using (6.68) modified for cylinders (by replacing Δx with $r\ln(1 + \Delta r/r)$, as above), the rate heat leaves the arm is $(2\pi rL)(K\ \Delta T)/(r\ln(1 + \Delta r/r)) \simeq (2\pi rLK\ \Delta T)/(\Delta r)$. The exact expression gives 118 W, while the approximation gives 112 W. In 2 min, the body loses (118 W)(120 s) = 14,200 J or 3.4 kcal.

Insulation I can be formally defined as $I = 1/h_\mathrm{t} = d/K$. It is sometimes expressed in terms of the clothing or Clo value, where 1 Clo is the insulation for a man in an average suit with no gloves or light gloves, wearing leather shoes and light socks [10, 22]. An insulation of 1 Clo = 0.155 m^2-°C/W (= 0.88 ft^2-°F/BTU). A nude person has 0 Clo insulation. A light working ensemble (undershorts, wool socks, cotton work shirt—shirt tail out and neck open—and work trousers) corresponds to 0.6 Clo. A typical light suit has 0.7–1.0 Clo insulation, while a heavy business suit has 1.2 Clo insulation. A polar weather suit made of heavy wool pile has 3–4 Clo insulation. Typical values of insulation are given in Table 6.44. The insulation provided by human clothing is about 0.25 °C-m^2 /W per cm of clothing thickness. The thermal conductivity of various body tissues is given in Table 6.36 and that of common materials in Table 6.37.

Heat Loss by Evaporation of Water: Sweating and Breathing

The amount of heat needed to evaporate 1 L of water is 540 kcal (near its boiling point); this is the heat of vaporization. When we sweat there is loss of heat only if the water is allowed to evaporate. (Wiping off the sweat will not cool you.) This normally accounts for ~7 kcal/h loss.

Table 6.44 The insulation from animal coats, fabrics, body components, and other materials, in units of m^2-°C/W per cm of insulation

Material	Insulation per cm
Cattle coat	0.07–0.10
Sheep coat	0.13–0.24
Husky dog coat	0.24
Pig coat	0.04
Wool	0.29
Goose down	0.38
Human clothing (average)	0.25
Air	0.36/0.39
Muscle, bone	0.01/0.024
Body fat	0.01/0.05
Water	0.01/0.017
Ice	0.01/0.004
Typical wood[a]	0.01/0.062

h_t is this 1/(insulation per cm × the thickness in cm). Thermal conductivity K is 1/(insulation per cm) or 0.01 × this value when expressed in units of W/m-°C.
[a]Balsa wood is higher; green lumber is smaller.
Using data from [2, 11]; also see [22]

We normally breathe in air that is cooler than body temperature. It gets warmed up and is then exhaled: this is also a source of body cooling. Also, we breathe in relatively dry air, and exhale air saturated with water vapor. Because this water vapor is formed by the evaporation of liquid water, this is yet another source of cooling.

Respiratory heat loss due to the lungs can be described by

$$-\left(\frac{dQ}{dt}\right)_l = \rho_{air} c_{p,\,air}(T_{exp} - T_{insp})\frac{dV_{air}}{dt} + \Delta H_{evap,\,water}(\rho_{exp,\,water} - \rho_{insp,\,water})\frac{dV_{air}}{dt}, \tag{6.70}$$

where ρ is the mass density (of air or of expired or inspired water vapor), $c_{p,\,air}$ is the specific heat of air (2.4×10^{-4} kcal/g-°C), T is the temperature of expired or inspired air, dV/dt is the rate of inhaling air, and $\Delta H_{evap,\,water}$ is the heat of evaporation of water (0.54 kcal/g). The first term accounts for body cooling by the heating of inspired air before expiration (exhalation) and the second term accounts for body cooling by the evaporation of water in the lungs that increases the water content of inspired air before expiration.

When it is hot, why does it feel even hotter than the ambient temperature when it is humid? Humidity decreases the rate of heat loss from the body by decreasing the rate of heat flow from the body to the air, decreasing the evaporation of sweat (because the vapor pressure of water in the air decreases the difference between the

vapor pressure above the sweat and in the air), and decreasing the rate of respiratory loss of heat due to the last term in (6.70). The *heat index* describes how hot it really feels at a given temperature and humidity [96, 97]. For example, when it is 90 °F (32°C), it feels like it is 95°F (35 °C) when the relative humidity is 50%, 109°F (43 °C) when it is 75%, and 132 °F (56°C) when it is 100%. (The relative humidity is the actual vapor pressure of water in air divided by the equilibrium vapor pressure of water (in contact with liquid) water, at the same temperature. It is usually expressed in per cent.) The likelihood of heat disorders, such as heat stroke and sunstroke, are characterized by warnings such as "caution" for heat index values of 80–90 °F (27–32 °C), "extreme caution" from 90 to 105 °F (32 to 41 °C), "danger" from 105 to 129 °F (41 to 54 °C), and "extreme danger" above 130 °F (54°C). The heat index is explored more in Problems 6.120–6.123.

The effect of air humidity can also be evaluated by considering three "temperatures": the usual *dry-bulb temperature*, the *wet-bulb temperature*, and the *dew point*. The wet-bulb temperature of an object is the temperature the object would cool to, by supplying the heat of vaporization, so there would be 100% relative humidity locally. The wet-bulb thermometer is a dry-bulb thermometer covered with an object soaked in water, so it indicates the maximum effect of evaporative cooling at the absolute humidity level. The dew point is the temperature at which condensation begins (100% relative humidity) at the absolute humidity level. For 100% relative humidity the three temperatures are the same; as it drops, the wet-bulb temperature decreases and the dew point decreases even faster. The wet-bulb temperature gives an indication of body temperature control due to evaporative cooling. For wet-bulb temperatures of ≈95 °F (35°C) and higher the body cannot cool itself enough solely due to evaporative cooling.

Humans can run long distances in part because we can lose heat by sweating, more than a liter of water an hour that can vaporize. We have 3–10 million sweat glands over our bodies. Without these, we could would need to rely on panting, as do animals with fur, who have sweat glands for cooling only in regions devoid of fur, such in the pads of their feet in dogs [44, 56, 91].

6.6 Body Temperature

The normal core body temperature is in the range of 36.5–37.5 °C (97.7–99.5 °F); for a given individual the core temperature generally increases during the waking hours by 0.5 °C (∼0.9 °F). Anesthesia lowers core body temperature by ∼1 °C, probably because the anesthetics interfere with how the body regulates temperature; it can also lead to temperature variations along the surfaces of the limbs of several degrees [37]. Temperatures measured in the rectum are 0.4 °C (∼0.7 °F) higher than those measured in the mouth [59].

The body is very good at controlling its temperature. Figure 6.20 shows that the nude body can maintain the core temperature between 97 and 100 °F (36 and 38 °C) for several hours with environmental temperatures ranging between 70 and 130 °F

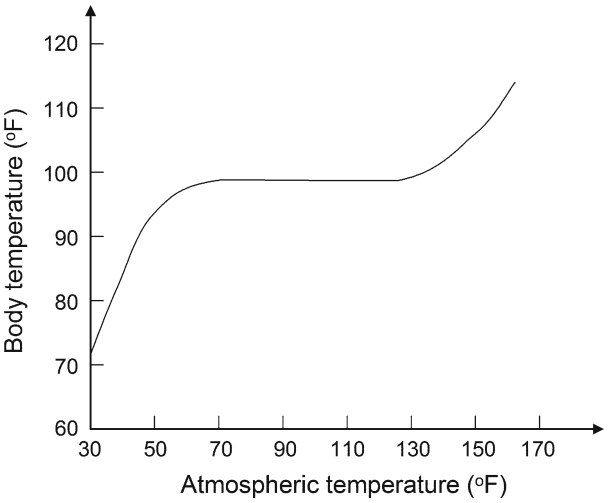


Fig. 6.20 Body control of core temperature for several hours with large variations in local environmental temperature, for dry air. Normal body temperature 98.6 °F (37 °C) is maintained for atmospheric temperatures from ~70 to 130 °F (~21 to 54 °C). The exact shape of this curve depends on the surroundings: humidity, air movement, thermal radiation to the body, and so on (based on [45])

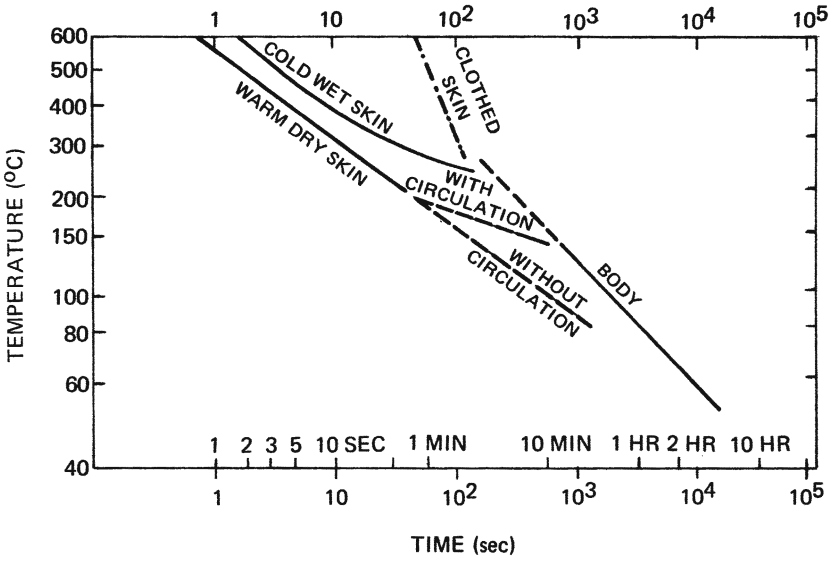


Fig. 6.21 Maximum tolerable elevated temperatures (of air) as a function of exposure times, for different body conditions. Tolerable conditions are below the lines for each skin condition and for the body, and intolerable conditions are above them (from [10])

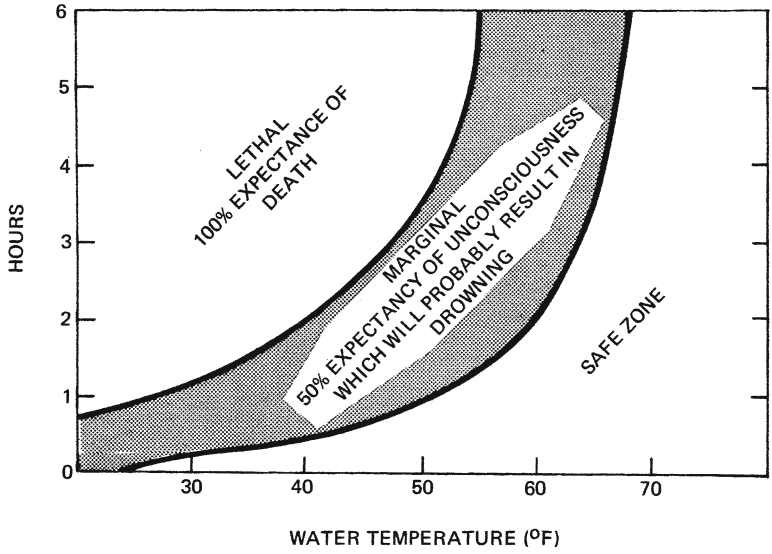


Fig. 6.22 Life expectancy of a person without an exposure suit in cold water for different times (from [10])

(21 and 54 °C) in dry air. However, we clearly have limits. Figure 6.21 shows that we can take only limited exposure to hot air, while Fig. 6.22 shows we can survive only limited exposure to cold water; in both cases the larger the temperature deviation from normal body temperature, the shorter the maximum possible exposure time until death. Figure 6.23 tracks how skin and inner body temperatures decrease during exposure to cold water, and how it recovers when the body is then placed in warm water. *How do the energetics of the body change to maintain normal temperature and to resist changes that can be induced by extreme environmental temperatures or by exercise?*

Each of the modes of body heat loss we have described is important for regulating the body temperature. You can control the convection route by changing your clothes. Some are controlled involuntarily by your body, such as the sweating and breathing modes. *What happens when the room temperature exceeds the body core and skin temperature?* The body is then heated by radiation. Convection needs to be re-evaluated. Exhaled air is no longer heated up. *What happens when you exercise and your metabolic rate increases?* The body has to respond in a way to increase the rate of losing body heat. All these modes of losing heat are important. A quantitative model of this regulation is presented in Chap. 13, as part of the discussion of feedback and control in the body.

Heat production varies with the local, environmental temperature, which we called T_{room} above. It is high at low temperatures, then decreases with increasing temperature until T_{c}, becomes constant, and then at T_{h} it increases again, as is shown in Fig. 6.24. The *critical temperature* T_{c} (or *lower critical temperature*) is therefore the

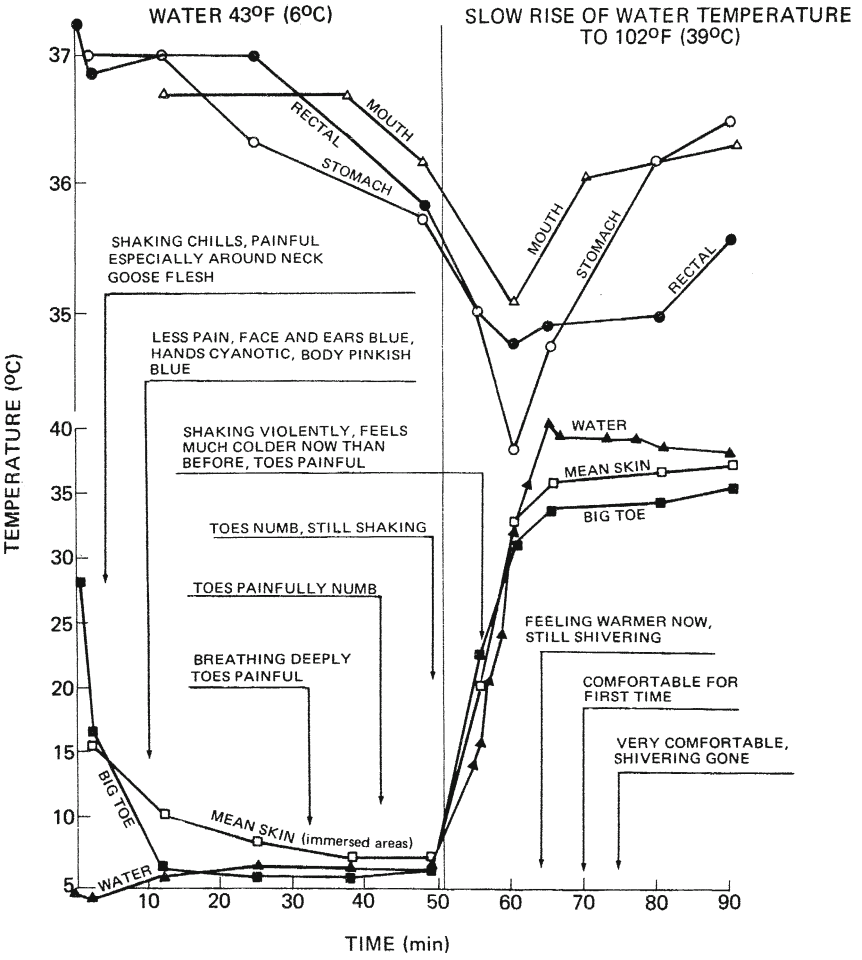


Fig. 6.23 Variation in skin and internal temperatures when a person is immersed in 6 °C (43°F) water for 52 min and then warmed slowly using 39 °C water (from [10])

environmental temperature at which heat production begins to rise when the environmental temperature falls. It is 26 °C for a naked adult man and 35 °C for a naked baby, assuming minimal air movement. Similarly, the *temperature of hyperthermal rise* T_h (or *upper critical temperature*) is the environmental temperature at which heat production begins to rise when the environmental temperature increases. Between T_c and T_h is the *zone of minimal metabolism* or the *thermoneutral zone*.

Below T_c, the heat loss increases due to nonevaporative pathways, such as radiation and convection. The increased metabolism is due to increases in muscle tension and shivering, which is technically the visible movement of subcutaneous muscle. This increases as the difference between body and environmental temperature increases,

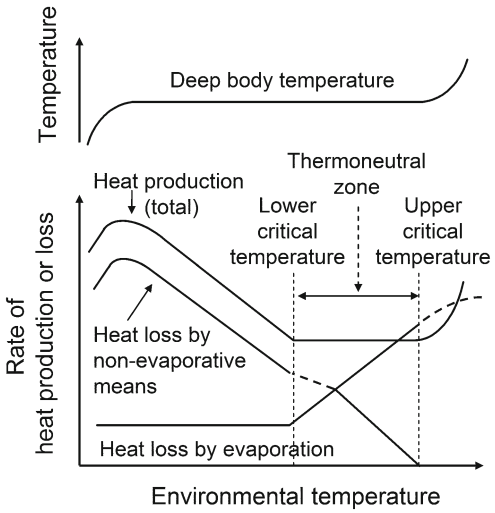


Fig. 6.24 Critical temperatures and relation between the environmental temperature and heat production in homeothermic animals (which are those having constant and relatively high body temperature, which means birds and mammals) (based on [11, 72])

until the *summit metabolism* is reached for that environmental temperature. The summit metabolism is about $21m_b^{0.75}$, in W with m_b in kg, for man (and generally about $25m_b^{0.75}$ for most mammals). This is about 7× the fasting rate of metabolism or 2–3× the field rate of energy expenditure. (It is still much lower than the peak metabolic rates you can achieve during cycling, running, and so on.) At lower temperatures, the core body temperature cannot be maintained and there is body cooling and hypothermia. Above T_h the heat loss is dominated by evaporation.

The critical temperature can be evaluated by equating heat loss and heat production by the metabolism. During exercise the work done and heat stored or liberated from tissues need to be considered also. The core body temperature can be related to the maximal aerobic capacity by

$$T_{\text{core}}(^\circ\text{C}) = 36.5 + 3.0 \times \frac{\text{oxygen consumption}}{\text{maximum possible oxygen consumption}}, \qquad (6.71)$$

because the oxygen consumption rates are proportional to the metabolic rates. Also, movement during exercise increases convective heat loss. A rise in body temperature of 1 °C increases heat production by ∼13%. Heat production is also induced by pyrogens, etc., and by trauma, bone fractures, burns, etc.

The temperature T in (6.35) is the average body temperature. The temperature of the body skin T_{skin}(34 °C = 307 K) is usually lower than that of the core (rectal) T_{core}(37 °C = 310 K)—that you measure with a thermometer. Table 6.45 gives the normal temperature of different body organs and blood vessels. The temperature of inner organs can differ by 0.2–1.2 °C and by 0.9 °C in a given organ under normal conditions [38]. There are 1.4 °C temperature variations in the brain, with the cortex being cooler than the basal regions. The average temperature of the body is the

Table 6.45 Core temperatures within the human body

Body region	Normal temperature (°C)
Skin	32–35
Scrotum	34.0
Liver	36.4–36.8
Oral cavity	36.5–36.6
Superior vena cava	36.65
Esophagus, lungs	36.75
Heart (right ventricle)	36.75
Aorta, inferior vena cava	36.75
Pulmonary artery and vein	36.75
Kidney	36.85
Spinal cord	36.95
Stomach, rectum (mean)	37.0
Rectum (range)	36.2–37.8
Brain, uterus	37.3

Using data from [38]

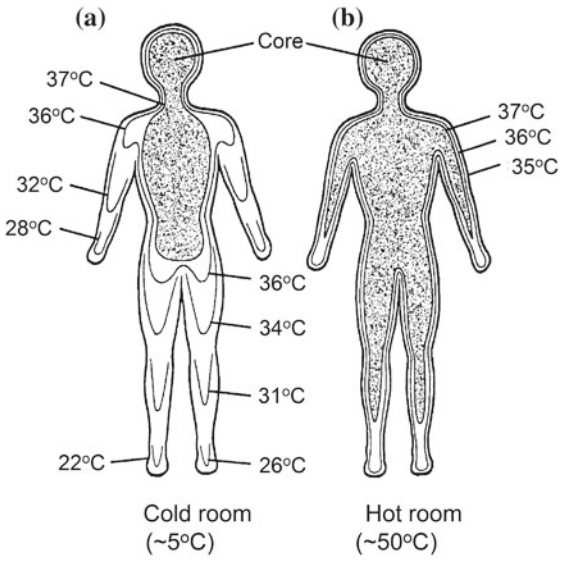


Fig. 6.25 Temperature in the body in cold and hot environments (From [38]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com)

weighted average of the two temperatures, $T_{av} = (80 \pm 5\%)T_{core} + (20 \pm 5\%)T_{skin}$. The temperature throughout the body varies with the temperature in the environment. As shown in Fig. 6.25, these variations can be very large in a cold room. These internal distributions can be modeled using the thermal properties of the body (Table 6.36) and the thermal transport models presented here.

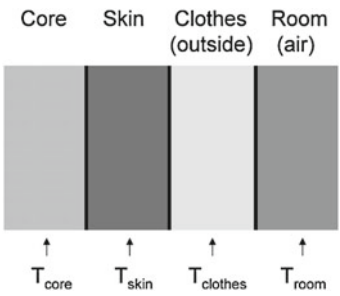


Fig. 6.26 Model of heat flow from the body core to air

Variations in temperatures within the body can be important, as is how these temperatures are measured. Under normal conditions, the temperatures of all of the regions in the heart and lungs are the same within ~0.1 °C [92]. During hypothermia, such as that caused by full body immersion in cold water, the temperature measured in the lower esophagus or the esophagus region near the heart seem to track the temperature of the heart well. Temperatures measured in the rectum change much slower during hypothermia than in these regions and can give provide temperatures that are falsely ~1 °C higher than than those near the heart. This can be significant because ventricular fibrillation (see Chap. 8) occurs when the body core temperature falls below ~27–28 °C [26, 92].

Let us interrelate the body temperature in the deep body or core, T_{core}, of the skin, T_{skin}, on the outside of our clothes, T_{clothes}, and in the air, T_{room}. This can viewed in terms of the model of Fig. 6.26 in which each region is a rectangular slab. More refined models would consider the body as a cylinder, as in Fig. 6.17b [88].

Following (6.66), we can model the flow between two of these adjacent regions by

$$(T_1 - T_2) = -I_{12}\frac{1}{A}\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right), \tag{6.72}$$

where the heat flow, $\mathrm{d}Q/\mathrm{d}t$, across a cross-sectional area A between regions 1 and 2 is controlled by the insulation between them

$$I_{12} = \frac{1}{\sum h_{x,12}}, \tag{6.73}$$

where $h_{x,12}$ are the relevant heat transfer coefficients between the two regions, such as those due to radiation, convection, conduction, and evaporation.

The difference between the temperature in the body core and on the skin is determined by the flow of heat from the core to the skin and the tissue insulation I_{t}, and is

$$T_{\text{core}} - T_{\text{skin}} = -\frac{I_{\text{t}}}{A}\left[\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{r}} + \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{c}} + \left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)_{\text{v}}\right], \tag{6.74}$$

where $(dQ/dt)_r$, $(dQ/dt)_c$, and $(dQ/dt)_v$ are, respectively, the flows of radiative heat loss, convective/conductive heat loss, and the loss of heat by vaporizing moisture on the skin surface. (The loss of heat from the lungs and respiratory passages is not included.) Because the heat flow is from the core to the skin for each, each term is negative and the skin is cooler than the core, as is expected. The insulation provided by human tissue has a maximum value of 0.10 °C-m^2/W in adults at cold temperatures (below the critical temperature) and a minimum value of 0.03 °C-m^2/W at high temperatures. In babies these high and low values are of 0.05 °C-m^2/W and 0.015 °C-m^2/W, respectively.

The difference between the temperature on the skin and the outside of one's clothing is

$$T_{\text{skin}} - T_{\text{clothes}} = -\frac{I_c}{A}\left[\left(\frac{dQ}{dt}\right)_r + \left(\frac{dQ}{dt}\right)_c\right], \tag{6.75}$$

where I_c is the insulation of the clothing (or fur for animals). The insulation provided per cm thickness of human clothing averages to 0.25 °C-m^2/W-cm (Table 6.44).

The difference between the temperature outside of one's clothing and the air is

$$T_{\text{clothes}} - T_{\text{room}} = -\frac{I_a}{A}\left[\left(\frac{dQ}{dt}\right)_r + \left(\frac{dQ}{dt}\right)_c\right], \tag{6.76}$$

where I_a is the insulation of the air (Table 6.42).

6.7 Summary

The energy used by the body and body temperature are determined by the body metabolism, which includes the use of food, the energy needed to operate the body, and the production of heat, and by the loss of heat by the body. The energy value of food can be analyzed and compared to the production of ATP by the body and the use of ATP in different activities. The body metabolic rate is composed of many components, each of which can be modeled. This includes analyzing the rate that energy is needed for the operation of body functions for a person who is inactive, which combine to form the BMR, and the rate that energy is needed for the body to engage in physical activity. The body metabolism and food intake can be linked to models of the steady-state body weight and of weight gain and loss. The body can lose heat by radiation, convection, conduction, and the evaporation of water, each of which can be modeled and linked to models that determine the body temperature.

Problems

Heat Capacity

6.1 A person weighing 60 kg drinks 600 mL (0.6 kg) of water. Say the water is at a temperature that is either 25 °C above or below the body temperature. How does the body temperature change? (Use the fact that the specific heat of the water is 1 kcal/kg-°C, so that it is supplying the body with an energy of 1 kcal/kg-°C × 0.6 kg × 25 °C, and that the specific heat of the body is 0.83 kcal/kg-°C.) Would such a warm drink (+25 °C) help with hypothermia? Would such a cold drink (−25 °C) help with a fever?

6.2 What is the heat capacity of a typical 20 μm diameter human cell? Assume it is a sphere and has a specific heat equal to the body average.

6.3 In a typical 20 μm diameter human cell there is approximately 5×10^{-6} J, 2.0×10^{-5} J, and 8×10^{-6} J of energy available respectively from carbohydrates, protein, and fat. If all of this energy were converted to heat, what would be the temperature rise in the cell? Use the results of Problem 6.2 and ignore heat losses.

6.4 A British thermal unit (BTU) is the amount of heat needed to raise the temperature of 1 lb of water by 1 °F. (This unit is therefore defined as being equal to the specific heat of water, which is 1.0 cal/g-°C.) The amount of heat needed to freeze water or melt ice (the latent heat of fusion (freezing)) is 80 cal/g = 335 J/g and that needed to vaporize water (the latent heat of vaporization) is 543 cal/g = 2,272 J/g.
(a) Show that 1 BTU = 1,055 J.
(b) Find the number of BTU required to melt a pound of ice.
(c) Find the number of BTU required to vaporize a pound of water.

Caloric Value

6.5 Consider the oxidation of the fat tributyrin: $C_3H_5O_3(OC_4H_7)_3 + 18.5O_2 \rightarrow 15CO_2 + 13H_2O + 1,941$ kcal.
(a) Find the energy release/g, calorific equivalent, and respiratory exchange ratio (RER) (or respiratory quotient (RQ)).
(b) Are these values representative of a fat? Why or why not?

6.6 Consider the oxidation of the alcohol (ethanol): $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O + 327$ kcal.
(a) Find the energy release/g, calorific equivalent, and respiratory exchange ratio (RER) (or respiratory quotient (RQ)).
(b) Compare these values to those for carbohydrates, proteins, and fats.

6.7 Determine the energy content in one fluid ounce of 86 proof liquor from the caloric content of alcohol. (200 proof means that the volume fraction of alcohol is 100%.)

6.8 Show that the volume of alcohol in a (1 oz) shot of liquor, a glass of wine, and a bottle of beer are all about the same, and therefore they have the same energy content (use Table 6.5).

6.9 Are the energy densities per unit mass for carbohydrates, proteins, and fats listed in Table 6.3 consistent with the values in Table 6.2?

6.10 (a) Derive general expressions for the percentage of energy coming from carbohydrates and fats and the heat energy per L oxygen consumed for an arbitrary RER between 0.71 and 1.00.
(b) Determine the percentage of kcal coming from carbohydrates and fats and the heat energy per L oxygen consumed for each of these RERs: 0.71, 0.75, 0.80, 085, 0.90, 0.95, and 1.00.

6.11 Show how rounding off the carbohydrate, protein, and fat masses in the standard donut to the nearest integral value of grams can easily lead to the difference in the calculated and listed caloric content. (Show that this is true even ignoring the likely rounding off of the total caloric content to the nearest tens of kcal—so 280 kcal could mean any value from 275.1 to 284.9 kcal.) Assume that carbohydrates and proteins have caloric content 4.0 kcal/g and fats have caloric content 9.0 kcal/g.

6.12 Repeat Problem 6.11, assuming that the mass contents are exact but now considering uncertainties in the caloric contents of the donut carbohydrates, proteins, and fats.

6.13 The Entenmann's ™ Rich Frosted Donut designated in the text as our standard donut was sold as a stand-alone item for many years and as of 2006 was still available in the donut variety pack with the same food content. In 2006, the Rich Frosted Donut in stand-alone packaging had somewhat different food content: 26 g fat, 29 g carbohydrate, 2 g protein and a total mass of 64 g. Compare its caloric content to that of our standard donut.

6.14 The Entenmann's™ Glazed Donut has 13 g fat, 34 g carbohydrate, 2 g protein, and a total mass of 60 g (as of 2006).
(a) Compare its caloric content as listed on the package (260 kcal) to that of our standard donut.
(b) Find percentage of calories that come from fat.
(c) Find the percentage of donut mass that does not come from fat, carbohydrate, or protein.

6.15 The package labeling of the 2014 version of the Rich Frosted Entenmann's™ Donut (standard packaging, not a variety pack) says that a donut contains 18 g of fat, 27 g of carbohydrate, and 2 g of protein, which accounts for 47 g of the stated total 55 g mass of the donut. Show that using the same method as in the text, this suggests there are 278 kcal (278 food calories) per donut. How can this be reconciled with the claim on the package that there are 270 kcal per donut?

6.16 Is the claim on the Entenmann's™ website (downloaded on May 19, 2015) that the 2015 version of the Rich Frosted Entenmann's™ Donut has 300 kcal per donut consistent with the information that this 60 g donut has 20 g of fat, 30 g of carbohydrate, and 2 g of protein?

6.17 For which foods in Table 6.5 do the caloric contents of carbohydrates, proteins, and fats not explain the total caloric value? Why?

6.18 We know that fats have about 9 kcal/g energy content and proteins and carbohydrates have about 4 kcal/g. Explain why the metabolizable energy per unit mass of the foods in Tables 6.5 and 6.6 are so different than these numbers? In particular why are the values for milk and carrots so low? What can we learn about oatmeal and flour? What do you learn about butter and nuts?

6.19 A certain fruit has a metabolizable energy of 0.4 kcal/g normally and 2.4 kcal/g after it has been dried. Assume the fruit consists of only water, sugar, and nonmetabolizable matter and find the fraction of each in the fruit and in the fruit after it is dried.

6.20 Determine the energy content per unit mass (kcal/g) for each nonalcoholic food in Table 6.5. Explain the variation in terms of relative amounts of carbohydrates/protein/fat, water, and "ash."

6.21 Determine the percentage of calories from fat for each nonalcoholic food in Table 6.5.

6.22 If nonfat milk is simply whole milk minus the fat, determine the caloric content, mass of carbohydrate, protein, and fat, and % water of 1 cup (245 g) of nonfat milk.

6.23 How much oil is there in French fries? More precisely, what fractions of the mass and caloric value of French fries are due to the oil and what fractions are due to the potato? Use the data given in Table 6.5. Ignore the skin of the potato and consider the possibility that water is lost in the frying process. Oil is all fat. How much water is lost in the frying process?

6.24 The yolk from a large raw egg has a mass of 17 g, of which 3 g is protein and 5 g is fat (and a trace amount is carbohydrate). It has a total of 59 kcal energy content and is 49% water. Determine the corresponding values for the white of the egg, using Table 6.5.

6.25 For the wine and beer listed in Table 6.5, calculate the number of kcal and mass (g) due to alcohol (ethanol) in each, and the fractional volume of alcohol in each, assuming ethanol has a mass density of 0.79 g/cm^3.

6.26 Determining the heat of combustion of proteins from their nitrogen content needs to be done with care [68].
(a) Approximately 60% of the nitrogen in potatoes is due to proteins and 40% to aspargin. Assume there are 6.25 g protein per gram N, with a heat of combustion

is 5.8 kcal/g protein, and 4.7 g asparagine per gram N, with a heat of combustion of 3.45 kcal/g asparagin. Show that this means there are 28.24 kcal in the 5.63 g of nitrogenous compounds from 1 g of N, which corresponds to an averaged value of 5.02 kcal/per g nitrogenous compound in potatoes.
(b) What heat of combustion would be determined for the nitrogenous material with 1 g N, if you use this averaged heat of combustion and the erroneous assumption that there are 6.25 g protein per gram N in this food?
(c) Repeat this for the erroneous assumption that all measured N is due to protein.

6.27 You eat a pound of food. You know that 10% of its weight is water, 5% of its weight is from nondigestible solids, and 30% of its calories come from fat. How many calories have you ingested?

6.28 In a hypothetical proposed diet, a person is supposed to derive $40\pm5\%$, $30\pm5\%$, and $30 \pm 5\%$, of her calories from carbohydrates, proteins, and fat, respectively.
(a) Use Table 6.5 to design two different diet plans for a 2,000 kcal/day diet, using very different foods in each plan.
(b) Which diet has a daily lighter mass, and therefore is perhaps better for backpacking or space trips? Why is this so?

6.29 (a) You are at a salad bar and want to intake 600 kcal. If you assume the food is 20% water (by mass) and 5% ash, how much food should you take (in oz and g) if you assume that of the remaining amount 50% is carbohydrate, 20% is protein and 30% is fat?
(b) For this amount of food, what would the caloric value be if nonwater, nonash content were really all carbohydrate, all protein, or all fat?

6.30 The energy released per ton of detonated TNT (trinitrotoluene) is 4.18×10^9 J. The same food energy is found in how many tons of our standard donuts?

6.31 A body consumes 0.3 L-atm. of oxygen every minute.
(a) Determine the rate of heat production if only carbohydrates are being consumed.
(b) Determine the rate of heat production if only fats are being consumed.

Metabolic Rates

6.32 The Gibbs free energy ΔG of a chemical reaction is the energy available when it occurs at a constant temperature T and constant pressure (1 atm.). The free energy of ATP hydrolysis (6.16) is $\Delta G^0 = -7.3$ kcal/mol under the standard conditions of $T = 298$ K, 1 atm., pH = 7.0, and equal 1 M (molar, mol/L) concentrations of all reactants and products. For other conditions, it is $\Delta G = \Delta G^0 + RT \ln(K/K^0)$, where R is the gas constant (1.987 cal/mol-K), T is the temperature for the condition, K is the equilibrium for the condition, and K^0 is the equilibrium constant for standard conditions [75]. The equilibrium constant is the product of the concentrations of the products (ADP and P_i here) divided by that of the reactants (ATP here).

(The concentrations of other products and reactants in (6.16), such as H_2O, are essentially the same for the different conditions and do not need to be included. Why?)
(a) Calculate the free energy at $T = 310\,\text{K}$ in the cell of a representative human erythrocyte (red blood cell) with concentrations of 2.25 mM (millimolar, millimol/L) for ATP, 0.25 mM for ADP, and 1.65 mM for P_i.
(b) In resting muscle, the concentration of free ADP is thought to be between 1 and 37 μM (micromolar). Calculate the free energy at $T = 310\,\text{K}$ if the ADP concentration is 25 μM (and other concentrations are the same as in (a)). How do these values in (a) and (b) compare to those given in the text? (Other than the difference in cell types, the major difference with part (a) is that the relevant concentration is that of free ADP and not total ADP, which includes free ADP and that bound to cellular proteins.)
(c) How does the use of such free energies under cellular conditions rather than those for standard conditions affect the calculation of the efficiency of converting glucose into ATP by the body relative to that of oxidizing glucose?

6.33 Find what the efficiency of body usage of glucose would be if in aerobic metabolism 1 mol of glucose produced 36–38 mol of ATP (the conversion factor often cited several years ago), instead of the currently accepted range 30–32 mol, as given in the chapter. Assume the energy released from ATP is 14.0 kcal/mol.

6.34 Assume that the oxidation of glucose extracts all the free energy possible. Find the maximum number of ATP molecules that could be formed if ATP hydrolysis were 100% efficient. Assume the energy released from ATP is 14.0 kcal/mol.

6.35 Say that in aerobic metabolism 1 mol of glucose can produce 32 mol of ATP, while under similar conditions 1 mol of carbohydrates from glycogen can produce 33 mol of ATP. Because glycogen is made by the body from glucose, is the body getting 1 additional mole of ATP for free by first converting glucose to glycogen and then using it or is something else really happening? Explain.

6.36 The standard free energy for the hydrolysis of ADP to AMP (adenosine monophosphate) is −7.8 kcal/mol. What is it for cellular conditions? Assume the conditions for an erythrocyte in Problem 6.32a and an AMP concentration of 0.2 mM [75].

6.37 Redo Problem 6.32 for palmitic acid.

6.38 (a) The molar mass of ATP is 507 g. If a 60 kg person has a metabolic rate of 2,000 kcal/day and ATP releases 14.0 kcal/mol during hydrolysis, what mass of ATP would be needed by that person per day if the ADP formed after ATP hydrolysis were not recycled back to ATP?
(b) How does this compare to the person's mass?
(c) How many cycles of ATP hydrolysis and re-creation does an average ATP molecule make per day and per minute if there are a total of 0.2 mol of ATP in the body?
(d) Repeat each part for ATP releasing 7.3 kcal/mol during hydrolysis.

6.39 (a) The molar mass of ATP is 507 g. How many kcal would be available to the body if all of the rest mass of one ATP molecule were converted to (useful) energy? Use $E = mc^2$.
(b) What mass of ATP (in g), would a person with a metabolic rate of 2,000 kcal/day use if all of the ATP were converted to energy?
(c) If a 1 GW nuclear power plant were to convert this ATP energy to electrical energy with 20% efficiency, how much ATP would it use (in g) per second and per year?
(d) In nuclear reactions, only of fraction of the rest mass is converted to energy. Repeat (a)–(c) if that fraction were 2.5×10^{-5}, which is the fraction of rest mass energy converted to kinetic energy in the alpha particle decay $^{241}\text{Am} \rightarrow \alpha + {}^{237}\text{Np}$.

6.40 Sketch on the same set of labeled axes the available rate of ATP usage (normalized to its maximum possible value) versus time for a person beginning moderate physical activities at $t = 0$, for each of the following cases:
(a) Only the phosphagen system is available.
(b) There is not much oxygen, so only the phosphagen and anaerobic glycolysis systems are available.
(c) There is much oxygen available, so the phosphagen and aerobic metabolism systems are used.

6.41 The maximum anaerobic peak power for men is 2.1 hp and the maximum metabolic rate during anaerobic glycolysis is 1.6 mol of ATP/min. Are these rates consistent? Why? (Assume 10 kcal per mole ATP here.)

6.42 Table 6.3 hows that ATP stores much less energy per unit volume and mass than other fuels. Why then is it so important?

6.43 Stearic acid is an 18-carbon chain fatty acid that undergoes 8 complete passes in the aerobic metabolism. How many moles of ATP are formed per mole of stearic acid?

6.44 Use the data in Table 6.22 to find what the maximum power output of a human (in W) would be if the entire human volume were composed of typical tissue cells operating at maximum power output. Does this exceed the listed maximum human body power output? Is this reasonable? Why?

6.45 (a) If the usual volume of a person were instead occupied only by tightly packed myosin muscle motors, what would the power consumed by the body be (in W) and how does this compare to the basal and maximum metabolic rates?
(b) What would be the rate of temperature rise of the body for such a volume, assuming the heat capacity is that of a normal human and there are no heat losses?

6.46 How many myosin crossbridges would be needed to power a 200 hp car? (Assume the data for the myosin molecules in Table 6.22.) How does this compare to the number of crossbridges in a typical human?

Basal Metabolic Rates

6.47 Use Kleiber's Law to compare the BMR per unit mass (with overall units kcal/kg-day) for a mouse (30 g), human (70 kg), and a blue whale (10^5 kg).

6.48 Compare the metabolic rate expected for the standard man in Chap. 1 using Kleiber's Law and the Harris–Benedict equations? Why are they different? Is this reasonable or not?

6.49 How many standard donuts must a 60 kg person eat daily to maintain that person's BMR, assuming Kleiber's Law.

6.50 How much food would you have to eat each day (in g) to balance a basal metabolic rate (BMR) of 1,700 kcal/day if you only ate (a) glucose or (b) (typical) fat?

6.51 Another formulation of the Harris–Benedict equations (6.28) and (6.29) utilizes rounded-off numbers:

$$\text{For men:} \quad \text{BMR} = 66 + 13.7m_\text{b} + 5H - 6.9Y \tag{6.77}$$

$$\text{For women:} \quad \text{BMR} = 665 + 9.6m_\text{b} + 1.7H - 4.7Y, \tag{6.78}$$

where the BMR is in kcal/day, m_b is the body mass in kg, H is the height in cm, and Y is the age in yr. How different are the two formulations for a 6-ft-tall, 25-yr-old male weighing 200 lb and a 5 ft 2 in., 50-yr-old female weighing 125 lb? Are these differences significant?

6.52 How much higher is the BMR for a pregnant women? Estimate this by considering the increase in her BMR from two factors. She must supply food to build the fetus along with the supporting structure in the womb and must supply the energy for the fetus' BMR. (She is literally eating for two.) Assume she is late in pregnancy and is gaining 4 lb a month of typical human tissue, assuming this does not include extra fluid retention outside of the womb. Assume the fetus has a BMR for a baby at birth that weighs 7 lb (and use the BMR charts for children under 3-yr-old).

6.53 A gestational weight gain of 12 kg during pregnancy is estimated to include 597 g of protein and 3.74 kg of fat. How much caloric energy does this require? (Use 5.65 kcal/g protein and 9.25 kcal/g fat [18].) What fraction of the total extra caloric intake of 77,000 kcal during pregnancy is this?

6.54 During the course of pregnancy the BMR of an initially 67 kg woman increases from 1,300 to 1,720 kcal/day, as she gains 18.5 kg [58]. Can this be understood by using either $m_\text{b}^{3/4}$ scaling or the relations in Table 6.19?

6.55 (a) How many standard donuts would an expectant mother need to eat to supply the total additional energies needs of ~90,000 kcal during the course of pregnancy? (b) How many donuts would she need to eat for each of the 280 days of gestation? (c) How would this answer change knowing that added caloric intake is needed only in the second half of pregnancy?

6.56 You tell an 80-kg-man who has been sitting and inactive for some time that he is no different from an 80 W incandescent light bulb. How accurate is your statement in terms of:
(a) total power emitted
(b) power emitted per unit volume
(c) approximate cost of energy output (in terms of US cents/kW-h)
(d) types of energy that are output
(e) types of energy that are input
(f) spectrum of any light (electromagnetic radiation) that is emitted
(g) flow of gases to and from each object?

6.57 If you were to make essentially the same statement as in Problem 6.56 concerning a light-emitting diode (LED) light bulb, what would be the wattage of that bulb (for the most current LED lights)?

6.58 Use Fig. 6.14 to find the metabolic power (in kcal/day) a starving person has by metabolizing stored body energy reserves during the fourth week of starvation. How does this compare to a typical basal metabolic rate?

6.59 Does the basal metabolic rate per unit volume for a human make quantitative sense, given the basal metabolic rates for different cells and the fraction of the body that contains cells (Table 6.22)?

Metabolic Rates During Activities

6.60 In a high jump a 70-kg person elevates his center of mass by 51 cm during an extension phase (with constant acceleration) that takes 0.25 s.
(a) What is the person's kinetic energy upon takeoff?
(b) If the muscle efficiency is 20%, how much chemical energy is used to make one jump? (Express your answer in joules and in kcal (the usual food calories).)
(c) What is the average power generated by the person during the jump in watts and horsepower (1 horsepower (hp) = 746 W)?
(d) How frequently would someone have to jump (during a 12 h awake cycle) to increase his/her (daily averaged) metabolism rate to twice the basal value (which is 1,500 kcal/day)?

6.61 Use Fig. 3.26 to find the metabolic rates (in kcal/h) for running at speeds of 2 and 4 m/s, each for 50 and 90 kg people? (Have you included the BMR?)

6.62 Convert the data on aerobic capacities in the paragraph describing the capacities of people from mL/kg-min to kcal/h for 70 kg men and 50 kg women.

6.63 The metabolic rate during walking, MR in cal/min, has been determined by measuring the rate of oxygen consumption, as a function of walking speed, v in m/min:

$$\text{MR} = 0.267v^2 + 2{,}160. \tag{6.79}$$

(a) Determine the walking speed at which the energy expended per unit distance is a minimum, in m/min, m/s, and mph (miles per hour) [50]. How does this compare to the minimum shown in Fig. 3.26?
(b) Determine the power consumption rate at this optimal speed in kcal/hr.

6.64 Determine how much energy a 70 kg person uses to run a marathon on level ground.

6.65 (a) Show that the rate of change of potential energy when walking or running at a speed v on a gradient at an angle θ (positive for uphill, negative for downhill) is $m_b g v \sin\theta$.
(b) Consider the metabolic needs for a slow walk by a 50 kg female, assuming the energy cost of walking above that of standing is 2.13 J/kg-m at 1.2 m/s (2.6 mph). Assume the work and power required going uphill is about 2.7× the increase in potential energy and when moving downhill less power is needed than that needed on level terrain by about 50% of the rate of decrease of potential energy. Find the average (total) metabolic need in kcal/min for this person walking at this constant speed: (i) on a level surface, (ii) on a 10% grade uphill, (iii) on a 10% grade downhill, and (iv) when half is on the 10% uphill grade and half on the 10% downhill grade (so the final elevation is the same as the initial). (v) How does the answer in (iv) compare to that in (i)?
(c) If you wanted to maintain your metabolic rate whether you were walking on the level surface, uphill, or downhill in part (b), how fast would you walk up and downhill, assuming that on the level terrain you were moving with the speed v in part (b).

6.66 Consider the accelerating sprinter in Chap. 3. What is the metabolic rate of the sprinter (in kcal/min) if the efficiency of converting metabolic energy to mechanical energy is 20%? How much metabolic energy does the sprinter consume in a 100 m dash?

6.67 (a) Determine the metabolic rate and oxygen consumption rates for each stage of the Bruce Protocol for a 75 kg male.
(b) How many kcal are burned in each stage and what is the total energy burned if the patient finishes all stages?

6.68 (a) Determine the metabolic rate and oxygen consumption rates for each stage of the Naughton Protocol for a 75 kg male.
(b) How many kcal are burned in each stage, and what is the total energy burned if the patient finishes all stages?

6.69 How do the metabolic rates at the highest stages of the Bruce and Naughton Protocol compare?

6.70 In indirect calorimetry the rates of O_2 consumption, dV_{O_2}/dt, and CO_2 production, dV_{CO_2}/dt, are determined while the subject is engaged in a physical activity.
(a) Describe how they can be determined by measuring the rates of inspiring and expiring volumes of air, dV_{air}/dt, and the fractions of O_2 and CO_2 in both the inspired and expired air.
(b) A subject is inspiring and expiring air at a rate of 100 L/min, the inspired air is 21% O_2, and the expired air is 16% O_2. Show that dV_{O_2}/dt is 5 L/min.
(c) What is the metabolic rate of the subject in kcal/min?
(d) In what type of activity could the subject be engaging?

6.71 From the discussion after (6.18), what is $p_a - p_v$ in the blood in the system (in mL of O_2/L of blood)?

6.72 (a) If the energy needed to supply the basal metabolism needs of a 70 kg male for one day were used to increase that person's gravitational potential energy with 100% efficiency, how high would that person be lifted?
(b) If a person can climb stairs with 20% mechanical efficiency, how high can the person climb using the daily metabolic energy described in part (a)? This corresponds to about how many floors?

6.73 In football, an offensive and defensive lineman, each 300 lb, face each other, crouched in a "stance," separated by 2 yards (which is a bit of an overestimate). When a "play" begins they rush at each other, each accelerating at $g/2$, and then they collide and spend the next 5–10 s pushing each other with, in this model, no motion. Consider one of the linemen, say the offensive lineman.
(a) How much kinetic energy does he develop before the collision (in J)?
(b) If he converts metabolic energy into this kinetic energy with 10% efficiency and plays 50 plays per game, how much metabolic energy does he use in a game (in kcal)?
(c) It is said that linemen use up to 10,000 kcal in a game. Does your answer in (b) support this statement? If not, why not?

6.74 As described in Chap. 7, a person tries to fly by flapping his arms. Say that at the beginning of each flap each arm starts with no kinetic energy and at the end of the flap each arm is moving at an average speed of 20 m/s.
(a) If this kinetic energy of the arms needs to be generated 3 times a second, and the body converts metabolic energy to this mechanical energy with 10% efficiency, how many kcal/min would a 70 kg body consume in this attempt to fly? (Use the anthropometric data in Chap. 1 to determine the mass of the arms.)
(b) Would it be difficult for a person to continue this fruitless and quite silly attempt to fly for a long time? Why?

6.75 A person lifts a 15-kg mass from the floor to over his head (a distance of 2 m). How many times does he have to do this to lose a pound of fat, assuming a muscle efficiency of 25%?

6.76 A 50 kg woman does 10 chin-ups in a minute (each raising her center of mass by 0.5 m). After 5 min how much mechanical work has she done (in J) and how much metabolic energy has she used (in kcal), assuming 25% muscle efficiency?

6.77 How much energy (gravitational potential energy) is required to lift a 70 kg person by 1 m (in kcal)? If your muscles can do this with 25% efficiency, how many times would you have to lift such a person to burn off the caloric content in one standard donut (280 kcal/donut)?

6.78 After a long run, you walk for 10 min at a leisurely pace. Is your metabolic rate the same as for walking in Table 6.21? Why?

6.79 You run for a half hour and expend 300 kcal/hr above the BMR during that time. After running, you huff and puff for another 5 min. Estimate the number of kcal you expend in excess of the BMR amount during this time. Is this significant?

6.80 Determine the minimum metabolic energy in kcal that a 50 kg person needs to expend to pump a 3-m long swing until the swing makes a 60° angle with the vertical, if muscle efficiency is 20%.

6.81 Short-term peak metabolic rates are much greater than the average rates that we can maintain for long times. Let us estimate them by assuming that we operate at this peak rate when we throw a $m = 1$ kg ball a distance of $d = 60$ m. This is done by throwing it at a speed of v at takeoff angle of 45° and ignoring air resistance. Also ignore the difference between the takeoff and landing in the trajectory of the ball.
(a) Show that the peak mechanical power is $P_{\text{mech}} = mv^2/t + mgv/\sqrt{2}$, where t is the time needed to launch the object by accelerating one's arms (at a constant rate) over a distance s. We will take $s = 1$ m.
(b) If the efficiency to do mechanical work is ϵ, show that the needed peak metabolic power is $P_{\text{met}} = m(gd)^{3/2}/2\epsilon s + mg^{3/2}d^{1/2}/\sqrt{2}\,\epsilon$.
(c) Using the given values, show that the peak metabolic rate is about 14,000 W, and therefore about 200× the BMR. (The BMR is roughly the same—and is usually higher—than the maximum long-term rate of doing mechanical work. They would be equal if the daily-average activity factor were 2.)

6.82 How would usual physical activity during pregnancy affect the MR, given what you know about the BMR?

Activity Factors

6.83 Estimate the average daily metabolic rate and activity factor of a 50 kg "homemaker." (Assume that f = MET.)

6.84 Calculate your activity factor f for a typical day. (Assume that f = MET.)

6.85 A student spends 6 h a day sleeping, 11 h sitting (attending classes (and awake), eating, playing video games, etc.), 0.5 h washing and dressing, 4.75 h walking, 1 h standing, 0.25 h playing basketball, and 0.5 h cycling. What are the student's MR and activity level? (Assume the BMR of a 55 kg, 20-yr-old female student and that f = MET.)

Weight Gain and Loss

6.86 Determine how many standard donuts a (70 kg) person should eat after playing basketball for 2 h to maintain the same energy reserve as the person
(a) had before the basketball playing or
(b) would have had after 2 h of sitting instead.

6.87 A 150 lb (68 kg) person consumes 2,500 kcal every day and maintains a steady weight. Now this person decides to eat an extra slice of bread every day (100 kcal) and eventually achieves a new steady-state weight.
(a) How much weight does that person gain (in steady state) if the person's entire metabolic rate scales, respectively, as (i) $m_{\mathrm{b}}^{3/4}$ or (ii) m_{b} (m_{b} = body mass)?
(b) For both scaling relationships, how much weight will that person lose if he/she ate one less slice of bread each day?
(c) If the person wants to eat that extra slice of bread everyday, but does not want to gain the extra weight, how long would that person have to run each day to keep that extra weight off?

6.88 How much weight do people gain with age? Consider a 30-yr-old, 6 ft (1.83 m), 150 lb (68 kg) male with an activity factor f = 1.5 and assume that the person's BMR scales as in (6.30)—which has a strong $m_{\mathrm{b}}^{3/4}$ dependence—and that the total metabolic rate MR equals this BMR times the activity factor. Find how much weight the person will gain (to his new steady-state weight):
(a) when he turns 50 (with the same food intake, activity level, etc.).
(b) if the activity level of this 30-year decreases from 1.5 to 1.4 (with the same food intake, etc.).
(c) if the 30-year old (f = 1.5) increases his food intake by 100 kcal/day (one big slice of bread a day).
(d) if each of (a)–(c) occurs (i.e., the 30-year-old becomes older (now 50), is less active (f now 1.4), and eats more (100 kcal/day more)).
(e) Do the separate changes in (a)–(c) sum to give that in (d)?
(f) If the person in (c) wants to eat that extra slice of bread everyday, but does not want to gain the extra weight, how long would that person have to run each day to keep that extra weight off?
You may assume that the specific stature remains constant, at its initial value, in parts (a)–(d).

6.89 If your daily metabolic rate exceeds your caloric intake by 1,000 kcal (which is a very large difference) for a week, how much weight will you lose? Assume that the change is due only to the use of body fat to make up this difference, with 9 kcal

per gram of body fat, so there is no other concomitant loss of weight. (Are there other concomitant weight losses? Some weight loss supplements may also lead to more trivial loss of body water.)

6.90 In one study [15], healthy, insulin-sensitive ∼50 year old men were fed a diet with energy content 6,200 kcal/day for seven days. Their usual energy expenditure is 2,030 kcal/day. They gained 3.5 kg in mass, all in fat. Does this make sense? Why?

6.91 If 5 lb (2.3 kg) of body fat were changed to 5 lb of body muscle, how would that decrease the level of other body fat because of the increased MR?

6.92 What is the mass in kg and weight in pounds of a hemisphere of human tissue with a diameter of 0.4 m (with the average density of such tissue being 0.9 g/cm^3)? (This is a model of the weight gain of a pregnant woman—aside from that due to water retention—or that due to a relatively small "beer belly.")

6.93 A change in body mass of 1 kg is caused by a change in metabolizable energy of about 3,500 kcal. This means that about 3,500 kcal of chemical energy is obtainable by oxidizing 1 kg of average body storage materials. Special relativity teaches that the total energy content of a mass m is $E = mc^2$. How much is this for 1 kg (in units of kcal)? How many times larger is this than the chemical energy?

6.94 The level of sodium ions in the body depends on salt intake and how the body controls the level. The body maintains a Na^+ concentration of about 10 mM (millimolar) for intracellular fluid (normally ∼5 L) and about 145 mM for extracellular fluid (normally ∼2.5 L, which is the blood plasma half of the 5 L of whole blood). Let us say that you eat a bag of potato chips that contains 2 g of sodium (as sodium ions) and let us assume that—for the moment—it all stays in the body, and, in particular, in these fluids. Now let us say that the body responds by trying to maintain the previous Na^+ concentrations, so you then drink (and retain) enough water to maintain these concentrations. How much weight will you gain, assuming all the new Na^+ (and therefore the added water) goes to either (a) the intracellular water, (b) the extracellular water, or (c) both, so as to keep the same ratio of intracellular and extracellular fluid volumes? (The recommended daily intake of Na is about 2.5 g.)

6.95 Repeat Problem 6.94 for eating a big bag of pretzels that contains 8 g of sodium.

Heat Loss

6.96 In (6.44) there was a net radiation deficit of 116 W for a wall (and air) temperature of 24 °C. Find the deficit for a wall temperature of 0 °C.

6.97 A standard person is in interstellar space. He emits thermal radiation based on his body surface temperature and receives thermal radiation from the 3 K black body background. How long would it take the person's temperature to decrease by 5 K? Assume no other sources of heat generation and loss, and ignore the fact that the person would not live very long for many other reasons. Also assume thermal transport in the body occurs on a much shorter time scale than the one you calculate here.

6.98 Consider a person sitting nude on a beach in Florida [19]. On a sunny day, visible radiation energy from the sun is absorbed by the person at a rate of 30 kcal/h or 34.9 W. The air temperature is a warm 30 °C and the individual's skin temperature is 32 °C. The effective body surface exposed to the sun is 0.9 m^2. (Assume this same area for sun absorption, radiative transfer, and convective loss. Is this a good assumption?)
(a) Find the net energy gain or loss from thermal radiation each hour. (Assume thermal radiative gain and loss according to (6.51) and an emissivity of 1.)
(b) If there is a 4 m/s breeze, find the energy lost by convection each h (use (6.61), with (6.63)).
(c) If the individual's metabolic rate is 80 kcal/h (93.0 W) and breathing accounts for a loss of 10 kcal/h (11.6 W), how much additional heat must be lost by evaporation to keep the body core temperature constant?

6.99 How different are (6.62) and (6.63) over the range where the latter is valid?

6.100 A racing cyclist produces heat at a rate 1,300 kcal/h in excess of the normal rate. If all of this excess heat is lost by perspiration and evaporative cooling, how many liters of water must the cyclist drink every hour to maintain body fluids (at the level she would have had without cycling)?

6.101 What happens when room temperature exceeds the body temperature—for both the black body and breathing "cooling" routes?

6.102 Compare the thermal conductivities of the materials in Table 6.37 to those of the parts of the body.

6.103 Use Table 6.37 to explain why is it bad to make clothing for cold weather out of aluminum foil, very good to make it out of cotton fabric, and great to make it out of layers of cloth separated by air?

6.104 (a) How thick does your clothing have to be on a cold day? Model a person by a cylinder 1.65 m high and 0.234 m in diameter. Assume that heat loss is 1,500 kcal/day and occurs at the circumference of the cylinder (but not top or bottom). Your skin temperature is 34 °C and the outside temperature is 0 °C. Assume the insulation values in Table 6.44 and that this is the only significant form of insulation. Use the ordinary heat flow equation for flat surfaces and planar regions (6.72).
(b) The radial heat flow in a cylinder of radius r can be modeled assuming flat surfaces if the thickness Δx is replaced by $r \ln(1 + \Delta r/r)$. How does this change the result in (a)?

6.105 The R-value is often used to characterize insulation (or thermal resistance) needed and used for insulating houses. It is formally the same as the term insulation defined in this chapter. When expressed in SI units, the insulation value is called the RSI-value, so if the insulation is 3 m^2-°C/W, the insulation value is RSI-3.
(a) In the US, the value of insulation in units of ft^2-°F-hour/BTU is used and called the R-value. Show that an R-value of 10 corresponds to an RSI-value of 1.761 and

that an RSI-value of 1 corresponds to an R-value of 5.6745.
(b) Find the R-value and RSI-value of 1 Clo.
(c) R-19 insulation is typical of modest insulation needed for a house. Approximately how many polar weather suits made of heavy wool pile does this correspond to? (These suits would be in series of course. Assume they are fully open, so you have a single, and not double, layer of insulation from each.)

6.106 Will wearing gloves or mittens keep your fingers warmer during cold weather?

6.107 (a) How much energy (kcal) do you lose every hour by thermal conduction when you stand barefoot on ice? Assume the flow is between the plane of the ice, which is at a temperature 0 °C, and a plane 5 mm deep into your body (across the skin), which is at 34 °C [2, 69]. (Why is this temperature reasonable?) Assume the coefficient of thermal conductivity is 0.3 W/m-°C. (Why is this value reasonable?) Also assume that each foot contacts the ice over an area of 0.02 m^2.
(b) You need to heat your body by this amount every hour to maintain your average body temperature. How much body fat (expressed in g and lb) would your body have to burn with 100% efficiency each hour to make up for this heat loss?
(c) How many donuts would you have to eat every hour to account for this heat loss (again assuming 100% efficiency in conversion to heat)?
(d) Assume that a burning a match provides 1 kJ of energy and determine how many matches would have to be burnt every hour to supply the body with heat to make up for this loss (assuming perfect transfer of energy from the match)?

6.108 Say you stand barefoot on ice [69]. Can blood flowing into your foot supply enough heat (by the convection of the flowing blood) to counter the loss of heat by thermal conduction from your foot to the ice, which we will assume here to be 20 W? Use (6.67). Assume that blood flows into the foot at a temperature of 36 °C and leaves it at 32 °C, so the foot is at 34 °C, and the specific heat of blood is 1 cal/g °C. The mass flow rate of blood $F_m = \rho A u$, where $\rho = 1.06\,\mathrm{g/cm^3}$ is the blood mass density, A is the cross-sectional area of the artery, and u is the linear speed of blood flow. Assume a major artery modeled as cylindrical tube of radius 2 mm with a blood speed of 35 cm/s.

6.109 Assume your core body temperature varies linearly with distance from 38 °C in your head to 31 °C in your feet (so $\mathrm{d}T/\mathrm{d}x$ is constant). (You might say that you are "hot-headed" and have "cold feet," which is not a very good combination.) Assume you are 1.8 m tall, have a mass of 80 kg and a constant transverse body cross-sectional area estimated from the data in Chap. 1, and the average thermal conductivity coefficient in your body is 0.3 W/m-°C. What is the heat flow in your body by thermal conduction and which way is the heat flowing [69]?

6.110 Model a person as a right circular cylinder [69]. In which case does the person lose heat faster: (i) the surface is 34 °C everywhere or (ii) half of the surface is at 36 °C and the other half is at 32 °C, assuming loss either by
(a) radiation or
(b) conduction.

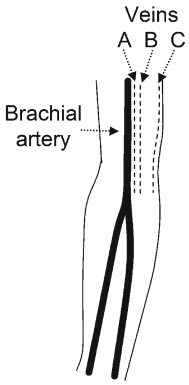


Fig. 6.27 Return blood flow in three hypothetical veins in the right arm, labeled A, B, and C)

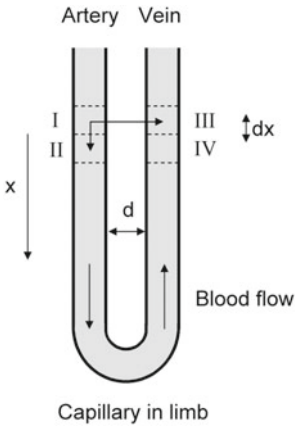


Fig. 6.28 Model of countercurrent heat exchange between an artery and vein. Heat from the artery flows either from region I to II by convection or from I to III by heat conduction (based on [69])

6.111 Use (6.64) to find a relation for the wind chill temperature in metric units, with temperature still in °C, but with wind speed in m/s.

6.112 Use (6.64) to find a relation for the wind chill temperature in English units, with temperature in °F and wind speed in mph.

6.113 Blood flow in which veins in Fig. 6.27 promotes
(a) relatively low heat loss to the outside world by direct thermal conduction
(b) high heat loss to the outside world by direct thermal conduction
(c) low heat loss by countercurrent flow?

6.114 (advanced problem) Determine the temperature profile in an artery ($T_a(x)$) and a vein ($T_v(x)$) that are close enough that they form a countercurrent heat exchange system [69], as in Figs. 6.18 and 6.19. Use the model in Fig. 6.28. The artery and vein are in such close proximity that there is rapid heat transfer between them. For simplicity, assume both vessels are identical and have cross-sectional area A. Blood flows in them with a speed v in the x and $-x$ directions, respectively, and has a specific heat c and mass density ρ.

(a) Explain why the conservation of energy implies that the rate of heat conduction from the warmer artery to the colder vein (from region I to III in Fig. 6.28) equals the loss of heat energy in the convection of blood flowing in the artery (from region I to II, because it gets colder along the x direction), and that this also equals the gain of heat energy in the convection of blood flowing in the vein (from region IV to III, because it gets warmer as it travels in the $-x$ direction).
(b) If the thermal conductivity is K, the distance between the vessels is d (approximately the combined thicknesses of the vessels), and the contact area is $w\,\mathrm{d}x$, where w is the width of the contact area along length $\mathrm{d}x$, show that the rate of heat conduction from the artery to the vein is

$$R_{\text{conduction}} = \frac{Ka}{d}(T_{\text{a}}(x) - T_{\text{v}}(x)) \tag{6.80}$$

(c) Show that the loss of heat flow in convection along the artery is

$$R_{\text{convection}} = -c\rho A v \frac{\mathrm{d}T_{\text{a}}(x)}{\mathrm{d}x} \tag{6.81}$$

(d) Combine (6.80) and (6.81) to obtain

$$\frac{\mathrm{d}T_{\text{a}}(x)}{\mathrm{d}x} = \frac{Kw}{c\rho A v d}(T_{\text{a}}(x) - T_{\text{v}}(x)) \tag{6.82}$$

and similarly for veins show that

$$\frac{\mathrm{d}T_{\text{v}}(x)}{\mathrm{d}x} = \frac{Kw}{c\rho A v d}(T_{\text{a}}(x) - T_{\text{v}}(x)) \tag{6.83}$$

(e) Solve these two equations to show that the temperature difference between the artery and vein is constant along x and that the temperature in each decreases linearly with x, by

$$T_{\text{a}}(x) = T_{\text{a}}(0) - \frac{Kw}{c\rho A v d}\,\Delta T\; x \tag{6.84}$$

and

$$T_{\text{v}}(x) = T_{\text{v}}(0) - \frac{Kw}{c\rho A v d}\,\Delta T\; x, \tag{6.85}$$

where $\Delta T = T_{\text{a}}(0) - T_{\text{v}}(0)$. (f) Sketch $T_{\text{a}}(x)$ and $T_{\text{v}}(x)$ versus x along a limb of length L, alternately if $(Kw/c\rho A v d)L$ is $\gg$ or $\ll 1$. What do these two very different conditions indicate?

6.115 Someone has a daily metabolic rate of 2,200 kcal, which is all converted to heat. If 21% of heat loss is through evaporation of water (on the skin and in the lungs), how much water is evaporated each day (in L)?

6.116 Derive (6.70).

6.117 Equation (6.70) explicitly neglects any heating or cooling of the water vapor. Is this a reasonable approximation? Why?

6.118 Calculate the values of the two terms in (6.70) in kcal/day units assuming the person inspires air at 20 °C (68 °F) that is very dry (0% relative humidity) at a rate of 6 L/min and exhales air (at the same rate of course) at 37 °C (98.6 °F) that is totally saturated with water vapor (100% relative humidity, 47.1 mmHg partial pressure, 44.0 g/m^3 water vapor). How significant is each loss to the overall heat loss by the body?

6.119 Repeat Problem 6.118 for the following three cases.
(a) For a person inhaling air at 20 °C that is totally saturated with water vapor (100% relative humidity, 17.5 mmHg partial pressure, 17.3 g/m^3 water vapor).
(b) For a person inhaling air at 40 °C (104 °F) that is very dry (0% relative humidity).
(c) For a person inhaling air at 40 °C that is totally saturated with water vapor (100% relative humidity, 55.3 mmHg partial pressure, 51.1 g/m^3 water vapor).
(d) Compare the overall cooling rate for these three cases with each other and that in Problem 6.118.

6.120 Explain each of the factors leading to the heat index.

6.121 The heat index, HI, has been fit by

$$\begin{aligned} \text{HI} = &-42.379 + 2.04901523T + 10.14333127R - 0.22475541TR \\ &- 6.83783 \times 10^{-3}T^2 - 5.481717 \times 10^{-2}R^2 + 1.22874 \times 10^{-3}T^2R \\ &+ 8.5282 \times 10^{-4}TR^2 - 1.99 \times 10^{-6}T^2R^2, \end{aligned} \tag{6.86}$$

with the HI and the ambient dry bulb temperature T in °F and the relative humidity R in % [96, 97]. (This assumes a wind speed of 5.65 mph (5 knots) and has an error of ±1.3 °F. Exposure to full sunshine can increase the HI by up to 15 °F.)
(a) Construct a table of the heat index in °F, with columns describing dry (bulk) temperatures ranging from 80–110 °F (in steps of 2 °F) and rows describing the relative humidity ranging from 40–100% (in steps of 5%).
(b) Mark the combinations of temperature and humidity in the table for each of the four health warning regions described in the text.

6.122 Repeat Problem 6.121 for 27–47 °C (in steps of 2 °C) and relative humidity ranging from 40–100% (in steps of 5%). (Convert the temperatures from °C to °F and then use (6.86).)

6.123 (a) Convert (6.86) from HI and T both being in °F to both being in °C.
(b) Repeat Problem 6.122, working directly in °C.

6.124 Consider two animals that we will model as cylinders [69]. Both have body temperatures of 35 °C and live in an environment of 0 °C. Animal A has a length of 1 m, diameter of 1 m, and a 1-cm thick layer of fat (outside the given diameter) for insulation. Animal B has a length of 6 m, diameter of 0.5 m, and a 4-cm thick layer of fat for insulation.

(a) Which animal has more thermal energy (heat)?
(b) Which animal loses heat at a faster rate? (Include losses around the diameter and at the ends, and use simple theory—i.e., assume planar flow and ignore the corrections needed for cylindrical flow.)
(c) Which animal loses a larger fraction of heat per unit time?
(d) Which animal do you expect to have the larger BMR per unit mass? Why?
(e) Are heat losses less for the more spherically shaped animals, such as animal A, or the more pencil-shaped animals, such as animal B? (Ignore the different thicknesses of the fat insulating layers.)

6.125 (a) Find the skin temperature assuming that 1 cm of fat, skin (with no thermal insulating properties assumed here), and 1 cm of clothing separate the body core at 37 °C and the outside at 20 °C. Take the thermal conductivity K of fat to be 5×10^{-4} cal/cm-s-°C and that of the clothing to be equal to that of air 6×10^{-5} cal/cm-s-°C [69].
(b) Find the heat loss per cm^2 per day.
(c) Find the heat loss per day over the entire body using a surface area of 1.5 m^2. How does this compare to the BMR? Does this make sense?

6.126 Repeat Problem 6.125 assuming the person is wearing no clothing.

References

1. T.C. Adam, E.S. Epel, Stress, eating and the reward system. Physiol. Behav. **91**, 449–458 (2007)
2. B.K. Ahlborn, *Zoological Physics: Quantitative Models, Body Design, Actions and Physical Limitations in Animals* (Springer, Berlin, 2004)
3. L.C. Aiello, P. Wheeler, The expensive-tissue hypothesis: the brain and the digestive system in human and primate evolution. Curr. Anthropol. **36**, 199–221 (1995)
4. R.M. Alexander, *The Human Machine* (Columbia University Press, New York, 1992)
5. E. Archer, R.P. Shook, D.M. Thomas, T.S. Church, P.T. Katzmarzyk, J.R. Hbert, K.L. McIver, G.A. Hand, C.J. Lavie, S.N. Blair, 45-year trends in womens use of time and household management energy expenditure. PLoS One **8**, e56620 (2013)
6. W.O. Atwater, F.G. Benedict, *Experiments on the Metabolism of Matter and Energy in the Human Body*. Agricultural Bulletin No. 136 (U.S. Department of Agriculture, Washington, 1903)
7. D.R. Bassett Jr, J. Pucher, R. Buehler, D.L. Thompson, S.E. Crouter, Walking, cycling, and obesity rates in Europe, North America, and Australia. J. Phys. Act. Health. **5**, 795–814 (2008)
8. G. Beccuti, S. Pannain, Sleep and obesity. Curr. Opin. Clin. Nutr. Metab. Care **14**, 402–412 (2011)
9. G.B. Benedek, F.M.H. Villars, *Physics with Illustrative Examples from Medicine and Biology, Three volumes*, 2nd edn. (AIP/Springer, New York/Berlin, 2000)
10. P.J. Berenson, W.G. Robertson: Temperature, in *Bioastronautics Data Book*, J.F. Parker Jr., V.R. West, eds. (NASA, Washington, 1973), Chap. 3, pp. 65–148
11. K. Blaxter, *Energy Metabolism in Animals and Man* (Cambridge University Press, Cambridge, 1989)
12. S.N. Blair, H.W. Kohl, C.E. Barlow, R.S. Paffenbarger, L.W. Gibbons, C.A. Macera, Changes in physical fitness and all-cause mortality: a prospective study of healthy and unhealthy men. J. Am. Med. Assoc. (JAMA) **273**, 1093–1098 (1995)

13. J.E. Blundell, N.A. King, Physical activity and regulation of food intake: current evidence. Med. Sci. Sports Exerc. **31**, S573–S589 (1999)
14. S.M. Boback, C.L Cox, B.D. Ott, R. Carmody, R.W. Wrangham, S.M. Secor: Cooking and grinding reduces the cost of meat digestion. Comp. Biochem. Physiol. A Mol. Integr. Physiol. **148**, 651–656 (2007)
15. G. Boden, C. Homko, C.A. Barrero, T.P. Stein, X. Chen, P. Cheung, C. Fecchio, S. Koller, S. Merali: Excessive caloric intake acutely causes oxidative stress, GLUT4 carbonylation, and insulin resistance in healthy men. Sci. Transl. Med. **7**, 304re7 (2015)
16. R.W. Bullard, Temperature regulation, in *Physiology*, 4th edn., ed. by E.E. Selkurt (Little Brown, Boston, 1976), p. 689
17. N.F. Butte, J.K. Moon, W.W. Wong, J.M. Hopkinson, E.O.B. Smith, Energy requirements from infancy to adulthood. Am. J. Clin. Nutr. **62**(5), 1047S–1052S (1995)
18. N.F. Butte, J.C. King, Energy requirements during pregnancy and lactation. Public Health Nutr. **8**, 1010–1027 (2005)
19. J.R. Cameron, J.G. Skofronick, R. Grant, *Physics of the Body*, 2nd edn. (Medical Physics, Madison, 1999)
20. N.A. Campbell, J.B. Reece, *Biology*, 7th edn. (Pearson-Benjamin Cummings, San Francisco, 2005)
21. R.N. Carmody, G.S. Weintraub, R.W. Wrangham, Energetic consequences of thermal and nonthermal food processing. Proc. Natl. Acad. Sci. (USA) **108**, 19199–19203 (2011)
22. K. Cena, J.A. Clark, Thermal insulation of animal coats and human clothing. Phys. Med. Biol. **23**, 565–591 (1978)
23. *Composition of Foods: Raw, Processed, Prepared-USDA National Nutrient Database for Standard Reference, Release 26, Documentation and User Guide*, revised November 2013 (U.S. Department of Agriculture, 2013) www.ars.usda.gov/sp2UserFiles/.../12354500/Data/SR26/sr26_doc.pdf
24. J.L. Clapperton, J.P. Joyce, K.L. Blaxter, Estimates of the contribution of solar radiation to the thermal exchanges of sheep at a latitude of 55°N. J. Agri. Sci. (Cambridge) **64**, 37–49 (1965)
25. C.M. Cook, D.A. Schoeller, Physical activity and weight control: conflicting findings. Curr. Opin. Clin. Nutr. Metab. Care **14**, 419–424 (2011)
26. K.E. Cooper, J.R. Kenyon, A comparison of temperatures measured in the rectum, sophagus, and on the surface of the aorta during hypothermia in man. Br. J. Surg. **44**, 616–619 (1957)
27. P.S. Dodds, D.H. Rothman, J.S. Weitz, Re-examination of the 3/4-law of metabolism. J. Theor. Biol. **209**, 9–27 (2001)
28. A.J. Donato, K. Tench, D.H. Glueck, D.R. Seals, I. Eskurza, H. Tanaka, Declines in physiological functional capacity with age: a longitudinal study in peak swimming performance. J. Appl. Physiol. **94**, 764–769 (2003)
29. H.M. Dunsworth, A.G. Warrener, T. Deacon, P.T. Ellison, H. Pontzer, Metabolic hypothesis for human altriciality. Proc. Natl. Acad. Soc. USA **109**, 15212–15216 (2012)
30. P.T. Ellison, *On Fertile Ground: A Natural History of Human Reproduction* (Harvard University Press, Cambrige, 2003)
31. P. Evenepoel, B. Geypens, A. Luypaerts, M. Hiele, Y. Ghoos, P. Rutgeerts, Digestibility of cooked and raw egg protein in humans as assessed by stable isotope techniques. J. Nutr. **128**, 1716–1722 (1998)
32. P. Evenepoel, D. Claus, B. Geypens, M. Hiele, K. Geboes, P. Rutgeerts, Y. Ghoos, Amount and fate of egg protein escaping assimilation in the small intestine of humans. Am. J. Physiol.—GI. **277**, G935–G943 (1999)
33. FAO/WHO/UNU: *Energy and Protein Requirements. Report of a Joint Experts Consultation*. WHO Technical Report Series No. 724 (World Health Organization, Geneva, 1985). http://www.fao.org/docrep/003/aa040e/aa040e00.htm
34. R. Floud, R.W. Fogel, B. Harris, S.C. Hong, *The Changing Body: Health, Nutrition, and Human Development in the Western World Since 1700* (Cambridge University Press, Cambridge, 2011)

35. M.L. Foss, S.J. Keteyian, *Fox's Physiological Basis for Exercise and Sport*, 6th edn. (McGraw-Hill, Boston, 1998)
36. E.L. Fox, D.K. Matthews, *Interval Training Conditioning for Sport and General Fitness* (Saunders, Philadelphia, 1974)
37. S.M. Frank, Y. Shir, S.N. Raja, L.A. Fleisher, C. Beattie, Core hypothermia and skin-surface temperature gradients. Epidural versus general anesthesia and the effects of age. Anesthesiology **80**, 502–508 (1994)
38. R.A. Freitas Jr., *Nanomedicine, Volume I: Basic Capabilities* (Landes Bioscience, Austin, 1999)
39. J.S. Garrow, Resting metabolic rate as a determinant of energy expenditure in man, in *Substrate and Energy Metabolism in Man*, ed., by J.S. Garrow, D. Halliday (Libbey, London, 1985), pp. 102–107
40. R.C. Garry, R. Passmore, G.M. Warnock, J.V.G.A. Durnin: Expenditure of energy and the consumption of food by miners and clerks, Fife, Scotland. Medical Research Council Special Report Series 289 (HMSO, London, 1955)
41. S.E. Gebhardt, R.G. Thomas, *Nutritive Value of Foods*, U.S. Department of Agriculture, Agricultural Research Service, Home and Garden Bulletin 72 (2002)
42. J.F. Gillooly, J.H. Brown, G.B. West, V.M. Savage, E.L. Charnov, Effects of size and temperature on metabolic rates. Science **293**, 2248–2251 (2001)
43. C.V. Gisolfi, C.B. Wenger, Temperature regulation during excerise: old concepts, new ideas. Exerc. Sport Sci. Rev. **12**, 339–372 (1984)
44. H. Gray, *The Organs of the Senses and the Common Integument, in Anatomy of the Human Body*, 20th edn. (Lea & Febiger, Philadelphia, 1918)
45. A.C. Guyton, J.E. Hall, *Human Physiology and Mechanisms of Disease*, 6th edn. (Saunders, Philadelphia, 1997)
46. K.A. Hammond, J. Diamond, Maximal sustained energy budgets in humans and animals. Nature **386**, 457–462 (1997)
47. W.P.T. James, E.C. Schofield, *Human Energy Requirements: A Manual for Planners and Nutritionists* (Oxford University Press, Oxford, 1990)
48. D.J. Johns, J. Hartmann-Boyce, S.A. Jebb, P. Aveyardl, Diet or exercise interventions vs combined behavioral weight management programs: A Systematic review and meta-analysis of direct comparisons. J. Acad. Nutr. Diet. **114**, 1557–1568 (2014)
49. W.V. Judy, Body temperature regulation, in *Physiology*, 5th edn., Chap. 28, ed. by E.E. Selkurt (Little Brown, Boston, 1984)
50. M.C.K. Khoo, *Physiological Control Systems: Analysis, Simulation, and Estimation* (Piscataway, IEEE, 2000)
51. M. Kleiber, Body size and metabolism. Hilgardia **6**, 315–353 (1932)
52. M. Kleiber, Body size and metabolic rate. Physiol. Rev. **27**, 511–541 (1947)
53. M. Kleiber, *The Fire of Life: An Introduction to Animal Energetics* (Wiley, New York, 1961)
54. T.M. Larsen, S.M. Dalskov, M. van Baak, S.A. Jebb, A. Papadaki, A.F. Pfeiffer, J.A. Martinez, T. Handjieva-Darlenska, M. Kunesová, M. Pihlsgård, S. Stender, C. Holst, W.H. Saris, A. Astrupl, Diets with high or low protein content and glycemic index for weight-loss maintenance. N. Eng. J. Med. **363**, 2102–2113 (2010)
55. A.L. Lehninger, *Biochemisty: The Molecular Basis of Cell Structure and Function*, 2nd edn. (Worth, New York, 1975)
56. D.E. Lieberman, *The Story of the Human Body: Evolution, Health, and Disease* (Pantheon, New York, 2013)
57. H. Lodish, A. Berk, P. Matsudaira, C.A. Kaiser, M. Krieger, M.P. Scott, S.L. Zipursky, J. Darnell, *Molecular Cell Biology*, 5th edn. (Freeman, New York, 2004)
58. M. Lof, H. Olausson, K. Bostrom, B. Janerot-Sjberg, A. Sohlstrom, E. Forsum, Changes in basal metabolic rate during pregnancy in relation to changes in body weight and composition, cardiac output, insulin-like growth factor I, and thyroid hormones and in relation to fetal growth. Am. J. Clin. Nutr. **81**, 678–685 (2005)
59. D.L. Longo, A.S. Fauci, D.L. Kasper, S.L. Hauser, J. Jameson, J. Loscalzo (eds.), *Harrison's Principles of Internal Medicine*, 18th edn. (McGraw-Hill, New York, 2011)

60. R. Margaria, P. Cerretelli, P. Aghemo, G. Sassi, Energy cost of running. J. Appl. Physiol. **18**, 367–370 (1963)
61. R.R. Markwald, E.L. Melanson, M.R. Smith, J. Higgins, L. Perreault, R.H. Eckel, K.P. Wright Jr, Impact of insufficient sleep on total daily energy expenditure, food intake, and weight gain. Proc. Nat. Acad. Sci. (USA) **110**, 5695–5700 (2013)
62. F.W. Marlowe, Hunter-gatherers and human evolution. Evol. Anthropol. **14**, 54–67 (2005)
63. A.G. Marshall, *Biophysical Chemistry* (Wiley, New York, 1978)
64. R.D. Mattes, P.M. Kris-Etherton, G.D. Foster, Impact of peanuts and tree nuts on body weight and healthy weight loss in adults. J. Nutr. **138**, 1741S–1745S (2008)
65. T.A. McMahon, Size and shape in biology. Science **179**, 1201–1204 (1973)
66. T.A. McMahon, *Muscles, Reflexes, and Locomotion* (Princeton University Press, Princeton, 1984)
67. A. McTiernan, B. Sorensen, M.L. Irwin, A. Morgan, Y. Yasui, R.E. Rudolph, C. Surawicz, J.W. Lampe, P.D. Lampe, K. Ayub, J.D. Potter, Exercise effect on weight and body fat in men and women. Obesity **15**, 1496–1512 (2007)
68. A.L. Merrill, B.K. Watt: *Energy Value of Foods-Basis and Derivation—Agriculture Handbook No. 74*, revised Feb. 1973 (U.S. Department of Agriculture, 1973) http://www.ars.usda.gov/SP2UserFiles/Place/12354500/Data/Classics/ah74.pdf
69. H.J. Metcalf, *Topics in Classical Biophysics* (Prentice-Hall, Englewood Cliffs, 1980)
70. J.L. Monteith, *Principles of Environmental Physics* (Edward Arnold, London, 1973)
71. L.E. Mount, *Adaptation to the Thermal Environment* (Edward Arnold, London, 1979)
72. L.E. Mount, in *Heat Loss from Animals and Man*, ed. by J.L. Monteith, L.E. Mount (Butterworths, London, 1974)
73. J. Moya, L. Phillips, J. Sanford, M. Wooton, A. Gregg, L. Schuda, A review of physiological and behavioral changes during pregnancy and lactation: potential exposure factors and data gaps. J. Expo. Sci. Environ. Epidemiol. **24**, 449–458 (2014)
74. M.W. Moyer, Supercharging brown fat to battle obesity: why turning down the thermostat could help win the battle of the bulge. Scientific American **311**(2) (2014). http://www.scientificamerican.com/article/supercharging-brown-fat-to-battle-obesity/
75. D.L. Nelson, M.M. Cox, *Lehninger Principles of Biochemisty*, 4th edn. (Worth, New York, 2005)
76. J.A. Novotny, S.K. Gebauer, D.J. Baer, Discrepancy between Atwater factor predicted and empirically measured energy value of almonds in human diets. Am. J. Clin. Nutr. **96**, 296–301 (2012)
77. R. Passmore, J.V.G.A. Durnin, *Energy, Work and Leisure* (Heinemann Educational Books, London, 1967)
78. C.C. Peterson, K.A. Nagy, J. Diamond, Sustained metabolic scope. Proc. Natl. Acad. Sci. USA **87**, 2324–2328 (1990)
79. A. Pietrobelli, D.B. Allison, S. Heshka, M. Heo, Z.M. Wang, A. Bertkau, B. Laferrere, M. Rosenbaum, J.F. Aloia, F.X. Pi-Sunyer, S.B. Heymsfield, Sexual dimorphism in the energy content of weight change. Int. J. Obes. **26**, 1339–1348 (2002)
80. M.H. Pimentel, D. Pimentel (eds.), *Food, Energy, and Society*, 3rd edn. (Edited by CRC Press, Boca Raton, 2007)
81. P. Poirier, J.-P. Desprs, Exercise in weight management of obesity. J Cardiol. Clin. **19**, 459–470 (2001)
82. H. Pontzer, D.A. Raichlen, B.M. Wood, A.Z.P. Mabulla, S.B. Racette, F.W. Marlowe, Hunter-Gatherer Energetics and Human Obesity PLoS One. **7**(7). doi:10.1371/journal.pone.0040503 (2012)
83. P.E. di Prampero, The energy cost of human locomotion on land and in water. Int. J. Sports Med. **7**, 55–72 (1986)
84. A.M. Prentice, R.G. Whitehead, S.B. Roberts, A.A. Paul, Long-term energy balance in childbearing Gambian women. Am. J. Clin. Nutr. **34**, 2790–2799 (1981)
85. A.M. Prentice, C.J. Spaaij, G.R. Goldberg, S.D. Poppitt, J.M. van Raaij, M. Totton, D. Swann, A.E. Black, Energy requirements of pregnant and lactating women. Eur. J. Clin. Nutr. **50**(Suppl 1), S82–S110 (1996)

86. L.G.C.E. Pugh, The influence of wind resistance in running and walking and the efficiency of work against horizontal and vertical forces. J. Physiol. (London) **48**, 518–522 (1971)
87. C. Reiff, K. Marlatt, D.R. Dengel, Difference in caloric expenditure in sitting versus standing desks. J. Phys. Act. Health. **9**, 1009–1011 (2012)
88. V.C. Rideout, *Mathematical and Computer Modeling of Physiological Systems* (Prentice-Hall, Englewood Cliffs, 1991)
89. R.M. Ross, A.S. Jackson, *Exercise Concepts, Calculations, and Computer Applications* (Benchmark, Carmel, 1990)
90. W.N. Schofield, Predicting basal metabolic rate, new standards and review of previous work. Hum. Nutr. Clin. Nutr. **39C**(Suppl. 1), 5–41 (1985)
91. G.G. Schwartz, L.A. Rosenblum, Allometry of primate hair density and the evolution of human hairlessness. Am. J. Phys. Anthropol. **55**, 9–12 (1981)
92. J.W. Severinghaus, Temperature gradients during hypothermia. Ann. New York Acad. Sci. **80**, 515–521 (1959)
93. J. Singh, A.M. Diaz, E. Prentice, W.A. Coward, J. Ashford, M. Sawyer, R.G. Whitehead, Energy expenditure of Gambian women during peak agricultural activity measured by the doubly-labelled water method. Brit. J. Nutr. **62**, 315–329 (1989)
94. D.W. Spady, P.R. Payne, D. Picou, J.C. Waterlow, Energy balance during recovery from malnutrition. Am. J. Clin. Nutr. **29**, 1073–1088 (1976)
95. A. Stalheim-Smith, G.K. Fitch, *Understanding Human Anatomy and Physiology* (West, St. Paul, MN, 1993)
96. R.G. Steadman, The assessment of sultriness. Part I. A temperature-humidity index based on human physiology and clothing science. J. Appl. Meteorol. **18**, 861–873 (1979)
97. R.B. Stull, *Meteorology for Scientists and Engineers*, 2nd edn. (Brooks Cole, Pacific Grove, 1999)
98. W.C. Swinbank, Long wave radiation from clear skies. Q. J. R. Meterol. Soc. **89**, 339–351 (1963)
99. H. Tanaka, D.R. Seals, Topical review: endurance exercise performance in Masters athletes: age-associated changes and underlying physiological mechanisms. J. Physiol. **586**, 55–63 (2008)
100. *USDA National Nutrient Database for Standard Reference, Release 26. Nutrient Data Laboratory* (U.S. Department of Agriculture, Agricultural Research Service, 2013) http://www.ars.usda.gov/ba/bhnrc/ndl
101. A.A. van der Klaauw, I.S. Farooqi, Review: the hunger genes: pathways to obesity. Cell **161**, 119–132 (2015)
102. D.T. Villareal, S. Chode, N. Parimi, D.R. Sinacore, T. Hilton, R. Armamento-Villareal, N. Napoli, C. Qualls, K. Shah, Weight loss, exercise, or both and physical function in obese older adults. N. Eng. J. Med. **364**, 1218–1229 (2011)
103. H.T. Waaler, Height, weight and mortality The Norwegian experience. Acta Medica Scand. **215**(S679), 1–56 (1984)
104. P.E. Wheeler, The thermoregulatory advantages of hominid bipedalism in open equatorial environments: The contribution of increased convective heat loss and cutaneous evaporative cooling. J. Hum. Evol. **21**, 107–115 (1991)
105. D.R. Wilkie, an as a source of mechanical power. Ergonomics **3**, 1–8 (1960)
106. J.H. Wilmore, D.L. Costill, *Physiology of Sport and Exercise*, 3rd edn. (Human Kinetics, Champaign, 2004)
107. D.A. Winter, *Biomechanics and Motor Control of Human Movement*, 3rd edn. (Wiley, New York, 2005)
108. M. Wishnofsky, Caloric equivalents of gained or lost weight. Am. J. Clin. Nutr. **6**, 542–546 (1958)